CLASS 10 MATHEMATICS MASTER PRACTICE WORKSHEET

2,800 Structurally Diverse Questions Across 14 Core Patterns

Designed for AI Interactive Quiz Generation, Timed Tests, and Board Exam Mastery

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WORKBOOK STRUCTURE & INSTRUCTIONS

This master worksheet contains **2,800 practice questions** systematically generated across 14 high-yield board exam patterns (200 questions per pattern). Each pattern is subdivided into **8 distinct question sub-types** (25 questions per sub-type):

- **Sub-type A**: Direct Computation & Algebraic Factoring
- **Sub-type B**: Reverse Synthesis & Reconstructing Equations
- **Sub-type C**: Parameter Evaluation & Unknown Coefficients (k, p, m)
- **Sub-type D**: Multiple Choice Questions with Distractors (Optimized for 5-Min AI Quizzes)
- **Sub-type E**: Conceptual, Discriminant Analysis & Property Verification
- **Sub-type F**: Real-World Application & Contextual Word Problems
- **Sub-type G**: Analytical Ratios & Graphical Geometry
- **Sub-type H**: Multi-Step Comparative Reasoning & Advanced Transformations

Part 1 contains Questions Q1 to Q2800. Part 2 contains the Complete Answer Key.

PART 1: PRACTICE QUESTIONS (Q1 – Q2800)

Pattern 1: Middle-Term Splitting with Radical Coefficients

Q1. [A] Find the zeroes of the quadratic polynomial $p(x) = \sqrt{2}x^2 + 3x + \sqrt{2}$ by middle-term splitting.

Q2. [A] Find the zeroes of the quadratic polynomial $p(x) = 2\sqrt{3}x^2 + 14x + 2\sqrt{3}$ by middle-term splitting.

Q3. [A] Find the zeroes of the quadratic polynomial $p(x) = 3\sqrt{5}x^2 + 48x + 3\sqrt{5}$ by middle-term splitting.

Q4. [A] Find the zeroes of the quadratic polynomial $p(x) = 4\sqrt{7}x^2 + 32x + 4\sqrt{7}$ by middle-term splitting.

Q5. [A] Find the zeroes of the quadratic polynomial $p(x) = 5\sqrt{2}x^2 + 25x + 5\sqrt{2}$ by middle-term splitting.

Q6. [A] Find the zeroes of the quadratic polynomial $p(x) = \sqrt{3}x^2 + 15x + 6\sqrt{3}$ by middle-term splitting.

Q7. [A] Find the zeroes of the quadratic polynomial $p(x) = 2\sqrt{5}x^2 + 17x + 7\sqrt{5}$ by middle-term splitting.

Q8. [A] Find the zeroes of the quadratic polynomial $p(x) = 3\sqrt{7}x^2 + 43x + \sqrt{7}$ by middle-term splitting.

Q9. [A] Find the zeroes of the quadratic polynomial $p(x) = 4\sqrt{2}x^2 + 26x + 2\sqrt{2}$ by middle-term splitting.

Q10. [A] Find the zeroes of the quadratic polynomial $p(x) = 5\sqrt{3}x^2 + 18x + 3\sqrt{3}$ by middle-term splitting.

Q11. [A] Find the zeroes of the quadratic polynomial $p(x) = \sqrt{5}x^2 + 14x + 4\sqrt{5}$ by middle-term splitting.

Q12. [A] Find the zeroes of the quadratic polynomial $p(x) = 2\sqrt{7}x^2 + 47x + 5\sqrt{7}$ by middle-term splitting.

Q13. [A] Find the zeroes of the quadratic polynomial $p(x) = 3\sqrt{2}x^2 + 12x + 6\sqrt{2}$ by middle-term splitting.

Q14. [A] Find the zeroes of the quadratic polynomial $p(x) = 4\sqrt{3}x^2 + 31x + 7\sqrt{3}$ by middle-term splitting.

Q15. [A] Find the zeroes of the quadratic polynomial $p(x) = 5\sqrt{5}x^2 + 76x + \sqrt{5}$ by middle-term splitting.

Q16. [A] Find the zeroes of the quadratic polynomial $p(x) = \sqrt{7}x^2 + 9x + 2\sqrt{7}$ by middle-term splitting.

Q17. [A] Find the zeroes of the quadratic polynomial $p(x) = 2\sqrt{2}x^2 + 11x + 3\sqrt{2}$ by middle-term splitting.

Q18. [A] Find the zeroes of the quadratic polynomial $p(x) = 3\sqrt{3}x^2 + 31x + 4\sqrt{3}$ by middle-term splitting.

Q19. [A] Find the zeroes of the quadratic polynomial $p(x) = 4\sqrt{5}x^2 + 25x + 5\sqrt{5}$ by middle-term splitting.

Q20. [A] Find the zeroes of the quadratic polynomial $p(x) = 5\sqrt{7}x^2 + 76x + 6\sqrt{7}$ by middle-term splitting.

Q21. [A] Find the zeroes of the quadratic polynomial $p(x) = \sqrt{2}x^2 + 13x + 7\sqrt{2}$ by middle-term splitting.

Q22. [A] Find the zeroes of the quadratic polynomial $p(x) = 2\sqrt{3}x^2 + 7x + \sqrt{3}$ by middle-term splitting.

Q23. [A] Find the zeroes of the quadratic polynomial

$p(x) = 3\sqrt{5}x^2 + 32x + 2\sqrt{5}$ by middle-term splitting.

Q24. [A] Find the zeroes of the quadratic polynomial $p(x) = 4\sqrt{7}x^2 + 87x + 3\sqrt{7}$ by middle-term splitting.

Q25. [A] Find the zeroes of the quadratic polynomial $p(x) = 5\sqrt{2}x^2 + 14x + 4\sqrt{2}$ by middle-term splitting.

Q26. [B] Form a quadratic polynomial with radical leading coefficient $\sqrt{3}$ whose zeroes are $x = -\frac{2\sqrt{3}}{3}$ and $x = -\frac{5\sqrt{3}}{3}$.

Q27. [B] Form a quadratic polynomial with radical leading coefficient $2\sqrt{5}$ whose zeroes are $x = -\frac{3\sqrt{5}}{5}$ and $x = -\frac{6\sqrt{5}}{10}$.

Q28. [B] Form a quadratic polynomial with radical leading coefficient $3\sqrt{7}$ whose zeroes are $x = -\frac{1\sqrt{7}}{7}$ and $x = -\frac{7\sqrt{7}}{21}$.

Q29. [B] Form a quadratic polynomial with radical leading coefficient $4\sqrt{2}$ whose zeroes are $x = -1\sqrt{2}$ and $x = -\frac{1\sqrt{2}}{8}$.

Q30. [B] Form a quadratic polynomial with radical leading coefficient $5\sqrt{3}$ whose zeroes are $x = -1\sqrt{3}$ and $x = -\frac{2\sqrt{3}}{15}$.

Q31. [B] Form a quadratic polynomial with radical leading coefficient $\sqrt{5}$ whose zeroes are $x = -\frac{1\sqrt{5}}{5}$ and $x = -\frac{3\sqrt{5}}{5}$.

Q32. [B] Form a quadratic polynomial with radical leading coefficient $2\sqrt{7}$ whose zeroes are $x = -\frac{2\sqrt{7}}{7}$ and $x = -\frac{4\sqrt{7}}{14}$.

Q33. [B] Form a quadratic polynomial with radical leading coefficient $3\sqrt{2}$ whose zeroes are $x = -\frac{3\sqrt{2}}{2}$ and $x = -\frac{5\sqrt{2}}{6}$.

Q34. [B] Form a quadratic polynomial with radical leading coefficient $4\sqrt{3}$ whose zeroes are $x = -\frac{1\sqrt{3}}{3}$ and $x = -\frac{6\sqrt{3}}{12}$.

Q35. [B] Form a quadratic polynomial with radical leading coefficient $5\sqrt{5}$ whose zeroes are $x = -\frac{2\sqrt{5}}{5}$ and $x = -\frac{7\sqrt{5}}{25}$.

Q36. [B] Form a quadratic polynomial with radical leading coefficient $\sqrt{7}$ whose zeroes are $x = -\frac{3\sqrt{7}}{7}$ and $x = -\frac{1\sqrt{7}}{7}$.

Q37. [B] Form a quadratic polynomial with radical leading coefficient $2\sqrt{2}$ whose zeroes are $x = -\frac{1\sqrt{2}}{2}$ and $x = -\frac{2\sqrt{2}}{4}$.

Q38. [B] Form a quadratic polynomial with radical leading coefficient $3\sqrt{3}$ whose zeroes are $x = -\frac{2\sqrt{3}}{3}$ and $x = -\frac{3\sqrt{3}}{9}$.

Q39. [B] Form a quadratic polynomial with radical leading coefficient $4\sqrt{5}$ whose zeroes are $x = -\frac{3\sqrt{5}}{5}$ and $x = -\frac{4\sqrt{5}}{20}$.

Q40. [B] Form a quadratic polynomial with radical leading coefficient $5\sqrt{7}$ whose zeroes are $x = -\frac{1\sqrt{7}}{7}$ and $x = -\frac{5\sqrt{7}}{35}$.

Q41. [B] Form a quadratic polynomial with radical leading coefficient $\sqrt{2}$ whose zeroes are $x = -1\sqrt{2}$ and $x = -\frac{6\sqrt{2}}{2}$.

Q42. [B] Form a quadratic polynomial with radical leading coefficient $2\sqrt{3}$ whose zeroes are $x = -1\sqrt{3}$ and $x = -\frac{7\sqrt{3}}{6}$.

Q43. [B] Form a quadratic polynomial with radical leading coefficient $3\sqrt{5}$ whose zeroes are $x = -\frac{1\sqrt{5}}{5}$ and $x = -\frac{1\sqrt{5}}{15}$.

Q44. [B] Form a quadratic polynomial with radical leading coefficient $4\sqrt{7}$ whose zeroes are $x = -\frac{2\sqrt{7}}{7}$ and $x = -\frac{2\sqrt{7}}{28}$.

Q45. [B] Form a quadratic polynomial with radical leading coefficient $5\sqrt{2}$ whose zeroes are $x = -\frac{3\sqrt{2}}{2}$ and $x = -\frac{3\sqrt{2}}{10}$.

Q46. [B] Form a quadratic polynomial with radical leading coefficient $\sqrt{3}$ whose zeroes are $x = -\frac{1\sqrt{3}}{3}$ and $x = -\frac{4\sqrt{3}}{3}$.

Q47. [B] Form a quadratic polynomial with radical leading coefficient $2\sqrt{5}$ whose zeroes are $x = -\frac{2\sqrt{5}}{5}$ and $x = -\frac{5\sqrt{5}}{10}$.

Q48. [B] Form a quadratic polynomial with radical leading coefficient $3\sqrt{7}$ whose zeroes are $x = -\frac{3\sqrt{7}}{7}$ and $x = -\frac{6\sqrt{7}}{21}$.

Q49. [B] Form a quadratic polynomial with radical leading coefficient $4\sqrt{2}$ whose zeroes are $x = -\frac{1\sqrt{2}}{2}$ and $x = -\frac{7\sqrt{2}}{8}$.

Q50. [B] Form a quadratic polynomial with radical leading coefficient $5\sqrt{3}$ whose zeroes are $x = -\frac{2\sqrt{3}}{3}$ and $x = -\frac{1\sqrt{3}}{15}$.

Q51. [C] If $x = -\frac{3\sqrt{5}}{5}$ is a zero of $p(x) = \sqrt{5}x^2 + kx + 2\sqrt{5}$, find the value of k and the second zero.

Q52. [C] If $x = -\frac{1\sqrt{7}}{7}$ is a zero of $p(x) = 2\sqrt{7}x^2 + kx + 3\sqrt{7}$, find the value of k and the second zero.

Q53. [C] If $x = -1\sqrt{2}$ is a zero of $p(x) = 3\sqrt{2}x^2 + kx + 4\sqrt{2}$, find the value of k and the second zero.

Q54. [C] If $x = -1\sqrt{3}$ is a zero of $p(x) = 4\sqrt{3}x^2 + kx + 5\sqrt{3}$, find the value of k and the second zero.

Q55. [C] If $x = -\frac{1\sqrt{5}}{5}$ is a zero of $p(x) = 5\sqrt{5}x^2 + kx + 6\sqrt{5}$, find the value of k and the second zero.

Q56. [C] If $x = -\frac{2\sqrt{7}}{7}$ is a zero of $p(x) = \sqrt{7}x^2 + kx + 7\sqrt{7}$, find the value of k and the second zero.

Q57. [C] If $x = -\frac{3\sqrt{2}}{2}$ is a zero of $p(x) = 2\sqrt{2}x^2 + kx + \sqrt{2}$, find the value of k and the second zero.

Q58. [C] If $x = -\frac{1\sqrt{3}}{3}$ is a zero of $p(x) = 3\sqrt{3}x^2 + kx + 2\sqrt{3}$, find the value of k and the second zero.

Q59. [C] If $x = -\frac{2\sqrt{5}}{5}$ is a zero of $p(x) = 4\sqrt{5}x^2 + kx + 3\sqrt{5}$, find the value of k and the second zero.

Q60. [C] If $x = -\frac{3\sqrt{7}}{7}$ is a zero of $p(x) = 5\sqrt{7}x^2 + kx + 4\sqrt{7}$, find the value of k and the second zero.

Q61. [C] If $x = -\frac{1\sqrt{2}}{2}$ is a zero of $p(x) = \sqrt{2}x^2 + kx + 5\sqrt{2}$, find the value of k and the second zero.

Q62. [C] If $x = -\frac{2\sqrt{3}}{3}$ is a zero of $p(x) = 2\sqrt{3}x^2 + kx + 6\sqrt{3}$, find the value of k and the second zero.

Q63. [C] If $x = -\frac{3\sqrt{5}}{5}$ is a zero of $p(x) = 3\sqrt{5}x^2 + kx + 7\sqrt{5}$, find the value of k and the second zero.

Q64. [C] If $x = -\frac{1\sqrt{7}}{7}$ is a zero of $p(x) = 4\sqrt{7}x^2 + kx + \sqrt{7}$, find the value of k and the second zero.

Q65. [C] If $x = -1\sqrt{2}$ is a zero of $p(x) = 5\sqrt{2}x^2 + kx + 2\sqrt{2}$, find the value of k and the second zero.

Q66. [C] If $x = -1\sqrt{3}$ is a zero of $p(x) = \sqrt{3}x^2 + kx + 3\sqrt{3}$, find the value of k and the second zero.

Q67. [C] If $x = -\frac{1\sqrt{5}}{5}$ is a zero of $p(x) = 2\sqrt{5}x^2 + kx + 4\sqrt{5}$, find the value of k and the second zero.

Q68. [C] If $x = -\frac{2\sqrt{7}}{7}$ is a zero of $p(x) = 3\sqrt{7}x^2 + kx + 5\sqrt{7}$, find the value of k and the second zero.

Q69. [C] If $x = -\frac{3\sqrt{2}}{2}$ is a zero of $p(x) = 4\sqrt{2}x^2 + kx + 6\sqrt{2}$, find the value of k and the second zero.

Q70. [C] If $x = -\frac{1\sqrt{3}}{3}$ is a zero of $p(x) = 5\sqrt{3}x^2 + kx + 7\sqrt{3}$, find the value of k and the second zero.

Q71. [C] If $x = -\frac{2\sqrt{5}}{5}$ is a zero of $p(x) = \sqrt{5}x^2 + kx + \sqrt{5}$, find the value of k and the second zero.

Q72. [C] If $x = -\frac{3\sqrt{7}}{7}$ is a zero of $p(x) = 2\sqrt{7}x^2 + kx + 2\sqrt{7}$,

find the value of k and the second zero.

Q73. [C] If $x = -\frac{1\sqrt{2}}{2}$ is a zero of $p(x) = 3\sqrt{2}x^2 + kx + 3\sqrt{2}$, find the value of k and the second zero.

Q74. [C] If $x = -\frac{2\sqrt{3}}{3}$ is a zero of $p(x) = 4\sqrt{3}x^2 + kx + 4\sqrt{3}$, find the value of k and the second zero.

Q75. [C] If $x = -\frac{3\sqrt{5}}{5}$ is a zero of $p(x) = 5\sqrt{5}x^2 + kx + 5\sqrt{5}$, find the value of k and the second zero.

Q76. [D] Which of the following represents the zeroes of $p(x) = \sqrt{7}x^2 + 13x + 6\sqrt{7}$?
(A) $x = -\frac{1\sqrt{7}}{7}, -\frac{6\sqrt{7}}{7}$ (B) $x = \sqrt{7}, -6$ (C) $x = -1, \sqrt{7}$
(D) $x = 1\sqrt{7}, 6$

Q77. [D] Which of the following represents the zeroes of $p(x) = 2\sqrt{2}x^2 + 15x + 7\sqrt{2}$?
(A) $x = -1\sqrt{2}, -\frac{7\sqrt{2}}{4}$ (B) $x = \sqrt{2}, -7$ (C) $x = -2, \sqrt{2}$
(D) $x = 2\sqrt{2}, 7$

Q78. [D] Which of the following represents the zeroes of $p(x) = 3\sqrt{3}x^2 + 28x + \sqrt{3}$?
(A) $x = -1\sqrt{3}, -\frac{1\sqrt{3}}{9}$ (B) $x = \sqrt{3}, -1$ (C) $x = -3, \sqrt{3}$
(D) $x = 3\sqrt{3}, 1$

Q79. [D] Which of the following represents the zeroes of $p(x) = 4\sqrt{5}x^2 + 22x + 2\sqrt{5}$?
(A) $x = -\frac{1\sqrt{5}}{5}, -\frac{2\sqrt{5}}{20}$ (B) $x = \sqrt{5}, -2$ (C) $x = -1, \sqrt{5}$
(D) $x = 1\sqrt{5}, 2$

Q80. [D] Which of the following represents the zeroes of $p(x) = 5\sqrt{7}x^2 + 73x + 3\sqrt{7}$?
(A) $x = -\frac{2\sqrt{7}}{7}, -\frac{3\sqrt{7}}{35}$ (B) $x = \sqrt{7}, -3$ (C) $x = -2, \sqrt{7}$
(D) $x = 2\sqrt{7}, 3$

Q81. [D] Which of the following represents the zeroes of $p(x) = \sqrt{2}x^2 + 10x + 4\sqrt{2}$?
(A) $x = -\frac{3\sqrt{2}}{2}, -\frac{4\sqrt{2}}{2}$ (B) $x = \sqrt{2}, -4$ (C) $x = -3, \sqrt{2}$
(D) $x = 3\sqrt{2}, 4$

Q82. [D] Which of the following represents the zeroes of $p(x) = 2\sqrt{3}x^2 + 11x + 5\sqrt{3}$?
(A) $x = -\frac{1\sqrt{3}}{3}, -\frac{5\sqrt{3}}{6}$ (B) $x = \sqrt{3}, -5$ (C) $x = -1, \sqrt{3}$
(D) $x = 1\sqrt{3}, 5$

Q83. [D] Which of the following represents the zeroes of $p(x) = 3\sqrt{5}x^2 + 36x + 6\sqrt{5}$?
(A) $x = -\frac{2\sqrt{5}}{5}, -\frac{6\sqrt{5}}{15}$ (B) $x = \sqrt{5}, -6$ (C) $x = -2, \sqrt{5}$
(D) $x = 2\sqrt{5}, 6$

Q84. [D] Which of the following represents the zeroes of $p(x) = 4\sqrt{7}x^2 + 91x + 7\sqrt{7}$?
(A) $x = -\frac{3\sqrt{7}}{7}, -\frac{7\sqrt{7}}{28}$ (B) $x = \sqrt{7}, -7$ (C) $x = -3, \sqrt{7}$
(D) $x = 3\sqrt{7}, 7$

Q85. [D] Which of the following represents the zeroes of $p(x) = 5\sqrt{2}x^2 + 11x + \sqrt{2}$?
(A) $x = -\frac{1\sqrt{2}}{2}, -\frac{1\sqrt{2}}{10}$ (B) $x = \sqrt{2}, -1$ (C) $x = -1, \sqrt{2}$
(D) $x = 1\sqrt{2}, 1$

Q86. [D] Which of the following represents the zeroes of $p(x) = \sqrt{3}x^2 + 8x + 2\sqrt{3}$?
(A) $x = -\frac{2\sqrt{3}}{3}, -\frac{2\sqrt{3}}{3}$ (B) $x = \sqrt{3}, -2$ (C) $x = -2, \sqrt{3}$
(D) $x = 2\sqrt{3}, 2$

Q87. [D] Which of the following represents the zeroes of $p(x) = 2\sqrt{5}x^2 + 33x + 3\sqrt{5}$?
(A) $x = -\frac{3\sqrt{5}}{5}, -\frac{3\sqrt{5}}{10}$ (B) $x = \sqrt{5}, -3$ (C) $x = -3, \sqrt{5}$
(D) $x = 3\sqrt{5}, 3$

Q88. [D] Which of the following represents the zeroes of $p(x) = 3\sqrt{7}x^2 + 25x + 4\sqrt{7}$?
(A) $x = -\frac{1\sqrt{7}}{7}, -\frac{4\sqrt{7}}{21}$ (B) $x = \sqrt{7}, -4$ (C) $x = -1, \sqrt{7}$
(D) $x = 1\sqrt{7}, 4$

Q89. [D] Which of the following represents the zeroes of $p(x) = 4\sqrt{2}x^2 + 21x + 5\sqrt{2}$?
(A) $x = -1\sqrt{2}, -\frac{5\sqrt{2}}{8}$ (B) $x = \sqrt{2}, -5$ (C) $x = -2, \sqrt{2}$
(D) $x = 2\sqrt{2}, 5$

Q90. [D] Which of the following represents the zeroes of $p(x) = 5\sqrt{3}x^2 + 51x + 6\sqrt{3}$?
(A) $x = -1\sqrt{3}, -\frac{6\sqrt{3}}{15}$ (B) $x = \sqrt{3}, -6$ (C) $x = -3, \sqrt{3}$
(D) $x = 3\sqrt{3}, 6$

Q91. [D] Which of the following represents the zeroes of $p(x) = \sqrt{5}x^2 + 12x + 7\sqrt{5}$?
(A) $x = -\frac{1\sqrt{5}}{5}, -\frac{7\sqrt{5}}{5}$ (B) $x = \sqrt{5}, -7$ (C) $x = -1, \sqrt{5}$
(D) $x = 1\sqrt{5}, 7$

Q92. [D] Which of the following represents the zeroes of $p(x) = 2\sqrt{7}x^2 + 29x + \sqrt{7}$?
(A) $x = -\frac{2\sqrt{7}}{7}, -\frac{1\sqrt{7}}{14}$ (B) $x = \sqrt{7}, -1$ (C) $x = -2, \sqrt{7}$
(D) $x = 2\sqrt{7}, 1$

Q93. [D] Which of the following represents the zeroes of $p(x) = 3\sqrt{2}x^2 + 20x + 2\sqrt{2}$?
(A) $x = -\frac{3\sqrt{2}}{2}, -\frac{2\sqrt{2}}{6}$ (B) $x = \sqrt{2}, -2$ (C) $x = -3, \sqrt{2}$
(D) $x = 3\sqrt{2}, 2$

Q94. [D] Which of the following represents the zeroes of $p(x) = 4\sqrt{3}x^2 + 15x + 3\sqrt{3}$?
(A) $x = -\frac{1\sqrt{3}}{3}, -\frac{3\sqrt{3}}{12}$ (B) $x = \sqrt{3}, -3$ (C) $x = -1, \sqrt{3}$
(D) $x = 1\sqrt{3}, 3$

Q95. [D] Which of the following represents the zeroes of $p(x) = 5\sqrt{5}x^2 + 54x + 4\sqrt{5}$?
(A) $x = -\frac{2\sqrt{5}}{5}, -\frac{4\sqrt{5}}{25}$ (B) $x = \sqrt{5}, -4$ (C) $x = -2, \sqrt{5}$
(D) $x = 2\sqrt{5}, 4$

Q96. [D] Which of the following represents the zeroes of $p(x) = \sqrt{7}x^2 + 26x + 5\sqrt{7}$?
(A) $x = -\frac{3\sqrt{7}}{7}, -\frac{5\sqrt{7}}{7}$ (B) $x = \sqrt{7}, -5$ (C) $x = -3, \sqrt{7}$
(D) $x = 3\sqrt{7}, 5$

Q97. [D] Which of the following represents the zeroes of $p(x) = 2\sqrt{2}x^2 + 10x + 6\sqrt{2}$?
(A) $x = -\frac{1\sqrt{2}}{2}, -\frac{6\sqrt{2}}{4}$ (B) $x = \sqrt{2}, -6$ (C) $x = -1, \sqrt{2}$
(D) $x = 1\sqrt{2}, 6$

Q98. [D] Which of the following represents the zeroes of $p(x) = 3\sqrt{3}x^2 + 25x + 7\sqrt{3}$?
(A) $x = -\frac{2\sqrt{3}}{3}, -\frac{7\sqrt{3}}{9}$ (B) $x = \sqrt{3}, -7$ (C) $x = -2, \sqrt{3}$
(D) $x = 2\sqrt{3}, 7$

Q99. [D] Which of the following represents the zeroes of $p(x) = 4\sqrt{5}x^2 + 61x + \sqrt{5}$?
(A) $x = -\frac{3\sqrt{5}}{5}, -\frac{1\sqrt{5}}{20}$ (B) $x = \sqrt{5}, -1$ (C) $x = -3, \sqrt{5}$

(D) $x = 3\sqrt{5}, 1$

Q100. [D] Which of the following represents the zeroes of $p(x) = 5\sqrt{7}x^2 + 37x + 2\sqrt{7}$?
(A) $x = -\frac{1\sqrt{7}}{7}, -\frac{2\sqrt{7}}{35}$ (B) $x = \sqrt{7}, -2$ (C) $x = -1, \sqrt{7}$
(D) $x = 1\sqrt{7}, 2$

Q101. [E] Find the zeroes of $p(x) = \sqrt{2}x^2 + 7x + 3\sqrt{2}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q102. [E] Find the zeroes of $p(x) = 2\sqrt{3}x^2 + 22x + 4\sqrt{3}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q103. [E] Find the zeroes of $p(x) = 3\sqrt{5}x^2 + 20x + 5\sqrt{5}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q104. [E] Find the zeroes of $p(x) = 4\sqrt{7}x^2 + 62x + 6\sqrt{7}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q105. [E] Find the zeroes of $p(x) = 5\sqrt{2}x^2 + 37x + 7\sqrt{2}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q106. [E] Find the zeroes of $p(x) = \sqrt{3}x^2 + 4x + \sqrt{3}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q107. [E] Find the zeroes of $p(x) = 2\sqrt{5}x^2 + 22x + 2\sqrt{5}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q108. [E] Find the zeroes of $p(x) = 3\sqrt{7}x^2 + 66x + 3\sqrt{7}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q109. [E] Find the zeroes of $p(x) = 4\sqrt{2}x^2 + 12x + 4\sqrt{2}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q110. [E] Find the zeroes of $p(x) = 5\sqrt{3}x^2 + 35x + 5\sqrt{3}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q111. [E] Find the zeroes of $p(x) = \sqrt{5}x^2 + 21x + 6\sqrt{5}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q112. [E] Find the zeroes of $p(x) = 2\sqrt{7}x^2 + 21x + 7\sqrt{7}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q113. [E] Find the zeroes of $p(x) = 3\sqrt{2}x^2 + 13x + \sqrt{2}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q114. [E] Find the zeroes of $p(x) = 4\sqrt{3}x^2 + 38x + 2\sqrt{3}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q115. [E] Find the zeroes of $p(x) = 5\sqrt{5}x^2 + 28x + 3\sqrt{5}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q116. [E] Find the zeroes of $p(x) = \sqrt{7}x^2 + 18x + 4\sqrt{7}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q117. [E] Find the zeroes of $p(x) = 2\sqrt{2}x^2 + 17x + 5\sqrt{2}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q118. [E] Find the zeroes of $p(x) = 3\sqrt{3}x^2 + 15x + 6\sqrt{3}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q119. [E] Find the zeroes of $p(x) = 4\sqrt{5}x^2 + 47x + 7\sqrt{5}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q120. [E] Find the zeroes of $p(x) = 5\sqrt{7}x^2 + 106x + \sqrt{7}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q121. [E] Find the zeroes of $p(x) = \sqrt{2}x^2 + 4x + 2\sqrt{2}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q122. [E] Find the zeroes of $p(x) = 2\sqrt{3}x^2 + 15x + 3\sqrt{3}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q123. [E] Find the zeroes of $p(x) = 3\sqrt{5}x^2 + 49x + 4\sqrt{5}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q124. [E] Find the zeroes of $p(x) = 4\sqrt{7}x^2 + 33x + 5\sqrt{7}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q125. [E] Find the zeroes of $p(x) = 5\sqrt{2}x^2 + 26x + 6\sqrt{2}$ and verify that Sum $= -b/a$ and Product $= c/a$.

Q126. [F] The area of a rectangular garden plot is given by $A(x) = \sqrt{3}x^2 + 16x + 7\sqrt{3}$ square meters. Find the dimensions x where $A(x) = 0$.

Q127. [F] The area of a rectangular garden plot is given by $A(x) = 2\sqrt{5}x^2 + 11x + \sqrt{5}$ square meters. Find the dimensions x where $A(x) = 0$.

Q128. [F] The area of a rectangular garden plot is given by $A(x) = 3\sqrt{7}x^2 + 44x + 2\sqrt{7}$ square meters. Find the dimensions x where $A(x) = 0$.

Q129. [F] The area of a rectangular garden plot is given by $A(x) = 4\sqrt{2}x^2 + 27x + 3\sqrt{2}$ square meters. Find the dimensions x where $A(x) = 0$.

Q130. [F] The area of a rectangular garden plot is given by $A(x) = 5\sqrt{3}x^2 + 19x + 4\sqrt{3}$ square meters. Find the dimensions x where $A(x) = 0$.

Q131. [F] The area of a rectangular garden plot is given by $A(x) = \sqrt{5}x^2 + 15x + 5\sqrt{5}$ square meters. Find the dimensions x where $A(x) = 0$.

Q132. [F] The area of a rectangular garden plot is given by $A(x) = 2\sqrt{7}x^2 + 48x + 6\sqrt{7}$ square meters. Find the dimensions x where $A(x) = 0$.

Q133. [F] The area of a rectangular garden plot is given by $A(x) = 3\sqrt{2}x^2 + 13x + 7\sqrt{2}$ square meters. Find the dimensions x where $A(x) = 0$.

Q134. [F] The area of a rectangular garden plot is given by $A(x) = 4\sqrt{3}x^2 + 25x + \sqrt{3}$ square meters. Find the dimensions x where $A(x) = 0$.

Q135. [F] The area of a rectangular garden plot is given by $A(x) = 5\sqrt{5}x^2 + 77x + 2\sqrt{5}$ square meters. Find the dimensions x where $A(x) = 0$.

Q136. [F] The area of a rectangular garden plot is given by $A(x) = \sqrt{7}x^2 + 10x + 3\sqrt{7}$ square meters. Find the dimensions x where $A(x) = 0$.

Q137. [F] The area of a rectangular garden plot is given by $A(x) = 2\sqrt{2}x^2 + 12x + 4\sqrt{2}$ square meters. Find the dimensions x where $A(x) = 0$.

Q138. [F] The area of a rectangular garden plot is given by $A(x) = 3\sqrt{3}x^2 + 32x + 5\sqrt{3}$ square meters. Find the dimensions x where $A(x) = 0$.

Q139. [F] The area of a rectangular garden plot is given by $A(x) = 4\sqrt{5}x^2 + 26x + 6\sqrt{5}$ square meters. Find the dimensions x where $A(x) = 0$.

Q140. [F] The area of a rectangular garden plot is given by $A(x) = 5\sqrt{7}x^2 + 77x + 7\sqrt{7}$ square meters. Find the dimensions x where $A(x) = 0$.

Q141. [F] The area of a rectangular garden plot is given by $A(x) = \sqrt{2}x^2 + 7x + \sqrt{2}$ square meters. Find the dimensions x where $A(x) = 0$.

Q142. [F] The area of a rectangular garden plot is given

by $A(x) = 2\sqrt{3}x^2 + 8x + 2\sqrt{3}$ square meters. Find the dimensions x where $A(x) = 0$.

Q143. [F] The area of a rectangular garden plot is given by $A(x) = 3\sqrt{5}x^2 + 33x + 3\sqrt{5}$ square meters. Find the dimensions x where $A(x) = 0$.

Q144. [F] The area of a rectangular garden plot is given by $A(x) = 4\sqrt{7}x^2 + 88x + 4\sqrt{7}$ square meters. Find the dimensions x where $A(x) = 0$.

Q145. [F] The area of a rectangular garden plot is given by $A(x) = 5\sqrt{2}x^2 + 15x + 5\sqrt{2}$ square meters. Find the dimensions x where $A(x) = 0$.

Q146. [F] The area of a rectangular garden plot is given by $A(x) = \sqrt{3}x^2 + 12x + 6\sqrt{3}$ square meters. Find the dimensions x where $A(x) = 0$.

Q147. [F] The area of a rectangular garden plot is given by $A(x) = 2\sqrt{5}x^2 + 37x + 7\sqrt{5}$ square meters. Find the dimensions x where $A(x) = 0$.

Q148. [F] The area of a rectangular garden plot is given by $A(x) = 3\sqrt{7}x^2 + 22x + \sqrt{7}$ square meters. Find the dimensions x where $A(x) = 0$.

Q149. [F] The area of a rectangular garden plot is given by $A(x) = 4\sqrt{2}x^2 + 18x + 2\sqrt{2}$ square meters. Find the dimensions x where $A(x) = 0$.

Q150. [F] The area of a rectangular garden plot is given by $A(x) = 5\sqrt{3}x^2 + 48x + 3\sqrt{3}$ square meters. Find the dimensions x where $A(x) = 0$.

Q151. [G] Calculate the discriminant D of the polynomial $p(x) = \sqrt{5}x^2 + 9x + 4\sqrt{5}$ and state the nature of its roots.

Q152. [G] Calculate the discriminant D of the polynomial $p(x) = 2\sqrt{7}x^2 + 33x + 5\sqrt{7}$ and state the nature of its roots.

Q153. [G] Calculate the discriminant D of the polynomial $p(x) = 3\sqrt{2}x^2 + 24x + 6\sqrt{2}$ and state the nature of its roots.

Q154. [G] Calculate the discriminant D of the polynomial $p(x) = 4\sqrt{3}x^2 + 19x + 7\sqrt{3}$ and state the nature of its roots.

Q155. [G] Calculate the discriminant D of the polynomial $p(x) = 5\sqrt{5}x^2 + 51x + \sqrt{5}$ and state the nature of its roots.

Q156. [G] Calculate the discriminant D of the polynomial $p(x) = \sqrt{7}x^2 + 23x + 2\sqrt{7}$ and state the nature of its roots.

Q157. [G] Calculate the discriminant D of the polynomial $p(x) = 2\sqrt{2}x^2 + 7x + 3\sqrt{2}$ and state the nature of its roots.

Q158. [G] Calculate the discriminant D of the polynomial $p(x) = 3\sqrt{3}x^2 + 22x + 4\sqrt{3}$ and state the nature of its roots.

Q159. [G] Calculate the discriminant D of the polynomial $p(x) = 4\sqrt{5}x^2 + 65x + 5\sqrt{5}$ and state the nature of its roots.

Q160. [G] Calculate the discriminant D of the polynomial $p(x) = 5\sqrt{7}x^2 + 41x + 6\sqrt{7}$ and state the nature of its roots.

Q161. [G] Calculate the discriminant D of the polynomial $p(x) = \sqrt{2}x^2 + 11x + 7\sqrt{2}$ and state the nature of its roots.

Q162. [G] Calculate the discriminant D of the polynomial $p(x) = 2\sqrt{3}x^2 + 19x + \sqrt{3}$ and state the nature of its roots.

Q163. [G] Calculate the discriminant D of the polynomial $p(x) = 3\sqrt{5}x^2 + 17x + 2\sqrt{5}$ and state the nature of its roots.

Q164. [G] Calculate the discriminant D of the polynomial $p(x) = 4\sqrt{7}x^2 + 59x + 3\sqrt{7}$ and state the nature of its roots.

Q165. [G] Calculate the discriminant D of the polynomial $p(x) = 5\sqrt{2}x^2 + 34x + 4\sqrt{2}$ and state the nature of its roots.

Q166. [G] Calculate the discriminant D of the polynomial $p(x) = \sqrt{3}x^2 + 8x + 5\sqrt{3}$ and state the nature of its roots.

Q167. [G] Calculate the discriminant D of the polynomial $p(x) = 2\sqrt{5}x^2 + 26x + 6\sqrt{5}$ and state the nature of its roots.

Q168. [G] Calculate the discriminant D of the polynomial $p(x) = 3\sqrt{7}x^2 + 70x + 7\sqrt{7}$ and state the nature of its roots.

Q169. [G] Calculate the discriminant D of the polynomial $p(x) = 4\sqrt{2}x^2 + 9x + \sqrt{2}$ and state the nature of its roots.

Q170. [G] Calculate the discriminant D of the polynomial $p(x) = 5\sqrt{3}x^2 + 32x + 2\sqrt{3}$ and state the nature of its roots.

Q171. [G] Calculate the discriminant D of the polynomial $p(x) = \sqrt{5}x^2 + 18x + 3\sqrt{5}$ and state the nature of its roots.

Q172. [G] Calculate the discriminant D of the polynomial $p(x) = 2\sqrt{7}x^2 + 18x + 4\sqrt{7}$ and state the nature of its roots.

Q173. [G] Calculate the discriminant D of the polynomial $p(x) = 3\sqrt{2}x^2 + 17x + 5\sqrt{2}$ and state the nature of its roots.

Q174. [G] Calculate the discriminant D of the polynomial $p(x) = 4\sqrt{3}x^2 + 42x + 6\sqrt{3}$ and state the nature of its roots.

Q175. [G] Calculate the discriminant D of the polynomial $p(x) = 5\sqrt{5}x^2 + 32x + 7\sqrt{5}$ and state the nature of its roots.

Q176. [H] If α and β are zeroes of $p(x) = \sqrt{7}x^2 + 15x + \sqrt{7}$, evaluate the ratio α/β.

Q177. [H] If α and β are zeroes of $p(x) = 2\sqrt{2}x^2 + 14x + 2\sqrt{2}$, evaluate the ratio α/β.

Q178. [H] If α and β are zeroes of $p(x) = 3\sqrt{3}x^2 + 12x + 3\sqrt{3}$, evaluate the ratio α/β.

Q179. [H] If α and β are zeroes of $p(x) = 4\sqrt{5}x^2 + 44x + 4\sqrt{5}$, evaluate the ratio α/β.

Q180. [H] If α and β are zeroes of $p(x) = 5\sqrt{7}x^2 + 110x + 5\sqrt{7}$, evaluate the ratio α/β.

Q181. [H] If α and β are zeroes of $p(x) = \sqrt{2}x^2 + 8x + 6\sqrt{2}$, evaluate the ratio α/β.

Q182. [H] If α and β are zeroes of $p(x) = 2\sqrt{3}x^2 + 19x + 7\sqrt{3}$, evaluate the ratio α/β.

Q183. [H] If α and β are zeroes of $p(x) = 3\sqrt{5}x^2 + 46x + \sqrt{5}$, evaluate the ratio α/β.

Q184. [H] If α and β are zeroes of $p(x) = 4\sqrt{7}x^2 + 30x + 2\sqrt{7}$, evaluate the ratio α/β.

Q185. [H] If α and β are zeroes of $p(x) = 5\sqrt{2}x^2 + 23x + 3\sqrt{2}$, evaluate the ratio α/β.

Q186. [H] If α and β are zeroes of $p(x) = \sqrt{3}x^2 + 13x + 4\sqrt{3}$, evaluate the ratio α/β.

Q187. [H] If α and β are zeroes of $p(x) = 2\sqrt{5}x^2 + 15x + 5\sqrt{5}$, evaluate the ratio α/β.

Q188. [H] If α and β are zeroes of $p(x) = 3\sqrt{7}x^2 + 48x + 6\sqrt{7}$, evaluate the ratio α/β.

Q189. [H] If α and β are zeroes of $p(x) =$

$4\sqrt{2}x^2 + 31x + 7\sqrt{2}$, evaluate the ratio α/β.

Q190. [H] If α and β are zeroes of $p(x) = 5\sqrt{3}x^2+16x+\sqrt{3}$, evaluate the ratio α/β.

Q191. [H] If α and β are zeroes of $p(x) = \sqrt{5}x^2+12x+2\sqrt{5}$, evaluate the ratio α/β.

Q192. [H] If α and β are zeroes of $p(x) = 2\sqrt{7}x^2 + 45x + 3\sqrt{7}$, evaluate the ratio α/β.

Q193. [H] If α and β are zeroes of $p(x) = 3\sqrt{2}x^2 + 10x + 4\sqrt{2}$, evaluate the ratio α/β.

Q194. [H] If α and β are zeroes of $p(x) = 4\sqrt{3}x^2 + 29x + 5\sqrt{3}$, evaluate the ratio α/β.

Q195. [H] If α and β are zeroes of $p(x) = 5\sqrt{5}x^2 + 81x + 6\sqrt{5}$, evaluate the ratio α/β.

Q196. [H] If α and β are zeroes of $p(x) = \sqrt{7}x^2+14x+7\sqrt{7}$, evaluate the ratio α/β.

Q197. [H] If α and β are zeroes of $p(x) = 2\sqrt{2}x^2+9x+\sqrt{2}$, evaluate the ratio α/β.

Q198. [H] If α and β are zeroes of $p(x) = 3\sqrt{3}x^2 + 29x + 2\sqrt{3}$, evaluate the ratio α/β.

Q199. [H] If α and β are zeroes of $p(x) = 4\sqrt{5}x^2 + 23x + 3\sqrt{5}$, evaluate the ratio α/β.

Q200. [H] If α and β are zeroes of $p(x) = 5\sqrt{7}x^2 + 74x + 4\sqrt{7}$, evaluate the ratio α/β.

Pattern 2: Condition-Based Reciprocal Relationships of Zeroes

Q201. [A] If one zero of the polynomial $p(x) = (k^2 + 4)x^2 + 10x + 4k$ is the reciprocal of the other, find the value of k.

Q202. [A] If one zero of the polynomial $p(x) = (k^2 + 9)x^2 + 11x + 6k$ is the reciprocal of the other, find the value of k.

Q203. [A] If one zero of the polynomial $p(x) = (k^2 + 16)x^2 + 12x + 8k$ is the reciprocal of the other, find the value of k.

Q204. [A] If one zero of the polynomial $p(x) = (k^2 + 25)x^2 + 13x + 10k$ is the reciprocal of the other, find the value of k.

Q205. [A] If one zero of the polynomial $p(x) = (k^2 + 36)x^2 + 14x + 12k$ is the reciprocal of the other, find the value of k.

Q206. [A] If one zero of the polynomial $p(x) = (k^2 + 49)x^2 + 15x + 14k$ is the reciprocal of the other, find the value of k.

Q207. [A] If one zero of the polynomial $p(x) = (k^2 + 4)x^2 + 16x + 4k$ is the reciprocal of the other, find the value of k.

Q208. [A] If one zero of the polynomial $p(x) = (k^2 + 9)x^2 + 17x + 6k$ is the reciprocal of the other, find the value of k.

Q209. [A] If one zero of the polynomial $p(x) = (k^2 + 16)x^2 + 18x + 8k$ is the reciprocal of the other, find the value of k.

Q210. [A] If one zero of the polynomial $p(x) = (k^2 + 25)x^2 + 19x + 10k$ is the reciprocal of the other, find the value of k.

Q211. [A] If one zero of the polynomial $p(x) = (k^2 + 36)x^2 + 20x + 12k$ is the reciprocal of the other, find the value of k.

Q212. [A] If one zero of the polynomial $p(x) = (k^2 + 49)x^2 + 21x + 14k$ is the reciprocal of the other, find the value of k.

Q213. [A] If one zero of the polynomial $p(x) = (k^2 + 4)x^2 + 22x + 4k$ is the reciprocal of the other, find the value of k.

Q214. [A] If one zero of the polynomial $p(x) = (k^2 + 9)x^2 + 23x + 6k$ is the reciprocal of the other, find the value of k.

Q215. [A] If one zero of the polynomial $p(x) = (k^2 + 16)x^2 + 24x + 8k$ is the reciprocal of the other, find the value of k.

Q216. [A] If one zero of the polynomial $p(x) = (k^2 + 25)x^2 + 10x + 10k$ is the reciprocal of the other, find the value of k.

Q217. [A] If one zero of the polynomial $p(x) = (k^2 + 36)x^2 + 11x + 12k$ is the reciprocal of the other, find the value of k.

Q218. [A] If one zero of the polynomial $p(x) = (k^2 + 49)x^2 + 12x + 14k$ is the reciprocal of the other, find the value of k.

Q219. [A] If one zero of the polynomial $p(x) = (k^2 + 4)x^2 + 13x + 4k$ is the reciprocal of the other, find the value of k.

Q220. [A] If one zero of the polynomial $p(x) = (k^2 + 9)x^2 + 14x + 6k$ is the reciprocal of the other, find the value of k.

Q221. [A] If one zero of the polynomial $p(x) = (k^2 + 16)x^2 + 15x + 8k$ is the reciprocal of the other, find the value of k.

Q222. [A] If one zero of the polynomial $p(x) = (k^2 + 25)x^2 + 16x + 10k$ is the reciprocal of the other, find the value of k.

Q223. [A] If one zero of the polynomial $p(x) = (k^2 + 36)x^2 + 17x + 12k$ is the reciprocal of the other, find the value of k.

Q224. [A] If one zero of the polynomial $p(x) = (k^2 + 49)x^2 + 18x + 14k$ is the reciprocal of the other, find the value of k.

Q225. [A] If one zero of the polynomial $p(x) = (k^2 + 4)x^2 + 19x + 4k$ is the reciprocal of the other, find the value of k.

Q226. [B] Find the value of a such that the zeroes of $p(x) = (a^2 + 3)x^2 + 20x + 6a - 3$ are reciprocals of each

other.

Q227. [B] Find the value of a such that the zeroes of $p(x) = (a^2+4)x^2 + 21x + 8a - 4$ are reciprocals of each other.

Q228. [B] Find the value of a such that the zeroes of $p(x) = (a^2+5)x^2 + 22x + 10a - 5$ are reciprocals of each other.

Q229. [B] Find the value of a such that the zeroes of $p(x) = (a^2+6)x^2 + 23x + 12a - 6$ are reciprocals of each other.

Q230. [B] Find the value of a such that the zeroes of $p(x) = (a^2+7)x^2 + 24x + 14a - 7$ are reciprocals of each other.

Q231. [B] Find the value of a such that the zeroes of $p(x) = (a^2+2)x^2 + 10x + 4a - 2$ are reciprocals of each other.

Q232. [B] Find the value of a such that the zeroes of $p(x) = (a^2+3)x^2 + 11x + 6a - 3$ are reciprocals of each other.

Q233. [B] Find the value of a such that the zeroes of $p(x) = (a^2+4)x^2 + 12x + 8a - 4$ are reciprocals of each other.

Q234. [B] Find the value of a such that the zeroes of $p(x) = (a^2+5)x^2 + 13x + 10a - 5$ are reciprocals of each other.

Q235. [B] Find the value of a such that the zeroes of $p(x) = (a^2+6)x^2 + 14x + 12a - 6$ are reciprocals of each other.

Q236. [B] Find the value of a such that the zeroes of $p(x) = (a^2+7)x^2 + 15x + 14a - 7$ are reciprocals of each other.

Q237. [B] Find the value of a such that the zeroes of $p(x) = (a^2+2)x^2 + 16x + 4a - 2$ are reciprocals of each other.

Q238. [B] Find the value of a such that the zeroes of $p(x) = (a^2+3)x^2 + 17x + 6a - 3$ are reciprocals of each other.

Q239. [B] Find the value of a such that the zeroes of $p(x) = (a^2+4)x^2 + 18x + 8a - 4$ are reciprocals of each other.

Q240. [B] Find the value of a such that the zeroes of $p(x) = (a^2+5)x^2 + 19x + 10a - 5$ are reciprocals of each other.

Q241. [B] Find the value of a such that the zeroes of $p(x) = (a^2+6)x^2 + 20x + 12a - 6$ are reciprocals of each other.

Q242. [B] Find the value of a such that the zeroes of $p(x) = (a^2+7)x^2 + 21x + 14a - 7$ are reciprocals of each other.

Q243. [B] Find the value of a such that the zeroes of $p(x) = (a^2+2)x^2 + 22x + 4a - 2$ are reciprocals of each other.

Q244. [B] Find the value of a such that the zeroes of $p(x) = (a^2+3)x^2 + 23x + 6a - 3$ are reciprocals of each other.

Q245. [B] Find the value of a such that the zeroes of $p(x) = (a^2+4)x^2 + 24x + 8a - 4$ are reciprocals of each other.

Q246. [B] Find the value of a such that the zeroes of $p(x) = (a^2+5)x^2 + 10x + 10a - 5$ are reciprocals of each other.

Q247. [B] Find the value of a such that the zeroes of $p(x) = (a^2+6)x^2 + 11x + 12a - 6$ are reciprocals of each other.

Q248. [B] Find the value of a such that the zeroes of $p(x) = (a^2+7)x^2 + 12x + 14a - 7$ are reciprocals of each other.

Q249. [B] Find the value of a such that the zeroes of $p(x) = (a^2+2)x^2 + 13x + 4a - 2$ are reciprocals of each other.

Q250. [B] Find the value of a such that the zeroes of $p(x) = (a^2+3)x^2 + 14x + 6a - 3$ are reciprocals of each other.

Q251. [C] Given $p(x) = (2k+1)x^2 - 15x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q252. [C] Given $p(x) = (2k+1)x^2 - 16x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q253. [C] Given $p(x) = (2k+1)x^2 - 17x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q254. [C] Given $p(x) = (2k+1)x^2 - 18x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q255. [C] Given $p(x) = (2k+1)x^2 - 19x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q256. [C] Given $p(x) = (2k+1)x^2 - 20x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q257. [C] Given $p(x) = (2k+1)x^2 - 21x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q258. [C] Given $p(x) = (2k+1)x^2 - 22x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q259. [C] Given $p(x) = (2k+1)x^2 - 23x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q260. [C] Given $p(x) = (2k+1)x^2 - 24x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q261. [C] Given $p(x) = (2k+1)x^2 - 10x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q262. [C] Given $p(x) = (2k+1)x^2 - 11x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q263. [C] Given $p(x) = (2k+1)x^2 - 12x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q264. [C] Given $p(x) = (2k+1)x^2 - 13x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q265. [C] Given $p(x) = (2k+1)x^2 - 14x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q266. [C] Given $p(x) = (2k+1)x^2 - 15x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q267. [C] Given $p(x) = (2k+1)x^2 - 16x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q268. [C] Given $p(x) = (2k+1)x^2 - 17x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q269. [C] Given $p(x) = (2k+1)x^2 - 18x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q270. [C] Given $p(x) = (2k+1)x^2 - 19x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q271. [C] Given $p(x) = (2k+1)x^2 - 20x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q272. [C] Given $p(x) = (2k+1)x^2 - 21x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q273. [C] Given $p(x) = (2k+1)x^2 - 22x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q274. [C] Given $p(x) = (2k+1)x^2 - 23x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q275. [C] Given $p(x) = (2k+1)x^2 - 24x + (k^2-1)$, determine k if the zeroes α and β satisfy $\alpha\beta = 1$.

Q276. [D] For what value of k are the roots of $(k^2+25)x^2 + 10x + 10k = 0$ reciprocals of each other?
(A) $k = 5$ (B) $k = -5$ (C) $k = 10$ (D) $k = 0$

Q277. [D] For what value of k are the roots of $(k^2+36)x^2 + 11x + 12k = 0$ reciprocals of each other?
(A) $k = 6$ (B) $k = -6$ (C) $k = 12$ (D) $k = 0$

Q278. [D] For what value of k are the roots of $(k^2+49)x^2 + 12x + 14k = 0$ reciprocals of each other?
(A) $k = 7$ (B) $k = -7$ (C) $k = 14$ (D) $k = 0$

Q279. [D] For what value of k are the roots of $(k^2+4)x^2 + 13x + 4k = 0$ reciprocals of each other?
(A) $k = 2$ (B) $k = -2$ (C) $k = 4$ (D) $k = 0$

Q280. [D] For what value of k are the roots of $(k^2+9)x^2 + 14x + 6k = 0$ reciprocals of each other?
(A) $k = 3$ (B) $k = -3$ (C) $k = 6$ (D) $k = 0$

Q281. [D] For what value of k are the roots of $(k^2+16)x^2 + 15x + 8k = 0$ reciprocals of each other?
(A) $k = 4$ (B) $k = -4$ (C) $k = 8$ (D) $k = 0$

Q282. [D] For what value of k are the roots of $(k^2+25)x^2 + 16x + 10k = 0$ reciprocals of each other?
(A) $k = 5$ (B) $k = -5$ (C) $k = 10$ (D) $k = 0$

Q283. [D] For what value of k are the roots of $(k^2+36)x^2 + 17x + 12k = 0$ reciprocals of each other?
(A) $k = 6$ (B) $k = -6$ (C) $k = 12$ (D) $k = 0$

Q284. [D] For what value of k are the roots of $(k^2+49)x^2 + 18x + 14k = 0$ reciprocals of each other?
(A) $k = 7$ (B) $k = -7$ (C) $k = 14$ (D) $k = 0$

Q285. [D] For what value of k are the roots of $(k^2+4)x^2 + 19x + 4k = 0$ reciprocals of each other?
(A) $k = 2$ (B) $k = -2$ (C) $k = 4$ (D) $k = 0$

Q286. [D] For what value of k are the roots of $(k^2+9)x^2 + 20x + 6k = 0$ reciprocals of each other?
(A) $k = 3$ (B) $k = -3$ (C) $k = 6$ (D) $k = 0$

Q287. [D] For what value of k are the roots of $(k^2+16)x^2 + 21x + 8k = 0$ reciprocals of each other?
(A) $k = 4$ (B) $k = -4$ (C) $k = 8$ (D) $k = 0$

Q288. [D] For what value of k are the roots of $(k^2+25)x^2 + 22x + 10k = 0$ reciprocals of each other?
(A) $k = 5$ (B) $k = -5$ (C) $k = 10$ (D) $k = 0$

Q289. [D] For what value of k are the roots of $(k^2+36)x^2 + 23x + 12k = 0$ reciprocals of each other?
(A) $k = 6$ (B) $k = -6$ (C) $k = 12$ (D) $k = 0$

Q290. [D] For what value of k are the roots of $(k^2+49)x^2 + 24x + 14k = 0$ reciprocals of each other?
(A) $k = 7$ (B) $k = -7$ (C) $k = 14$ (D) $k = 0$

Q291. [D] For what value of k are the roots of $(k^2+4)x^2 + 10x + 4k = 0$ reciprocals of each other?
(A) $k = 2$ (B) $k = -2$ (C) $k = 4$ (D) $k = 0$

Q292. [D] For what value of k are the roots of $(k^2+9)x^2 + 11x + 6k = 0$ reciprocals of each other?
(A) $k = 3$ (B) $k = -3$ (C) $k = 6$ (D) $k = 0$

Q293. [D] For what value of k are the roots of $(k^2+16)x^2 + 12x + 8k = 0$ reciprocals of each other?
(A) $k = 4$ (B) $k = -4$ (C) $k = 8$ (D) $k = 0$

Q294. [D] For what value of k are the roots of $(k^2+25)x^2 + 13x + 10k = 0$ reciprocals of each other?
(A) $k = 5$ (B) $k = -5$ (C) $k = 10$ (D) $k = 0$

Q295. [D] For what value of k are the roots of $(k^2+36)x^2 + 14x + 12k = 0$ reciprocals of each other?
(A) $k = 6$ (B) $k = -6$ (C) $k = 12$ (D) $k = 0$

Q296. [D] For what value of k are the roots of $(k^2+49)x^2 + 15x + 14k = 0$ reciprocals of each other?
(A) $k = 7$ (B) $k = -7$ (C) $k = 14$ (D) $k = 0$

Q297. [D] For what value of k are the roots of $(k^2+4)x^2 + 16x + 4k = 0$ reciprocals of each other?
(A) $k = 2$ (B) $k = -2$ (C) $k = 4$ (D) $k = 0$

Q298. [D] For what value of k are the roots of $(k^2+9)x^2 + 17x + 6k = 0$ reciprocals of each other?
(A) $k = 3$ (B) $k = -3$ (C) $k = 6$ (D) $k = 0$

Q299. [D] For what value of k are the roots of $(k^2+16)x^2 + 18x + 8k = 0$ reciprocals of each other?
(A) $k = 4$ (B) $k = -4$ (C) $k = 8$ (D) $k = 0$

Q300. [D] For what value of k are the roots of $(k^2+25)x^2 + 19x + 10k = 0$ reciprocals of each other?
(A) $k = 5$ (B) $k = -5$ (C) $k = 10$ (D) $k = 0$

Q301. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q302. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

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Q306. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

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Q309. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q310. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q311. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q312. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q313. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q314. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q315. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q316. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q317. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q318. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q319. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q320. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q321. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q322. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q323. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q324. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q325. [E] Is it possible for the polynomial $p(x) = 3x^2 + 5x + 7$ to have reciprocal zeroes for any real x? Explain using the coefficient ratio.

Q326. [F] A signal frequency equation $f(x) = (p^2 + 7)x^2 + 15x + 14p$ requires reciprocal resonance roots. Solve for p.

Q327. [F] A signal frequency equation $f(x) = (p^2 + 2)x^2 + 16x + 4p$ requires reciprocal resonance roots. Solve for p.

Q328. [F] A signal frequency equation $f(x) = (p^2 + 3)x^2 + 17x + 6p$ requires reciprocal resonance roots. Solve for p.

Q329. [F] A signal frequency equation $f(x) = (p^2 + 4)x^2 + 18x + 8p$ requires reciprocal resonance roots. Solve for p.

Q330. [F] A signal frequency equation $f(x) = (p^2 + 5)x^2 + 19x + 10p$ requires reciprocal resonance roots. Solve for p.

Q331. [F] A signal frequency equation $f(x) = (p^2 + 6)x^2 + 20x + 12p$ requires reciprocal resonance roots. Solve for p.

Q332. [F] A signal frequency equation $f(x) = (p^2 + 7)x^2 + 21x + 14p$ requires reciprocal resonance roots. Solve for p.

Q333. [F] A signal frequency equation $f(x) = (p^2 + 2)x^2 + 22x + 4p$ requires reciprocal resonance roots. Solve for p.

Q334. [F] A signal frequency equation $f(x) = (p^2 + 3)x^2 + 23x + 6p$ requires reciprocal resonance roots. Solve for p.

Q335. [F] A signal frequency equation $f(x) = (p^2 + 4)x^2 + 24x + 8p$ requires reciprocal resonance roots. Solve for p.

Q336. [F] A signal frequency equation $f(x) = (p^2 + 5)x^2 + 10x + 10p$ requires reciprocal resonance roots. Solve for p.

Q337. [F] A signal frequency equation $f(x) = (p^2 + 6)x^2 + 11x + 12p$ requires reciprocal resonance roots. Solve for p.

Q338. [F] A signal frequency equation $f(x) = (p^2 + 7)x^2 + 12x + 14p$ requires reciprocal resonance roots. Solve for p.

Q339. [F] A signal frequency equation $f(x) = (p^2 + 2)x^2 + 13x + 4p$ requires reciprocal resonance roots. Solve for p.

Q340. [F] A signal frequency equation $f(x) =$

$(p^2 + 3)x^2 + 14x + 6p$ requires reciprocal resonance roots. Solve for p.

Q341. [F] A signal frequency equation $f(x) = (p^2 + 4)x^2 + 15x + 8p$ requires reciprocal resonance roots. Solve for p.

Q342. [F] A signal frequency equation $f(x) = (p^2 + 5)x^2 + 16x + 10p$ requires reciprocal resonance roots. Solve for p.

Q343. [F] A signal frequency equation $f(x) = (p^2 + 6)x^2 + 17x + 12p$ requires reciprocal resonance roots. Solve for p.

Q344. [F] A signal frequency equation $f(x) = (p^2 + 7)x^2 + 18x + 14p$ requires reciprocal resonance roots. Solve for p.

Q345. [F] A signal frequency equation $f(x) = (p^2 + 2)x^2 + 19x + 4p$ requires reciprocal resonance roots. Solve for p.

Q346. [F] A signal frequency equation $f(x) = (p^2 + 3)x^2 + 20x + 6p$ requires reciprocal resonance roots. Solve for p.

Q347. [F] A signal frequency equation $f(x) = (p^2 + 4)x^2 + 21x + 8p$ requires reciprocal resonance roots. Solve for p.

Q348. [F] A signal frequency equation $f(x) = (p^2 + 5)x^2 + 22x + 10p$ requires reciprocal resonance roots. Solve for p.

Q349. [F] A signal frequency equation $f(x) = (p^2 + 6)x^2 + 23x + 12p$ requires reciprocal resonance roots. Solve for p.

Q350. [F] A signal frequency equation $f(x) = (p^2 + 7)x^2 + 24x + 14p$ requires reciprocal resonance roots. Solve for p.

Q351. [G] If α and $1/\alpha$ are zeroes of $(m^2+4)x^2+10x+2m^2$, verify if $m = 2$ is the unique solution.

Q352. [G] If α and $1/\alpha$ are zeroes of $(m^2+9)x^2+11x+2m^2$, verify if $m = 3$ is the unique solution.

Q353. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 16)x^2 + 12x + 2m^2$, verify if $m = 4$ is the unique solution.

Q354. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 25)x^2 + 13x + 2m^2$, verify if $m = 5$ is the unique solution.

Q355. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 36)x^2 + 14x + 2m^2$, verify if $m = 6$ is the unique solution.

Q356. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 49)x^2 + 15x + 2m^2$, verify if $m = 7$ is the unique solution.

Q357. [G] If α and $1/\alpha$ are zeroes of $(m^2+4)x^2+16x+2m^2$, verify if $m = 2$ is the unique solution.

Q358. [G] If α and $1/\alpha$ are zeroes of $(m^2+9)x^2+17x+2m^2$, verify if $m = 3$ is the unique solution.

Q359. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 16)x^2 + 18x + 2m^2$, verify if $m = 4$ is the unique solution.

Q360. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 25)x^2 + 19x + 2m^2$, verify if $m = 5$ is the unique solution.

Q361. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 36)x^2 + 20x + 2m^2$, verify if $m = 6$ is the unique solution.

Q362. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 49)x^2 + 21x + 2m^2$, verify if $m = 7$ is the unique solution.

Q363. [G] If α and $1/\alpha$ are zeroes of $(m^2+4)x^2+22x+2m^2$, verify if $m = 2$ is the unique solution.

Q364. [G] If α and $1/\alpha$ are zeroes of $(m^2+9)x^2+23x+2m^2$, verify if $m = 3$ is the unique solution.

Q365. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 16)x^2 + 24x + 2m^2$, verify if $m = 4$ is the unique solution.

Q366. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 25)x^2 + 10x + 2m^2$, verify if $m = 5$ is the unique solution.

Q367. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 36)x^2 + 11x + 2m^2$, verify if $m = 6$ is the unique solution.

Q368. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 49)x^2 + 12x + 2m^2$, verify if $m = 7$ is the unique solution.

Q369. [G] If α and $1/\alpha$ are zeroes of $(m^2+4)x^2+13x+2m^2$, verify if $m = 2$ is the unique solution.

Q370. [G] If α and $1/\alpha$ are zeroes of $(m^2+9)x^2+14x+2m^2$, verify if $m = 3$ is the unique solution.

Q371. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 16)x^2 + 15x + 2m^2$, verify if $m = 4$ is the unique solution.

Q372. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 25)x^2 + 16x + 2m^2$, verify if $m = 5$ is the unique solution.

Q373. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 36)x^2 + 17x + 2m^2$, verify if $m = 6$ is the unique solution.

Q374. [G] If α and $1/\alpha$ are zeroes of $(m^2 + 49)x^2 + 18x + 2m^2$, verify if $m = 7$ is the unique solution.

Q375. [G] If α and $1/\alpha$ are zeroes of $(m^2+4)x^2+19x+2m^2$, verify if $m = 2$ is the unique solution.

Q376. [H] If zeroes of $(k - 1)x^2 + 10x + (k^2 + 1)$ are reciprocals, find the positive integer value of k.

Q377. [H] If zeroes of $(k - 1)x^2 + 10x + (k^2 + 1)$ are reciprocals, find the positive integer value of k.

Q378. [H] If zeroes of $(k - 1)x^2 + 10x + (k^2 + 1)$ are reciprocals, find the positive integer value of k.

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Q383. [H] If zeroes of $(k - 1)x^2 + 10x + (k^2 + 1)$ are reciprocals, find the positive integer value of k.

Q384. [H] If zeroes of $(k - 1)x^2 + 10x + (k^2 + 1)$ are reciprocals, find the positive integer value of k.

Q385. [H] If zeroes of $(k - 1)x^2 + 10x + (k^2 + 1)$ are reciprocals, find the positive integer value of k.

Q386. [H] If zeroes of $(k - 1)x^2 + 10x + (k^2 + 1)$ are

reciprocals, find the positive integer value of k.

Q387. [H] If zeroes of $(k-1)x^2+10x+(k^2+1)$ are reciprocals, find the positive integer value of k.

Q388. [H] If zeroes of $(k-1)x^2+10x+(k^2+1)$ are reciprocals, find the positive integer value of k.

Q389. [H] If zeroes of $(k-1)x^2+10x+(k^2+1)$ are reciprocals, find the positive integer value of k.

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Q393. [H] If zeroes of $(k-1)x^2+10x+(k^2+1)$ are reciprocals, find the positive integer value of k.

Q394. [H] If zeroes of $(k-1)x^2+10x+(k^2+1)$ are reciprocals, find the positive integer value of k.

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Q399. [H] If zeroes of $(k-1)x^2+10x+(k^2+1)$ are reciprocals, find the positive integer value of k.

Q400. [H] If zeroes of $(k-1)x^2+10x+(k^2+1)$ are reciprocals, find the positive integer value of k.

Pattern 3: Symmetric Transformations of Zeroes

Q401. [A] If α,β are zeroes of $p(x)=x^2-3x+k$, find k such that $\alpha^2+\beta^2=7$.

Q402. [A] If α,β are zeroes of $p(x)=x^2-5x+k$, find k such that $\alpha^2+\beta^2=21$.

Q403. [A] If α,β are zeroes of $p(x)=x^2-7x+k$, find k such that $\alpha^2+\beta^2=43$.

Q404. [A] If α,β are zeroes of $p(x)=x^2-9x+k$, find k such that $\alpha^2+\beta^2=73$.

Q405. [A] If α,β are zeroes of $p(x)=x^2-11x+k$, find k such that $\alpha^2+\beta^2=111$.

Q406. [A] If α,β are zeroes of $p(x)=x^2-3x+k$, find k such that $\alpha^2+\beta^2=-3$.

Q407. [A] If α,β are zeroes of $p(x)=x^2-5x+k$, find k such that $\alpha^2+\beta^2=11$.

Q408. [A] If α,β are zeroes of $p(x)=x^2-7x+k$, find k such that $\alpha^2+\beta^2=33$.

Q409. [A] If α,β are zeroes of $p(x)=x^2-9x+k$, find k such that $\alpha^2+\beta^2=63$.

Q410. [A] If α,β are zeroes of $p(x)=x^2-11x+k$, find k such that $\alpha^2+\beta^2=119$.

Q411. [A] If α,β are zeroes of $p(x)=x^2-3x+k$, find k such that $\alpha^2+\beta^2=5$.

Q412. [A] If α,β are zeroes of $p(x)=x^2-5x+k$, find k such that $\alpha^2+\beta^2=19$.

Q413. [A] If α,β are zeroes of $p(x)=x^2-7x+k$, find k such that $\alpha^2+\beta^2=41$.

Q414. [A] If α,β are zeroes of $p(x)=x^2-9x+k$, find k such that $\alpha^2+\beta^2=71$.

Q415. [A] If α,β are zeroes of $p(x)=x^2-11x+k$, find k such that $\alpha^2+\beta^2=109$.

Q416. [A] If α,β are zeroes of $p(x)=x^2-3x+k$, find k such that $\alpha^2+\beta^2=-5$.

Q417. [A] If α,β are zeroes of $p(x)=x^2-5x+k$, find k such that $\alpha^2+\beta^2=9$.

Q418. [A] If α,β are zeroes of $p(x)=x^2-7x+k$, find k such that $\alpha^2+\beta^2=31$.

Q419. [A] If α,β are zeroes of $p(x)=x^2-9x+k$, find k such that $\alpha^2+\beta^2=79$.

Q420. [A] If α,β are zeroes of $p(x)=x^2-11x+k$, find k such that $\alpha^2+\beta^2=117$.

Q421. [A] If α,β are zeroes of $p(x)=x^2-3x+k$, find k such that $\alpha^2+\beta^2=3$.

Q422. [A] If α,β are zeroes of $p(x)=x^2-5x+k$, find k such that $\alpha^2+\beta^2=17$.

Q423. [A] If α,β are zeroes of $p(x)=x^2-7x+k$, find k such that $\alpha^2+\beta^2=39$.

Q424. [A] If α,β are zeroes of $p(x)=x^2-9x+k$, find k such that $\alpha^2+\beta^2=69$.

Q425. [A] If α,β are zeroes of $p(x)=x^2-11x+k$, find k such that $\alpha^2+\beta^2=107$.

Q426. [B] If α,β are zeroes of $p(x)=x^2-3x+k$ such that $\alpha-\beta=1$, find k.

Q427. [B] If α,β are zeroes of $p(x)=x^2-5x+k$ such that $\alpha-\beta=1$, find k.

Q428. [B] If α,β are zeroes of $p(x)=x^2-7x+k$ such that $\alpha-\beta=1$, find k.

Q429. [B] If α,β are zeroes of $p(x)=x^2-9x+k$ such that $\alpha-\beta=1$, find k.

Q430. [B] If α,β are zeroes of $p(x)=x^2-11x+k$ such that $\alpha-\beta=1$, find k.

Q431. [B] If α,β are zeroes of $p(x)=x^2-3x+k$ such that $\alpha-\beta=1$, find k.

Q432. [B] If α,β are zeroes of $p(x)=x^2-5x+k$ such that $\alpha-\beta=1$, find k.

Q433. [B] If α,β are zeroes of $p(x)=x^2-7x+k$ such that $\alpha-\beta=1$, find k.

Q434. [B] If α,β are zeroes of $p(x)=x^2-9x+k$ such that $\alpha-\beta=1$, find k.

Q435. [B] If α, β are zeroes of $p(x) = x^2 - 11x + k$ such that $\alpha - \beta = 1$, find k.

Q436. [B] If α, β are zeroes of $p(x) = x^2 - 3x + k$ such that $\alpha - \beta = 1$, find k.

Q437. [B] If α, β are zeroes of $p(x) = x^2 - 5x + k$ such that $\alpha - \beta = 1$, find k.

Q438. [B] If α, β are zeroes of $p(x) = x^2 - 7x + k$ such that $\alpha - \beta = 1$, find k.

Q439. [B] If α, β are zeroes of $p(x) = x^2 - 9x + k$ such that $\alpha - \beta = 1$, find k.

Q440. [B] If α, β are zeroes of $p(x) = x^2 - 11x + k$ such that $\alpha - \beta = 1$, find k.

Q441. [B] If α, β are zeroes of $p(x) = x^2 - 3x + k$ such that $\alpha - \beta = 1$, find k.

Q442. [B] If α, β are zeroes of $p(x) = x^2 - 5x + k$ such that $\alpha - \beta = 1$, find k.

Q443. [B] If α, β are zeroes of $p(x) = x^2 - 7x + k$ such that $\alpha - \beta = 1$, find k.

Q444. [B] If α, β are zeroes of $p(x) = x^2 - 9x + k$ such that $\alpha - \beta = 1$, find k.

Q445. [B] If α, β are zeroes of $p(x) = x^2 - 11x + k$ such that $\alpha - \beta = 1$, find k.

Q446. [B] If α, β are zeroes of $p(x) = x^2 - 3x + k$ such that $\alpha - \beta = 1$, find k.

Q447. [B] If α, β are zeroes of $p(x) = x^2 - 5x + k$ such that $\alpha - \beta = 1$, find k.

Q448. [B] If α, β are zeroes of $p(x) = x^2 - 7x + k$ such that $\alpha - \beta = 1$, find k.

Q449. [B] If α, β are zeroes of $p(x) = x^2 - 9x + k$ such that $\alpha - \beta = 1$, find k.

Q450. [B] If α, β are zeroes of $p(x) = x^2 - 11x + k$ such that $\alpha - \beta = 1$, find k.

Q451. [C] If α, β are zeroes of $x^2 - p(x+1) - c$, prove that $(\alpha + 1)(\beta + 1) = 1 - c$.

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Q471. [C] If α, β are zeroes of $x^2 - p(x+1) - c$, prove that $(\alpha + 1)(\beta + 1) = 1 - c$.

Q472. [C] If α, β are zeroes of $x^2 - p(x+1) - c$, prove that $(\alpha + 1)(\beta + 1) = 1 - c$.

Q473. [C] If α, β are zeroes of $x^2 - p(x+1) - c$, prove that $(\alpha + 1)(\beta + 1) = 1 - c$.

Q474. [C] If α, β are zeroes of $x^2 - p(x+1) - c$, prove that $(\alpha + 1)(\beta + 1) = 1 - c$.

Q475. [C] If α, β are zeroes of $x^2 - p(x+1) - c$, prove that $(\alpha + 1)(\beta + 1) = 1 - c$.

Q476. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q477. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q478. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q479. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q480. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q481. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q482. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?

(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q483. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q484. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q485. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q486. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q487. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q488. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q489. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q490. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q491. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q492. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q493. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q494. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q495. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q496. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q497. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q498. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q499. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q500. [D] Which expression equals $\frac{1}{\alpha} + \frac{1}{\beta}$ for $p(x) = ax^2 + bx + c$?
(A) $-b/c$ (B) $-b/a$ (C) c/a (D) b/c

Q501. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q502. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q503. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q504. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

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Q506. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q507. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q508. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q509. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q510. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q511. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q512. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q513. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q514. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q515. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q516. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q517. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q518. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q519. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

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Q521. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q522. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q523. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q524. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q525. [E] If α, β are zeroes of $2x^2 + 5x + k$, evaluate k when $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

Q526. [F] The roots α, β of a stability equation $x^2 - 3x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q527. [F] The roots α, β of a stability equation $x^2 - 5x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q528. [F] The roots α, β of a stability equation $x^2 - 7x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q529. [F] The roots α, β of a stability equation $x^2 - 9x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q530. [F] The roots α, β of a stability equation $x^2 - 11x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q531. [F] The roots α, β of a stability equation $x^2 - 3x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q532. [F] The roots α, β of a stability equation $x^2 - 5x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q533. [F] The roots α, β of a stability equation $x^2 - 7x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q534. [F] The roots α, β of a stability equation $x^2 - 9x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q535. [F] The roots α, β of a stability equation $x^2 - 11x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

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Q539. [F] The roots α, β of a stability equation $x^2 - 9x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q540. [F] The roots α, β of a stability equation $x^2 - 11x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q541. [F] The roots α, β of a stability equation $x^2 - 3x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q542. [F] The roots α, β of a stability equation $x^2 - 5x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q543. [F] The roots α, β of a stability equation $x^2 - 7x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q544. [F] The roots α, β of a stability equation $x^2 - 9x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q545. [F] The roots α, β of a stability equation $x^2 - 11x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q546. [F] The roots α, β of a stability equation $x^2 - 3x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q547. [F] The roots α, β of a stability equation $x^2 - 5x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q548. [F] The roots α, β of a stability equation $x^2 - 7x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q549. [F] The roots α, β of a stability equation $x^2 - 9x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q550. [F] The roots α, β of a stability equation $x^2 - 11x + k = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 3$. Find k.

Q551. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q552. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

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Q558. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

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Q561. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q562. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q563. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q564. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q565. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q566. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q567. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q568. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q569. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q570. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q571. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q572. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q573. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q574. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q575. [G] If α, β are zeroes of $p(x) = x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$.

Q576. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 3x + 5$.

Q577. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 5x + 6$.

Q578. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 7x + 7$.

Q579. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 9x + 8$.

Q580. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 11x + 9$.

Q581. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 3x + 1$.

Q582. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 5x + 2$.

Q583. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 7x + 3$.

Q584. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 9x + 4$.

Q585. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 11x + 5$.

Q586. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 3x + 6$.

Q587. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 5x + 7$.

Q588. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 7x + 8$.

Q589. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 9x + 9$.

Q590. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 11x + 1$.

Q591. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 3x + 2$.

Q592. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 5x + 3$.

Q593. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 7x + 4$.

Q594. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 9x + 5$.

Q595. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 11x + 6$.

Q596. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 3x + 7$.

Q597. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 5x + 8$.

Q598. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 7x + 9$.

Q599. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 9x + 1$.

Q600. [H] Evaluate $\alpha^3 + \beta^3$ for the polynomial $p(x) = x^2 - 11x + 2$.

Pattern 4: Bi-Quadratic Polynomials with Radical Zeroes

Q601. [A] Find all four zeroes of $p(x) = x^4 - 1x^3 + (0)x^2 + 2x - 4$, given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q602. [A] Find all four zeroes of $p(x) = x^4 - 2x^3 + (0)x^2 + 6x - 9$, given two zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

Q603. [A] Find all four zeroes of $p(x) = x^4 - 3x^3 + (-1)x^2 + 15x - 20$, given two zeroes are $\sqrt{5}$ and $-\sqrt{5}$.

Q604. [A] Find all four zeroes of $p(x) = x^4 - 4x^3 + (3)x^2 + 8x - 10$, given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q605. [A] Find all four zeroes of $p(x) = x^4 - 1x^3 + (3)x^2 + 3x - 18$, given two zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

Q606. [A] Find all four zeroes of $p(x) = x^4 - 2x^3 + (-3)x^2 + 10x - 10$, given two zeroes are $\sqrt{5}$ and $-\sqrt{5}$.

Q607. [A] Find all four zeroes of $p(x) = x^4 - 3x^3 + (1)x^2 + 6x - 6$, given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q608. [A] Find all four zeroes of $p(x) = x^4 - 4x^3 + (1)x^2 + 12x - 12$, given two zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

Q609. [A] Find all four zeroes of $p(x) = x^4 - 1x^3 + (0)x^2 + 5x - 25$, given two zeroes are $\sqrt{5}$ and $-\sqrt{5}$.

Q610. [A] Find all four zeroes of $p(x) = x^4 - 2x^3 + (4)x^2 + 4x - 12$, given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q611. [A] Find all four zeroes of $p(x) = x^4 - 3x^3 + (-1)x^2 + 9x - 6$, given two zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

Q612. [A] Find all four zeroes of $p(x) = x^4 - 4x^3 + (-2)x^2 + 20x - 15$, given two zeroes are $\sqrt{5}$ and $-\sqrt{5}$.

Q613. [A] Find all four zeroes of $p(x) = x^4 - 1x^3 + (2)x^2 + 2x - 8$, given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q614. [A] Find all four zeroes of $p(x) = x^4 - 2x^3 + (2)x^2 + 6x - 15$, given two zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

Q615. [A] Find all four zeroes of $p(x) = x^4 - 3x^3 + (1)x^2 + 15x - 30$, given two zeroes are $\sqrt{5}$ and $-\sqrt{5}$.

Q616. [A] Find all four zeroes of $p(x) = x^4 - 4x^3 + (0)x^2 + 8x - 4$, given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q617. [A] Find all four zeroes of $p(x) = x^4 - 1x^3 + (0)x^2 + 3x - 9$, given two zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

Q618. [A] Find all four zeroes of $p(x) = x^4 - 2x^3 + (-1)x^2 + 10x - 20$, given two zeroes are $\sqrt{5}$ and $-\sqrt{5}$.

Q619. [A] Find all four zeroes of $p(x) = x^4 - 3x^3 + (3)x^2 + 6x - 10$, given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q620. [A] Find all four zeroes of $p(x) = x^4 - 4x^3 + (3)x^2 + 12x - 18$, given two zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

Q621. [A] Find all four zeroes of $p(x) = x^4 - 1x^3 + (-3)x^2 + 5x - 10$, given two zeroes are $\sqrt{5}$ and $-\sqrt{5}$.

Q622. [A] Find all four zeroes of $p(x) = x^4 - 2x^3 + (1)x^2 + 4x - 6$, given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q623. [A] Find all four zeroes of $p(x) = x^4 - 3x^3 + (1)x^2 + 9x - 12$, given two zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

Q624. [A] Find all four zeroes of $p(x) = x^4 - 4x^3 + (0)x^2 + 20x - 25$, given two zeroes are $\sqrt{5}$ and $-\sqrt{5}$.

Q625. [A] Find all four zeroes of $p(x) = x^4 - 1x^3 + (4)x^2 + 2x - 12$, given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q626. [B] Given that $x = \pm\sqrt{3}$ are zeroes of $p(x) = x^4 - 2x^3 + (-1)x^2 + 6x - 6$, factorize $p(x)$ completely.

Q627. [B] Given that $x = \pm\sqrt{5}$ are zeroes of $p(x) = x^4 - 3x^3 + (-2)x^2 + 15x - 15$, factorize $p(x)$ completely.

Q628. [B] Given that $x = \pm\sqrt{2}$ are zeroes of $p(x) = x^4 - 4x^3 + (2)x^2 + 8x - 8$, factorize $p(x)$ completely.

Q629. [B] Given that $x = \pm\sqrt{3}$ are zeroes of $p(x) = x^4 - 1x^3 + (2)x^2 + 3x - 15$, factorize $p(x)$ completely.

Q630. [B] Given that $x = \pm\sqrt{5}$ are zeroes of $p(x) = x^4 - 2x^3 + (1)x^2 + 10x - 30$, factorize $p(x)$ completely.

Q631. [B] Given that $x = \pm\sqrt{2}$ are zeroes of $p(x) = x^4 - 3x^3 + (0)x^2 + 6x - 4$, factorize $p(x)$ completely.

Q632. [B] Given that $x = \pm\sqrt{3}$ are zeroes of $p(x) = x^4 - 4x^3 + (0)x^2 + 12x - 9$, factorize $p(x)$ completely.

Q633. [B] Given that $x = \pm\sqrt{5}$ are zeroes of $p(x) = x^4 - 1x^3 + (-1)x^2 + 5x - 20$, factorize $p(x)$ completely.

Q634. [B] Given that $x = \pm\sqrt{2}$ are zeroes of $p(x) = x^4 - 2x^3 + (3)x^2 + 4x - 10$, factorize $p(x)$ completely.

Q635. [B] Given that $x = \pm\sqrt{3}$ are zeroes of $p(x) = x^4 - 3x^3 + (3)x^2 + 9x - 18$, factorize $p(x)$ completely.

Q636. [B] Given that $x = \pm\sqrt{5}$ are zeroes of $p(x) = x^4 - 4x^3 + (-3)x^2 + 20x - 10$, factorize $p(x)$ completely.

Q637. [B] Given that $x = \pm\sqrt{2}$ are zeroes of $p(x) = x^4 - 1x^3 + (1)x^2 + 2x - 6$, factorize $p(x)$ completely.

Q638. [B] Given that $x = \pm\sqrt{3}$ are zeroes of $p(x) = x^4 - 2x^3 + (1)x^2 + 6x - 12$, factorize $p(x)$ completely.

Q639. [B] Given that $x = \pm\sqrt{5}$ are zeroes of $p(x) = x^4 - 3x^3 + (0)x^2 + 15x - 25$, factorize $p(x)$ completely.

Q640. [B] Given that $x = \pm\sqrt{2}$ are zeroes of $p(x) = x^4 - 4x^3 + (4)x^2 + 8x - 12$, factorize $p(x)$ completely.

Q641. [B] Given that $x = \pm\sqrt{3}$ are zeroes of $p(x) = x^4 - 1x^3 + (-1)x^2 + 3x - 6$, factorize $p(x)$ completely.

Q642. [B] Given that $x = \pm\sqrt{5}$ are zeroes of $p(x) = x^4 - 2x^3 + (-2)x^2 + 10x - 15$, factorize $p(x)$ completely.

Q643. [B] Given that $x = \pm\sqrt{2}$ are zeroes of $p(x) = x^4 - 3x^3 + (2)x^2 + 6x - 8$, factorize $p(x)$ completely.

Q644. [B] Given that $x = \pm\sqrt{3}$ are zeroes of $p(x) = x^4 - 4x^3 + (2)x^2 + 12x - 15$, factorize $p(x)$ completely.

Q645. [B] Given that $x = \pm\sqrt{5}$ are zeroes of $p(x) = x^4 - 1x^3 + (1)x^2 + 5x - 30$, factorize $p(x)$ completely.

Q646. [B] Given that $x = \pm\sqrt{2}$ are zeroes of $p(x) = x^4 - 2x^3 + (0)x^2 + 4x - 4$, factorize $p(x)$ completely.

Q647. [B] Given that $x = \pm\sqrt{3}$ are zeroes of $p(x) = x^4 - 3x^3 + (0)x^2 + 9x - 9$, factorize $p(x)$ completely.

Q648. [B] Given that $x = \pm\sqrt{5}$ are zeroes of $p(x) = x^4 - 4x^3 + (-1)x^2 + 20x - 20$, factorize $p(x)$ completely.

Q649. [B] Given that $x = \pm\sqrt{2}$ are zeroes of $p(x) = x^4 - 1x^3 + (3)x^2 + 2x - 10$, factorize $p(x)$ completely.

Q650. [B] Given that $x = \pm\sqrt{3}$ are zeroes of $p(x) = x^4 - 2x^3 + (3)x^2 + 6x - 18$, factorize $p(x)$ completely.

Q651. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{5}$, form the quadratic factor associated with these zeroes.

Q652. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{2}$, form the quadratic factor associated with these zeroes.

Q653. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{3}$, form the quadratic factor associated with these zeroes.

Q654. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{5}$, form the quadratic factor associated with these zeroes.

Q655. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{2}$, form the quadratic factor associated with these zeroes.

Q656. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{3}$, form the quadratic factor associated with these zeroes.

Q657. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{5}$, form the quadratic factor associated with these zeroes.

Q658. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{2}$, form the quadratic factor associated with these zeroes.

Q659. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{3}$, form the quadratic factor associated with these zeroes.

Q660. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{5}$, form the quadratic factor associated with these zeroes.

Q661. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{2}$, form the quadratic factor associated with these zeroes.

Q662. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{3}$, form the quadratic factor associated with these zeroes.

Q663. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{5}$, form the quadratic factor associated with these zeroes.

Q664. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{2}$, form the quadratic factor associated with these zeroes.

Q665. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{3}$, form the quadratic factor associated with these zeroes.

Q666. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{5}$, form the quadratic factor associated with these zeroes.

Q667. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{2}$, form the quadratic factor associated with these zeroes.

Q668. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{3}$, form the quadratic factor associated with these zeroes.

Q669. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{5}$, form the quadratic factor associated with these zeroes.

Q670. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{2}$, form the quadratic factor associated with these zeroes.

Q671. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{3}$, form the quadratic factor associated with these zeroes.

Q672. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{5}$, form the quadratic factor associated with these zeroes.

Q673. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{2}$, form the quadratic factor associated with these zeroes.

Q674. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{3}$, form the quadratic factor associated with these zeroes.

Q675. [C] If two zeroes of a biquadratic polynomial $p(x)$ are $2 \pm \sqrt{5}$, form the quadratic factor associated with these zeroes.

Q676. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q677. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q678. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q679. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q680. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q681. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q682. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q683. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q684. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q685. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q686. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q687. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q688. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q689. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q690. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q691. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q692. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1,

what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q693. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q694. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q695. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q696. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q697. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q698. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q699. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q700. [D] If two zeroes of $x^4 - 5x^2 + 4 = 0$ are 1 and -1, what are the remaining zeroes?
(A) $2, -2$ (B) $3, -3$ (C) $\sqrt{2}, -\sqrt{2}$ (D) $0, 4$

Q701. [E] Verify whether $x = \sqrt{3}$ is a zero of $p(x) = x^4 - 3x^2$ and find its multiplicity.

Q702. [E] Verify whether $x = \sqrt{5}$ is a zero of $p(x) = x^4 - 5x^2$ and find its multiplicity.

Q703. [E] Verify whether $x = \sqrt{2}$ is a zero of $p(x) = x^4 - 2x^2$ and find its multiplicity.

Q704. [E] Verify whether $x = \sqrt{3}$ is a zero of $p(x) = x^4 - 3x^2$ and find its multiplicity.

Q705. [E] Verify whether $x = \sqrt{5}$ is a zero of $p(x) = x^4 - 5x^2$ and find its multiplicity.

Q706. [E] Verify whether $x = \sqrt{2}$ is a zero of $p(x) = x^4 - 2x^2$ and find its multiplicity.

Q707. [E] Verify whether $x = \sqrt{3}$ is a zero of $p(x) = x^4 - 3x^2$ and find its multiplicity.

Q708. [E] Verify whether $x = \sqrt{5}$ is a zero of $p(x) = x^4 - 5x^2$ and find its multiplicity.

Q709. [E] Verify whether $x = \sqrt{2}$ is a zero of $p(x) = x^4 - 2x^2$ and find its multiplicity.

Q710. [E] Verify whether $x = \sqrt{3}$ is a zero of $p(x) = x^4 - 3x^2$ and find its multiplicity.

Q711. [E] Verify whether $x = \sqrt{5}$ is a zero of $p(x) = x^4 - 5x^2$ and find its multiplicity.

Q712. [E] Verify whether $x = \sqrt{2}$ is a zero of $p(x) = x^4 - 2x^2$ and find its multiplicity.

Q713. [E] Verify whether $x = \sqrt{3}$ is a zero of $p(x) = x^4 - 3x^2$ and find its multiplicity.

Q714. [E] Verify whether $x = \sqrt{5}$ is a zero of $p(x) = x^4 - 5x^2$ and find its multiplicity.

Q715. [E] Verify whether $x = \sqrt{2}$ is a zero of $p(x) = x^4 - 2x^2$ and find its multiplicity.

Q716. [E] Verify whether $x = \sqrt{3}$ is a zero of $p(x) = x^4 - 3x^2$ and find its multiplicity.

Q717. [E] Verify whether $x = \sqrt{5}$ is a zero of $p(x) = x^4 - 5x^2$ and find its multiplicity.

Q718. [E] Verify whether $x = \sqrt{2}$ is a zero of $p(x) = x^4 - 2x^2$ and find its multiplicity.

Q719. [E] Verify whether $x = \sqrt{3}$ is a zero of $p(x) = x^4 - 3x^2$ and find its multiplicity.

Q720. [E] Verify whether $x = \sqrt{5}$ is a zero of $p(x) = x^4 - 5x^2$ and find its multiplicity.

Q721. [E] Verify whether $x = \sqrt{2}$ is a zero of $p(x) = x^4 - 2x^2$ and find its multiplicity.

Q722. [E] Verify whether $x = \sqrt{3}$ is a zero of $p(x) = x^4 - 3x^2$ and find its multiplicity.

Q723. [E] Verify whether $x = \sqrt{5}$ is a zero of $p(x) = x^4 - 5x^2$ and find its multiplicity.

Q724. [E] Verify whether $x = \sqrt{2}$ is a zero of $p(x) = x^4 - 2x^2$ and find its multiplicity.

Q725. [E] Verify whether $x = \sqrt{3}$ is a zero of $p(x) = x^4 - 3x^2$ and find its multiplicity.

Q726. [F] A beam deflection equation $y(x) = x^4 - 7x^2 + 10$ has root symmetry. Find all real zeroes.

Q727. [F] A beam deflection equation $y(x) = x^4 - 5x^2 + 6$ has root symmetry. Find all real zeroes.

Q728. [F] A beam deflection equation $y(x) = x^4 - 7x^2 + 12$ has root symmetry. Find all real zeroes.

Q729. [F] A beam deflection equation $y(x) = x^4 - 6x^2 + 5$ has root symmetry. Find all real zeroes.

Q730. [F] A beam deflection equation $y(x) = x^4 - 4x^2 + 4$ has root symmetry. Find all real zeroes.

Q731. [F] A beam deflection equation $y(x) = x^4 - 6x^2 + 9$ has root symmetry. Find all real zeroes.

Q732. [F] A beam deflection equation $y(x) = x^4 - 9x^2 + 20$ has root symmetry. Find all real zeroes.

Q733. [F] A beam deflection equation $y(x) = x^4 - 3x^2 + 2$ has root symmetry. Find all real zeroes.

Q734. [F] A beam deflection equation $y(x) = x^4 - 5x^2 + 6$ has root symmetry. Find all real zeroes.

Q735. [F] A beam deflection equation $y(x) = x^4 - 8x^2 + 15$ has root symmetry. Find all real zeroes.

Q736. [F] A beam deflection equation $y(x) = x^4 - 6x^2 + 8$ has root symmetry. Find all real zeroes.

Q737. [F] A beam deflection equation $y(x) = x^4 - 4x^2 + 3$ has root symmetry. Find all real zeroes.

Q738. [F] A beam deflection equation $y(x) = x^4 - 7x^2 + 10$ has root symmetry. Find all real zeroes.

Q739. [F] A beam deflection equation $y(x) = x^4 - 5x^2 + 6$

has root symmetry. Find all real zeroes.

Q740. [F] A beam deflection equation $y(x) = x^4 - 7x^2 + 12$ has root symmetry. Find all real zeroes.

Q741. [F] A beam deflection equation $y(x) = x^4 - 6x^2 + 5$ has root symmetry. Find all real zeroes.

Q742. [F] A beam deflection equation $y(x) = x^4 - 4x^2 + 4$ has root symmetry. Find all real zeroes.

Q743. [F] A beam deflection equation $y(x) = x^4 - 6x^2 + 9$ has root symmetry. Find all real zeroes.

Q744. [F] A beam deflection equation $y(x) = x^4 - 9x^2 + 20$ has root symmetry. Find all real zeroes.

Q745. [F] A beam deflection equation $y(x) = x^4 - 3x^2 + 2$ has root symmetry. Find all real zeroes.

Q746. [F] A beam deflection equation $y(x) = x^4 - 5x^2 + 6$ has root symmetry. Find all real zeroes.

Q747. [F] A beam deflection equation $y(x) = x^4 - 8x^2 + 15$ has root symmetry. Find all real zeroes.

Q748. [F] A beam deflection equation $y(x) = x^4 - 6x^2 + 8$ has root symmetry. Find all real zeroes.

Q749. [F] A beam deflection equation $y(x) = x^4 - 4x^2 + 3$ has root symmetry. Find all real zeroes.

Q750. [F] A beam deflection equation $y(x) = x^4 - 7x^2 + 10$ has root symmetry. Find all real zeroes.

Q751. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q752. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q753. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q754. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q755. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q756. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

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Q759. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q760. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q761. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

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Q763. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

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Q766. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q767. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q768. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q769. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

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Q772. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q773. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q774. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q775. [G] Obtain all zeroes of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ given two zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q776. [H] Show that if $\alpha = 2 + \sqrt{3}$ is a zero of a polynomial with rational coefficients, then $2 - \sqrt{3}$ must also be a zero.

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Q789. [H] Show that if $\alpha = 2 + \sqrt{3}$ is a zero of a polynomial with rational coefficients, then $2 - \sqrt{3}$ must also be a zero.

Q790. [H] Show that if $\alpha = 2 + \sqrt{3}$ is a zero of a polynomial

with rational coefficients, then $2-\sqrt{3}$ must also be a zero.

Q791. [H] Show that if $\alpha = 2+\sqrt{3}$ is a zero of a polynomial with rational coefficients, then $2-\sqrt{3}$ must also be a zero.

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Q799. [H] Show that if $\alpha = 2+\sqrt{3}$ is a zero of a polynomial with rational coefficients, then $2-\sqrt{3}$ must also be a zero.

Q800. [H] Show that if $\alpha = 2+\sqrt{3}$ is a zero of a polynomial with rational coefficients, then $2-\sqrt{3}$ must also be a zero.

Pattern 5: Geometrical & Graphical Zeroes

Q801. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(-3, 0)$ and does not cross it. State the number and values of its zeroes.

Q802. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(-2, 0)$ and does not cross it. State the number and values of its zeroes.

Q803. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(-1, 0)$ and does not cross it. State the number and values of its zeroes.

Q804. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(0, 0)$ and does not cross it. State the number and values of its zeroes.

Q805. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(1, 0)$ and does not cross it. State the number and values of its zeroes.

Q806. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(2, 0)$ and does not cross it. State the number and values of its zeroes.

Q807. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(3, 0)$ and does not cross it. State the number and values of its zeroes.

Q808. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(-3, 0)$ and does not cross it. State the number and values of its zeroes.

Q809. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(-2, 0)$ and does not cross it. State the number and values of its zeroes.

Q810. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(-1, 0)$ and does not cross it. State the number and values of its zeroes.

Q811. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(0, 0)$ and does not cross it. State the number and values of its zeroes.

Q812. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(1, 0)$ and does not cross it. State the number and values of its zeroes.

Q813. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(2, 0)$ and does not cross it. State the number and values of its zeroes.

Q814. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(3, 0)$ and does not cross it. State the number and values of its zeroes.

Q815. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(-3, 0)$ and does not cross it. State the number and values of its zeroes.

Q816. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(-2, 0)$ and does not cross it. State the number and values of its zeroes.

Q817. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(-1, 0)$ and does not cross it. State the number and values of its zeroes.

Q818. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(0, 0)$ and does not cross it. State the number and values of its zeroes.

Q819. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(1, 0)$ and does not cross it. State the number and values of its zeroes.

Q820. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(2, 0)$ and does not cross it. State the number and values of its zeroes.

Q821. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(3, 0)$ and does not cross it. State the number and values of its zeroes.

Q822. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(-3, 0)$ and does not cross it. State the number and values of its zeroes.

Q823. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(-2, 0)$ and does not cross it. State the number and values of its zeroes.

Q824. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(-1, 0)$ and does not cross it. State the number and values of its zeroes.

Q825. [A] A parabola $y = ax^2 + bx + c$ touches the x-axis at $(0, 0)$ and does not cross it. State the number and values of its zeroes.

Q826. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q827. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of

$p(x)$ and list its zeroes.

Q828. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q829. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q830. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q831. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q832. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

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Q836. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q837. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q838. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q839. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q840. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q841. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

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Q843. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q844. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q845. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q846. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q847. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q848. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q849. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q850. [B] If the graph of $y = p(x)$ intersects the x-axis at $(-2, 0)$, $(1, 0)$, and $(4, 0)$, determine the minimum degree of $p(x)$ and list its zeroes.

Q851. [C] A quadratic graph $y = ax^2 + bx + c$ lies entirely above the x-axis. How many real zeroes does it possess and what is the sign of D?

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Q875. [C] A quadratic graph $y = ax^2 + bx + c$ lies entirely above the x-axis. How many real zeroes does it possess and what is the sign of D?

Q876. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis?
(A) 3 (B) 2 (C) 1 (D) 4

Q877. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis?
(A) 3 (B) 2 (C) 1 (D) 4

Q878. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis?
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Q888. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis?
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Q889. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis?
(A) 3 (B) 2 (C) 1 (D) 4

Q890. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis?
(A) 3 (B) 2 (C) 1 (D) 4

Q891. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis?
(A) 3 (B) 2 (C) 1 (D) 4

Q892. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis?
(A) 3 (B) 2 (C) 1 (D) 4

Q893. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis?
(A) 3 (B) 2 (C) 1 (D) 4

Q894. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis?
(A) 3 (B) 2 (C) 1 (D) 4

Q895. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis?
(A) 3 (B) 2 (C) 1 (D) 4

Q896. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis?
(A) 3 (B) 2 (C) 1 (D) 4

Q897. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis?
(A) 3 (B) 2 (C) 1 (D) 4

Q898. [D] What is the maximum number of points a cubic

polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis? (A) 3 (B) 2 (C) 1 (D) 4

Q899. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis? (A) 3 (B) 2 (C) 1 (D) 4

Q900. [D] What is the maximum number of points a cubic polynomial $y = ax^3 + bx^2 + cx + d$ can intersect the x-axis? (A) 3 (B) 2 (C) 1 (D) 4

Q901. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q902. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q903. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q904. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q905. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q906. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q907. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

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Q909. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q910. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q911. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

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Q913. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q914. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q915. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q916. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q917. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q918. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q919. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q920. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q921. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q922. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q923. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q924. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q925. [E] The parabola $y = ax^2 + bx + c$ is symmetric about $x = 2$. If one x-intercept is at $x = -1$, find the other intercept visually using symmetry.

Q926. [F] An archway modeled by $y = -x^2 + 4$ meets the ground at $y = 0$. Find the width of the arch at ground level.

Q927. [F] An archway modeled by $y = -x^2 + 4$ meets the ground at $y = 0$. Find the width of the arch at ground level.

Q928. [F] An archway modeled by $y = -x^2 + 4$ meets the ground at $y = 0$. Find the width of the arch at ground level.

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Q934. [F] An archway modeled by $y = -x^2 + 4$ meets the ground at $y = 0$. Find the width of the arch at ground level.

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Q936. [F] An archway modeled by $y = -x^2 + 4$ meets the ground at $y = 0$. Find the width of the arch at ground level.

Q937. [F] An archway modeled by $y = -x^2 + 4$ meets the ground at $y = 0$. Find the width of the arch at ground level.

Q938. [F] An archway modeled by $y = -x^2 + 4$ meets the ground at $y = 0$. Find the width of the arch at ground level.

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Q940. [F] An archway modeled by $y = -x^2 + 4$ meets the ground at $y = 0$. Find the width of the arch at ground level.

Q941. [F] An archway modeled by $y = -x^2 + 4$ meets the ground at $y = 0$. Find the width of the arch at ground level.

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Q947. [F] An archway modeled by $y = -x^2 + 4$ meets the ground at $y = 0$. Find the width of the arch at ground level.

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Q949. [F] An archway modeled by $y = -x^2 + 4$ meets the ground at $y = 0$. Find the width of the arch at ground level.

Q950. [F] An archway modeled by $y = -x^2 + 4$ meets the ground at $y = 0$. Find the width of the arch at ground level.

Q951. [G] If $y = p(x)$ crosses the y-axis at $(0, 5)$ and never touches the x-axis, determine the number of real zeroes of $p(x)$.

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Q975. [G] If $y = p(x)$ crosses the y-axis at $(0, 5)$ and never touches the x-axis, determine the number of real zeroes of $p(x)$.

Q976. [H] Identify the zeroes of $p(x) = x^3 - 4x$ by analyzing its horizontal axis intersection points.

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Q1000. [H] Identify the zeroes of $p(x) = x^3 - 4x$ by analyzing its horizontal axis intersection points.

Pattern 6: Interchanged Coefficients / Large System Values

Q1001. [A] Solve the system of equations using the addition-subtraction method:
$100x + 200y = 300$
$200x + 100y = 300$

Q1002. [A] Solve the system of equations using the addition-subtraction method:
$101x + 201y = 604$
$201x + 101y = 604$

Q1003. [A] Solve the system of equations using the addition-subtraction method:
$102x + 202y = 912$
$202x + 102y = 912$

Q1004. [A] Solve the system of equations using the addition-subtraction method:
$103x + 203y = 1224$
$203x + 103y = 1224$

Q1005. [A] Solve the system of equations using the addition-subtraction method:
$104x + 204y = 724$
$204x + 104y = 1124$

Q1006. [A] Solve the system of equations using the addition-subtraction method:
$105x + 205y = 515$
$205x + 105y = 415$

Q1007. [A] Solve the system of equations using the addition-subtraction method:
$106x + 206y = 830$
$206x + 106y = 730$

Q1008. [A] Solve the system of equations using the addition-subtraction method:
$107x + 207y = 1149$
$207x + 107y = 1049$

Q1009. [A] Solve the system of equations using the addition-subtraction method:
$108x + 208y = 640$
$208x + 108y = 940$

Q1010. [A] Solve the system of equations using the addition-subtraction method:
$109x + 209y = 963$
$209x + 109y = 1263$

Q1011. [A] Solve the system of equations using the addition-subtraction method:
$110x + 210y = 740$
$210x + 110y = 540$

Q1012. [A] Solve the system of equations using the addition-subtraction method:
$111x + 211y = 1066$
$211x + 111y = 866$

Q1013. [A] Solve the system of equations using the addition-subtraction method:
$112x + 212y = 548$
$212x + 112y = 748$

Q1014. [A] Solve the system of equations using the addition-subtraction method:
$113x + 213y = 878$
$213x + 113y = 1078$

Q1015. [A] Solve the system of equations using the addition-subtraction method:
$114x + 214y = 1212$
$214x + 114y = 1412$

Q1016. [A] Solve the system of equations using the addition-subtraction method:
$115x + 215y = 975$
$215x + 115y = 675$

Q1017. [A] Solve the system of equations using the addition-subtraction method:
$116x + 216y = 448$
$216x + 116y = 548$

Q1018. [A] Solve the system of equations using the addition-subtraction method:
$117x + 217y = 785$
$217x + 117y = 885$

Q1019. [A] Solve the system of equations using the addition-subtraction method:
$118x + 218y = 1126$
$218x + 118y = 1226$

Q1020. [A] Solve the system of equations using the addition-subtraction method:
$119x + 219y = 1471$
$219x + 119y = 1571$

Q1021. [A] Solve the system of equations using the addition-subtraction method:
$120x + 220y = 340$
$220x + 120y = 340$

Q1022. [A] Solve the system of equations using the addition-subtraction method:
$121x + 221y = 684$
$221x + 121y = 684$

Q1023. [A] Solve the system of equations using the addition-subtraction method:
$122x + 222y = 1032$
$222x + 122y = 1032$

Q1024. [A] Solve the system of equations using the addition-subtraction method:
$123x + 223y = 1384$
$223x + 123y = 1384$

Q1025. [A] Solve the system of equations using the addition-subtraction method:
$124x + 224y = 844$
$224x + 124y = 1244$

Q1026. [B] For the system $125x + 225y = 575$ and $225x + 125y = 475$, find the value of $(x + y)$ without individually solving for x and y.

Q1027. [B] For the system $126x + 226y = 930$ and $226x + 126y = 830$, find the value of $(x + y)$ without individually solving for x and y.

Q1028. [B] For the system $127x + 227y = 1289$ and $227x + 127y = 1189$, find the value of $(x + y)$ without individually solving for x and y.

Q1029. [B] For the system $128x + 228y = 740$ and $228x + 128y = 1040$, find the value of $(x + y)$ without individually solving for x and y.

Q1030. [B] For the system $129x + 229y = 1103$ and $229x + 129y = 1403$, find the value of $(x + y)$ without individually solving for x and y.

Q1031. [B] For the system $130x + 230y = 820$ and $230x + 130y = 620$, find the value of $(x + y)$ without individually solving for x and y.

Q1032. [B] For the system $131x + 231y = 1186$ and $231x + 131y = 986$, find the value of $(x + y)$ without individually solving for x and y.

Q1033. [B] For the system $132x + 232y = 628$ and $232x + 132y = 828$, find the value of $(x + y)$ without individually solving for x and y.

Q1034. [B] For the system $133x + 233y = 998$ and $233x + 133y = 1198$, find the value of $(x + y)$ without individually solving for x and y.

Q1035. [B] For the system $134x + 234y = 1372$ and $234x + 134y = 1572$, find the value of $(x + y)$ without individually solving for x and y.

Q1036. [B] For the system $135x + 235y = 1075$ and $235x + 135y = 775$, find the value of $(x + y)$ without individually solving for x and y.

Q1037. [B] For the system $136x + 236y = 508$ and $236x + 136y = 608$, find the value of $(x + y)$ without

individually solving for x and y.

Q1038. [B] For the system $137x + 237y = 885$ and $237x + 137y = 985$, find the value of $(x + y)$ without individually solving for x and y.

Q1039. [B] For the system $138x + 238y = 1266$ and $238x + 138y = 1366$, find the value of $(x + y)$ without individually solving for x and y.

Q1040. [B] For the system $139x + 239y = 1651$ and $239x + 139y = 1751$, find the value of $(x + y)$ without individually solving for x and y.

Q1041. [B] For the system $140x + 240y = 380$ and $240x + 140y = 380$, find the value of $(x + y)$ without individually solving for x and y.

Q1042. [B] For the system $141x + 241y = 764$ and $241x + 141y = 764$, find the value of $(x + y)$ without individually solving for x and y.

Q1043. [B] For the system $142x + 242y = 1152$ and $242x + 142y = 1152$, find the value of $(x + y)$ without individually solving for x and y.

Q1044. [B] For the system $143x + 243y = 1544$ and $243x + 143y = 1544$, find the value of $(x + y)$ without individually solving for x and y.

Q1045. [B] For the system $144x + 244y = 964$ and $244x + 144y = 1364$, find the value of $(x + y)$ without individually solving for x and y.

Q1046. [B] For the system $145x + 245y = 635$ and $245x + 145y = 535$, find the value of $(x + y)$ without individually solving for x and y.

Q1047. [B] For the system $146x + 246y = 1030$ and $246x + 146y = 930$, find the value of $(x + y)$ without individually solving for x and y.

Q1048. [B] For the system $147x + 247y = 1429$ and $247x + 147y = 1329$, find the value of $(x + y)$ without individually solving for x and y.

Q1049. [B] For the system $148x + 248y = 840$ and $248x + 148y = 1140$, find the value of $(x + y)$ without individually solving for x and y.

Q1050. [B] For the system $149x + 249y = 1243$ and $249x + 149y = 1543$, find the value of $(x + y)$ without individually solving for x and y.

Q1051. [C] For the system $100x + 250y = 850$ and $250x + 100y = 550$, evaluate the difference $(x - y)$.

Q1052. [C] For the system $101x + 251y = 1206$ and $251x + 101y = 906$, evaluate the difference $(x - y)$.

Q1053. [C] For the system $102x + 252y = 558$ and $252x + 102y = 858$, evaluate the difference $(x - y)$.

Q1054. [C] For the system $103x + 253y = 918$ and $253x + 103y = 1218$, evaluate the difference $(x - y)$.

Q1055. [C] For the system $104x + 254y = 1282$ and $254x + 104y = 1582$, evaluate the difference $(x - y)$.

Q1056. [C] For the system $105x + 255y = 1125$ and $255x + 105y = 675$, evaluate the difference $(x - y)$.

Q1057. [C] For the system $106x + 256y = 468$ and $256x + 106y = 618$, evaluate the difference $(x - y)$.

Q1058. [C] For the system $107x + 257y = 835$ and $257x + 107y = 985$, evaluate the difference $(x - y)$.

Q1059. [C] For the system $108x + 258y = 1206$ and $258x + 108y = 1356$, evaluate the difference $(x - y)$.

Q1060. [C] For the system $109x + 259y = 1581$ and $259x + 109y = 1731$, evaluate the difference $(x - y)$.

Q1061. [C] For the system $110x + 200y = 310$ and $200x + 110y = 310$, evaluate the difference $(x - y)$.

Q1062. [C] For the system $111x + 201y = 624$ and $201x + 111y = 624$, evaluate the difference $(x - y)$.

Q1063. [C] For the system $112x + 202y = 942$ and $202x + 112y = 942$, evaluate the difference $(x - y)$.

Q1064. [C] For the system $113x + 203y = 1264$ and $203x + 113y = 1264$, evaluate the difference $(x - y)$.

Q1065. [C] For the system $114x + 204y = 774$ and $204x + 114y = 1134$, evaluate the difference $(x - y)$.

Q1066. [C] For the system $115x + 205y = 525$ and $205x + 115y = 435$, evaluate the difference $(x - y)$.

Q1067. [C] For the system $116x + 206y = 850$ and $206x + 116y = 760$, evaluate the difference $(x - y)$.

Q1068. [C] For the system $117x + 207y = 1179$ and $207x + 117y = 1089$, evaluate the difference $(x - y)$.

Q1069. [C] For the system $118x + 208y = 680$ and $208x + 118y = 950$, evaluate the difference $(x - y)$.

Q1070. [C] For the system $119x + 209y = 1013$ and $209x + 119y = 1283$, evaluate the difference $(x - y)$.

Q1071. [C] For the system $120x + 210y = 750$ and $210x + 120y = 570$, evaluate the difference $(x - y)$.

Q1072. [C] For the system $121x + 211y = 1086$ and $211x + 121y = 906$, evaluate the difference $(x - y)$.

Q1073. [C] For the system $122x + 212y = 578$ and $212x + 122y = 758$, evaluate the difference $(x - y)$.

Q1074. [C] For the system $123x + 213y = 918$ and $213x + 123y = 1098$, evaluate the difference $(x - y)$.

Q1075. [C] For the system $124x + 214y = 1262$ and $214x + 124y = 1442$, evaluate the difference $(x - y)$.

Q1076. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1077. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1078. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1079. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1080. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?

(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1081. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1082. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1083. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1084. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1085. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1086. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1087. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1088. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1089. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1090. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1091. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1092. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1093. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1094. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1095. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1096. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1097. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1098. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1099. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1100. [D] What is the value of x for $152x - 378y = -74$ and $-378x + 152y = -604$?
(A) $x = 2$ (B) $x = 1$ (C) $x = 3$ (D) $x = -2$

Q1101. [E] Verify whether $x = 1, y = 1$ satisfies $100x + 240y = 340$ and $240x + 100y = 340$.

Q1102. [E] Verify whether $x = 2, y = 2$ satisfies $101x + 241y = 684$ and $241x + 101y = 684$.

Q1103. [E] Verify whether $x = 3, y = 3$ satisfies $102x + 242y = 1032$ and $242x + 102y = 1032$.

Q1104. [E] Verify whether $x = 4, y = 4$ satisfies $103x + 243y = 1384$ and $243x + 103y = 1384$.

Q1105. [E] Verify whether $x = 5, y = 1$ satisfies $104x + 244y = 764$ and $244x + 104y = 1324$.

Q1106. [E] Verify whether $x = 1, y = 2$ satisfies $105x + 245y = 595$ and $245x + 105y = 455$.

Q1107. [E] Verify whether $x = 2, y = 3$ satisfies $106x + 246y = 950$ and $246x + 106y = 810$.

Q1108. [E] Verify whether $x = 3, y = 4$ satisfies $107x + 247y = 1309$ and $247x + 107y = 1169$.

Q1109. [E] Verify whether $x = 4, y = 1$ satisfies $108x + 248y = 680$ and $248x + 108y = 1100$.

Q1110. [E] Verify whether $x = 5, y = 2$ satisfies $109x + 249y = 1043$ and $249x + 109y = 1463$.

Q1111. [E] Verify whether $x = 1, y = 3$ satisfies $110x + 250y = 860$ and $250x + 110y = 580$.

Q1112. [E] Verify whether $x = 2, y = 4$ satisfies $111x + 251y = 1226$ and $251x + 111y = 946$.

Q1113. [E] Verify whether $x = 3, y = 1$ satisfies $112x + 252y = 588$ and $252x + 112y = 868$.

Q1114. [E] Verify whether $x = 4, y = 2$ satisfies $113x + 253y = 958$ and $253x + 113y = 1238$.

Q1115. [E] Verify whether $x = 5, y = 3$ satisfies $114x + 254y = 1332$ and $254x + 114y = 1612$.

Q1116. [E] Verify whether $x = 1, y = 4$ satisfies $115x + 255y = 1135$ and $255x + 115y = 715$.

Q1117. [E] Verify whether $x = 2, y = 1$ satisfies $116x + 256y = 488$ and $256x + 116y = 628$.

Q1118. [E] Verify whether $x = 3, y = 2$ satisfies $117x + 257y = 865$ and $257x + 117y = 1005$.

Q1119. [E] Verify whether $x = 4, y = 3$ satisfies $118x + 258y = 1246$ and $258x + 118y = 1386$.

Q1120. [E] Verify whether $x = 5, y = 4$ satisfies $119x + 259y = 1631$ and $259x + 119y = 1771$.

Q1121. [E] Verify whether $x = 1, y = 1$ satisfies $120x + 200y = 320$ and $200x + 120y = 320$.

Q1122. [E] Verify whether $x = 2, y = 2$ satisfies

$121x + 201y = 644$ and $201x + 121y = 644$.

Q1123. [E] Verify whether $x = 3, y = 3$ satisfies $122x + 202y = 972$ and $202x + 122y = 972$.

Q1124. [E] Verify whether $x = 4, y = 4$ satisfies $123x + 203y = 1304$ and $203x + 123y = 1304$.

Q1125. [E] Verify whether $x = 5, y = 1$ satisfies $124x + 204y = 824$ and $204x + 124y = 1144$.

Q1126. [F] Two wholesale currency transactions are modeled by $125x + 205y = 535$ and $205x + 125y = 455$. Calculate the unit rates x and y.

Q1127. [F] Two wholesale currency transactions are modeled by $126x + 206y = 870$ and $206x + 126y = 790$. Calculate the unit rates x and y.

Q1128. [F] Two wholesale currency transactions are modeled by $127x + 207y = 1209$ and $207x + 127y = 1129$. Calculate the unit rates x and y.

Q1129. [F] Two wholesale currency transactions are modeled by $128x + 208y = 720$ and $208x + 128y = 960$. Calculate the unit rates x and y.

Q1130. [F] Two wholesale currency transactions are modeled by $129x + 209y = 1063$ and $209x + 129y = 1303$. Calculate the unit rates x and y.

Q1131. [F] Two wholesale currency transactions are modeled by $130x + 210y = 760$ and $210x + 130y = 600$. Calculate the unit rates x and y.

Q1132. [F] Two wholesale currency transactions are modeled by $131x + 211y = 1106$ and $211x + 131y = 946$. Calculate the unit rates x and y.

Q1133. [F] Two wholesale currency transactions are modeled by $132x + 212y = 608$ and $212x + 132y = 768$. Calculate the unit rates x and y.

Q1134. [F] Two wholesale currency transactions are modeled by $133x + 213y = 958$ and $213x + 133y = 1118$. Calculate the unit rates x and y.

Q1135. [F] Two wholesale currency transactions are modeled by $134x + 214y = 1312$ and $214x + 134y = 1472$. Calculate the unit rates x and y.

Q1136. [F] Two wholesale currency transactions are modeled by $135x + 215y = 995$ and $215x + 135y = 755$. Calculate the unit rates x and y.

Q1137. [F] Two wholesale currency transactions are modeled by $136x + 216y = 488$ and $216x + 136y = 568$. Calculate the unit rates x and y.

Q1138. [F] Two wholesale currency transactions are modeled by $137x + 217y = 845$ and $217x + 137y = 925$. Calculate the unit rates x and y.

Q1139. [F] Two wholesale currency transactions are modeled by $138x + 218y = 1206$ and $218x + 138y = 1286$. Calculate the unit rates x and y.

Q1140. [F] Two wholesale currency transactions are modeled by $139x + 219y = 1571$ and $219x + 139y = 1651$. Calculate the unit rates x and y.

Q1141. [F] Two wholesale currency transactions are modeled by $140x + 220y = 360$ and $220x + 140y = 360$. Calculate the unit rates x and y.

Q1142. [F] Two wholesale currency transactions are modeled by $141x + 221y = 724$ and $221x + 141y = 724$. Calculate the unit rates x and y.

Q1143. [F] Two wholesale currency transactions are modeled by $142x + 222y = 1092$ and $222x + 142y = 1092$. Calculate the unit rates x and y.

Q1144. [F] Two wholesale currency transactions are modeled by $143x + 223y = 1464$ and $223x + 143y = 1464$. Calculate the unit rates x and y.

Q1145. [F] Two wholesale currency transactions are modeled by $144x + 224y = 944$ and $224x + 144y = 1264$. Calculate the unit rates x and y.

Q1146. [F] Two wholesale currency transactions are modeled by $145x + 225y = 595$ and $225x + 145y = 515$. Calculate the unit rates x and y.

Q1147. [F] Two wholesale currency transactions are modeled by $146x + 226y = 970$ and $226x + 146y = 890$. Calculate the unit rates x and y.

Q1148. [F] Two wholesale currency transactions are modeled by $147x + 227y = 1349$ and $227x + 147y = 1269$. Calculate the unit rates x and y.

Q1149. [F] Two wholesale currency transactions are modeled by $148x + 228y = 820$ and $228x + 148y = 1060$. Calculate the unit rates x and y.

Q1150. [F] Two wholesale currency transactions are modeled by $149x + 229y = 1203$ and $229x + 149y = 1443$. Calculate the unit rates x and y.

Q1151. [G] Express y in terms of x for the equation $100x + 230y = 790$.

Q1152. [G] Express y in terms of x for the equation $101x + 231y = 1126$.

Q1153. [G] Express y in terms of x for the equation $102x + 232y = 538$.

Q1154. [G] Express y in terms of x for the equation $103x + 233y = 878$.

Q1155. [G] Express y in terms of x for the equation $104x + 234y = 1222$.

Q1156. [G] Express y in terms of x for the equation $105x + 235y = 1045$.

Q1157. [G] Express y in terms of x for the equation $106x + 236y = 448$.

Q1158. [G] Express y in terms of x for the equation $107x + 237y = 795$.

Q1159. [G] Express y in terms of x for the equation $108x + 238y = 1146$.

Q1160. [G] Express y in terms of x for the equation $109x + 239y = 1501$.

Q1161. [G] Express y in terms of x for the equation $110x + 240y = 350$.

Q1162. [G] Express y in terms of x for the equation

$111x + 241y = 704$.

Q1163. [G] Express y in terms of x for the equation $112x + 242y = 1062$.

Q1164. [G] Express y in terms of x for the equation $113x + 243y = 1424$.

Q1165. [G] Express y in terms of x for the equation $114x + 244y = 814$.

Q1166. [G] Express y in terms of x for the equation $115x + 245y = 605$.

Q1167. [G] Express y in terms of x for the equation $116x + 246y = 970$.

Q1168. [G] Express y in terms of x for the equation $117x + 247y = 1339$.

Q1169. [G] Express y in terms of x for the equation $118x + 248y = 720$.

Q1170. [G] Express y in terms of x for the equation $119x + 249y = 1093$.

Q1171. [G] Express y in terms of x for the equation $120x + 250y = 870$.

Q1172. [G] Express y in terms of x for the equation $121x + 251y = 1246$.

Q1173. [G] Express y in terms of x for the equation $122x + 252y = 618$.

Q1174. [G] Express y in terms of x for the equation $123x + 253y = 998$.

Q1175. [G] Express y in terms of x for the equation $124x + 254y = 1382$.

Q1176. [H] Evaluate $x^2 - y^2$ for the linear system $125x + 255y = 1145$ and $255x + 125y = 755$.

Q1177. [H] Evaluate $x^2 - y^2$ for the linear system $126x + 256y = 508$ and $256x + 126y = 638$.

Q1178. [H] Evaluate $x^2 - y^2$ for the linear system $127x + 257y = 895$ and $257x + 127y = 1025$.

Q1179. [H] Evaluate $x^2 - y^2$ for the linear system $128x + 258y = 1286$ and $258x + 128y = 1416$.

Q1180. [H] Evaluate $x^2 - y^2$ for the linear system $129x + 259y = 1681$ and $259x + 129y = 1811$.

Q1181. [H] Evaluate $x^2 - y^2$ for the linear system $130x + 200y = 330$ and $200x + 130y = 330$.

Q1182. [H] Evaluate $x^2 - y^2$ for the linear system $131x + 201y = 664$ and $201x + 131y = 664$.

Q1183. [H] Evaluate $x^2 - y^2$ for the linear system $132x + 202y = 1002$ and $202x + 132y = 1002$.

Q1184. [H] Evaluate $x^2 - y^2$ for the linear system $133x + 203y = 1344$ and $203x + 133y = 1344$.

Q1185. [H] Evaluate $x^2 - y^2$ for the linear system $134x + 204y = 874$ and $204x + 134y = 1154$.

Q1186. [H] Evaluate $x^2 - y^2$ for the linear system $135x + 205y = 545$ and $205x + 135y = 475$.

Q1187. [H] Evaluate $x^2 - y^2$ for the linear system $136x + 206y = 890$ and $206x + 136y = 820$.

Q1188. [H] Evaluate $x^2 - y^2$ for the linear system $137x + 207y = 1239$ and $207x + 137y = 1169$.

Q1189. [H] Evaluate $x^2 - y^2$ for the linear system $138x + 208y = 760$ and $208x + 138y = 970$.

Q1190. [H] Evaluate $x^2 - y^2$ for the linear system $139x + 209y = 1113$ and $209x + 139y = 1323$.

Q1191. [H] Evaluate $x^2 - y^2$ for the linear system $140x + 210y = 770$ and $210x + 140y = 630$.

Q1192. [H] Evaluate $x^2 - y^2$ for the linear system $141x + 211y = 1126$ and $211x + 141y = 986$.

Q1193. [H] Evaluate $x^2 - y^2$ for the linear system $142x + 212y = 638$ and $212x + 142y = 778$.

Q1194. [H] Evaluate $x^2 - y^2$ for the linear system $143x + 213y = 998$ and $213x + 143y = 1138$.

Q1195. [H] Evaluate $x^2 - y^2$ for the linear system $144x + 214y = 1362$ and $214x + 144y = 1502$.

Q1196. [H] Evaluate $x^2 - y^2$ for the linear system $145x + 215y = 1005$ and $215x + 145y = 795$.

Q1197. [H] Evaluate $x^2 - y^2$ for the linear system $146x + 216y = 508$ and $216x + 146y = 578$.

Q1198. [H] Evaluate $x^2 - y^2$ for the linear system $147x + 217y = 875$ and $217x + 147y = 945$.

Q1199. [H] Evaluate $x^2 - y^2$ for the linear system $148x + 218y = 1246$ and $218x + 148y = 1316$.

Q1200. [H] Evaluate $x^2 - y^2$ for the linear system $149x + 219y = 1621$ and $219x + 149y = 1691$.

Pattern 7: Fixed Charges & Variable Rate Constraints

Q1201. [A] A taxi service charges a fixed charge plus a rate per km. For 3 km, the charge is Rs. 26, and for 7 km, the charge is Rs. 34. Find the fixed charge and rate per km.

Q1202. [A] A taxi service charges a fixed charge plus a rate per km. For 4 km, the charge is Rs. 37, and for 8 km, the charge is Rs. 49. Find the fixed charge and rate per km.

Q1203. [A] A taxi service charges a fixed charge plus a rate per km. For 5 km, the charge is Rs. 50, and for 9 km, the charge is Rs. 66. Find the fixed charge and rate per km.

Q1204. [A] A taxi service charges a fixed charge plus a rate per km. For 6 km, the charge is Rs. 65, and for 10 km, the charge is Rs. 85. Find the fixed charge and rate per km.

Q1205. [A] A taxi service charges a fixed charge plus a rate per km. For 3 km, the charge is Rs. 58, and for 11 km, the charge is Rs. 106. Find the fixed charge and rate per km.

Q1206. [A] A taxi service charges a fixed charge plus a

rate per km. For 4 km, the charge is Rs. 53, and for 7 km, the charge is Rs. 59. Find the fixed charge and rate per km.

Q1207. [A] A taxi service charges a fixed charge plus a rate per km. For 5 km, the charge is Rs. 65, and for 8 km, the charge is Rs. 74. Find the fixed charge and rate per km.

Q1208. [A] A taxi service charges a fixed charge plus a rate per km. For 6 km, the charge is Rs. 79, and for 9 km, the charge is Rs. 91. Find the fixed charge and rate per km.

Q1209. [A] A taxi service charges a fixed charge plus a rate per km. For 3 km, the charge is Rs. 75, and for 10 km, the charge is Rs. 110. Find the fixed charge and rate per km.

Q1210. [A] A taxi service charges a fixed charge plus a rate per km. For 4 km, the charge is Rs. 89, and for 11 km, the charge is Rs. 131. Find the fixed charge and rate per km.

Q1211. [A] A taxi service charges a fixed charge plus a rate per km. For 5 km, the charge is Rs. 30, and for 7 km, the charge is Rs. 34. Find the fixed charge and rate per km.

Q1212. [A] A taxi service charges a fixed charge plus a rate per km. For 6 km, the charge is Rs. 43, and for 8 km, the charge is Rs. 49. Find the fixed charge and rate per km.

Q1213. [A] A taxi service charges a fixed charge plus a rate per km. For 3 km, the charge is Rs. 42, and for 9 km, the charge is Rs. 66. Find the fixed charge and rate per km.

Q1214. [A] A taxi service charges a fixed charge plus a rate per km. For 4 km, the charge is Rs. 55, and for 10 km, the charge is Rs. 85. Find the fixed charge and rate per km.

Q1215. [A] A taxi service charges a fixed charge plus a rate per km. For 5 km, the charge is Rs. 70, and for 11 km, the charge is Rs. 106. Find the fixed charge and rate per km.

Q1216. [A] A taxi service charges a fixed charge plus a rate per km. For 6 km, the charge is Rs. 57, and for 7 km, the charge is Rs. 59. Find the fixed charge and rate per km.

Q1217. [A] A taxi service charges a fixed charge plus a rate per km. For 3 km, the charge is Rs. 59, and for 8 km, the charge is Rs. 74. Find the fixed charge and rate per km.

Q1218. [A] A taxi service charges a fixed charge plus a rate per km. For 4 km, the charge is Rs. 71, and for 9 km, the charge is Rs. 91. Find the fixed charge and rate per km.

Q1219. [A] A taxi service charges a fixed charge plus a rate per km. For 5 km, the charge is Rs. 85, and for 10 km, the charge is Rs. 110. Find the fixed charge and rate per km.

Q1220. [A] A taxi service charges a fixed charge plus a rate per km. For 6 km, the charge is Rs. 101, and for 11 km, the charge is Rs. 131. Find the fixed charge and rate per km.

Q1221. [A] A taxi service charges a fixed charge plus a rate per km. For 3 km, the charge is Rs. 26, and for 7 km, the charge is Rs. 34. Find the fixed charge and rate per km.

Q1222. [A] A taxi service charges a fixed charge plus a rate per km. For 4 km, the charge is Rs. 37, and for 8 km, the charge is Rs. 49. Find the fixed charge and rate per km.

Q1223. [A] A taxi service charges a fixed charge plus a rate per km. For 5 km, the charge is Rs. 50, and for 9 km, the charge is Rs. 66. Find the fixed charge and rate per km.

Q1224. [A] A taxi service charges a fixed charge plus a rate per km. For 6 km, the charge is Rs. 65, and for 10 km, the charge is Rs. 85. Find the fixed charge and rate per km.

Q1225. [A] A taxi service charges a fixed charge plus a rate per km. For 3 km, the charge is Rs. 58, and for 11 km, the charge is Rs. 106. Find the fixed charge and rate per km.

Q1226. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 53 for 7 days and Suzi pays Rs. 59 for 10 days.

Q1227. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 65 for 8 days and Suzi pays Rs. 74 for 11 days.

Q1228. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 79 for 9 days and Suzi pays Rs. 91 for 12 days.

Q1229. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 75 for 6 days and Suzi pays Rs. 110 for 13 days.

Q1230. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 89 for 7 days and Suzi pays Rs. 131 for 14 days.

Q1231. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 30 for 8 days and Suzi pays Rs. 34 for 10 days.

Q1232. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 43 for 9 days and Suzi pays Rs. 49 for 11 days.

Q1233. [B] Form a pair of linear equations representing a

library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 42 for 6 days and Suzi pays Rs. 66 for 12 days.

Q1234. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 55 for 7 days and Suzi pays Rs. 85 for 13 days.

Q1235. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 70 for 8 days and Suzi pays Rs. 106 for 14 days.

Q1236. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 57 for 9 days and Suzi pays Rs. 59 for 10 days.

Q1237. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 59 for 6 days and Suzi pays Rs. 74 for 11 days.

Q1238. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 71 for 7 days and Suzi pays Rs. 91 for 12 days.

Q1239. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 85 for 8 days and Suzi pays Rs. 110 for 13 days.

Q1240. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 101 for 9 days and Suzi pays Rs. 131 for 14 days.

Q1241. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 26 for 6 days and Suzi pays Rs. 34 for 10 days.

Q1242. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 37 for 7 days and Suzi pays Rs. 49 for 11 days.

Q1243. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 50 for 8 days and Suzi pays Rs. 66 for 12 days.

Q1244. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 65 for 9 days and Suzi pays Rs. 85 for 13 days.

Q1245. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 58 for 6 days and Suzi pays Rs. 106 for 14 days.

Q1246. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 53 for 7 days and Suzi pays Rs. 59 for 10 days.

Q1247. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 65 for 8 days and Suzi pays Rs. 74 for 11 days.

Q1248. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 79 for 9 days and Suzi pays Rs. 91 for 12 days.

Q1249. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 75 for 6 days and Suzi pays Rs. 110 for 13 days.

Q1250. [B] Form a pair of linear equations representing a library's fixed 3-day charge x and additional daily fee y if Aftab pays Rs. 89 for 7 days and Suzi pays Rs. 131 for 14 days.

Q1251. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 5 days and pays Rs. 30, write the linear equation.

Q1252. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 6 days and pays Rs. 43, write the linear equation.

Q1253. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 3 days and pays Rs. 42, write the linear equation.

Q1254. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 4 days and pays Rs. 55, write the linear equation.

Q1255. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 5 days and pays Rs. 70, write the linear equation.

Q1256. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 6 days and pays Rs. 57, write the linear equation.

Q1257. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 3 days and pays Rs. 59, write the linear equation.

Q1258. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 4 days and pays Rs. 71, write the linear equation.

Q1259. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 5 days and pays Rs. 85, write the linear equation.

Q1260. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 6 days and pays Rs. 101, write the linear equation.

Q1261. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 3 days and pays Rs. 26, write the linear equation.

Q1262. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 4 days and pays Rs. 37, write the linear equation.

Q1263. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 5 days and pays Rs. 50, write the linear equation.

Q1264. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 6 days and pays Rs. 65, write the linear equation.

Q1265. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 3 days and pays Rs. 58, write the linear equation.

Q1266. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 4 days and pays Rs. 53, write the linear equation.

Q1267. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 5 days and pays Rs. 65, write the linear equation.

Q1268. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 6 days and pays Rs. 79, write the linear equation.

Q1269. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 3 days and pays Rs. 75, write the linear equation.

Q1270. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 4 days and pays Rs. 89, write the linear equation.

Q1271. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 5 days and pays Rs. 30, write the linear equation.

Q1272. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 6 days and pays Rs. 43, write the linear equation.

Q1273. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 3 days and pays Rs. 42, write the linear equation.

Q1274. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 4 days and pays Rs. 55, write the linear equation.

Q1275. [C] A boarding house charges a fixed monthly fee of Rs. x and Rs. y per day for food. If student A takes food for 5 days and pays Rs. 70, write the linear equation.

Q1276. [D] If fixed charge is Rs. 45 and per-km rate is Rs. 2, how much will a passenger pay for a trip of 15 km?
(A) Rs. 75 (B) Rs. 30 (C) Rs. 65 (D) Rs. 90

Q1277. [D] If fixed charge is Rs. 50 and per-km rate is Rs. 3, how much will a passenger pay for a trip of 15 km?
(A) Rs. 95 (B) Rs. 45 (C) Rs. 80 (D) Rs. 100

Q1278. [D] If fixed charge is Rs. 55 and per-km rate is Rs. 4, how much will a passenger pay for a trip of 15 km?
(A) Rs. 115 (B) Rs. 60 (C) Rs. 95 (D) Rs. 110

Q1279. [D] If fixed charge is Rs. 60 and per-km rate is Rs. 5, how much will a passenger pay for a trip of 15 km?
(A) Rs. 135 (B) Rs. 75 (C) Rs. 110 (D) Rs. 120

Q1280. [D] If fixed charge is Rs. 65 and per-km rate is Rs. 6, how much will a passenger pay for a trip of 15 km?
(A) Rs. 155 (B) Rs. 90 (C) Rs. 125 (D) Rs. 130

Q1281. [D] If fixed charge is Rs. 20 and per-km rate is Rs. 2, how much will a passenger pay for a trip of 15 km?
(A) Rs. 50 (B) Rs. 30 (C) Rs. 40 (D) Rs. 40

Q1282. [D] If fixed charge is Rs. 25 and per-km rate is Rs. 3, how much will a passenger pay for a trip of 15 km?
(A) Rs. 70 (B) Rs. 45 (C) Rs. 55 (D) Rs. 50

Q1283. [D] If fixed charge is Rs. 30 and per-km rate is Rs. 4, how much will a passenger pay for a trip of 15 km?
(A) Rs. 90 (B) Rs. 60 (C) Rs. 70 (D) Rs. 60

Q1284. [D] If fixed charge is Rs. 35 and per-km rate is Rs. 5, how much will a passenger pay for a trip of 15 km?
(A) Rs. 110 (B) Rs. 75 (C) Rs. 85 (D) Rs. 70

Q1285. [D] If fixed charge is Rs. 40 and per-km rate is Rs. 6, how much will a passenger pay for a trip of 15 km?
(A) Rs. 130 (B) Rs. 90 (C) Rs. 100 (D) Rs. 80

Q1286. [D] If fixed charge is Rs. 45 and per-km rate is Rs. 2, how much will a passenger pay for a trip of 15 km?
(A) Rs. 75 (B) Rs. 30 (C) Rs. 65 (D) Rs. 90

Q1287. [D] If fixed charge is Rs. 50 and per-km rate is Rs. 3, how much will a passenger pay for a trip of 15 km?
(A) Rs. 95 (B) Rs. 45 (C) Rs. 80 (D) Rs. 100

Q1288. [D] If fixed charge is Rs. 55 and per-km rate is Rs. 4, how much will a passenger pay for a trip of 15 km?
(A) Rs. 115 (B) Rs. 60 (C) Rs. 95 (D) Rs. 110

Q1289. [D] If fixed charge is Rs. 60 and per-km rate is Rs. 5, how much will a passenger pay for a trip of 15 km?
(A) Rs. 135 (B) Rs. 75 (C) Rs. 110 (D) Rs. 120

Q1290. [D] If fixed charge is Rs. 65 and per-km rate is Rs. 6, how much will a passenger pay for a trip of 15 km?
(A) Rs. 155 (B) Rs. 90 (C) Rs. 125 (D) Rs. 130

Q1291. [D] If fixed charge is Rs. 20 and per-km rate is Rs. 2, how much will a passenger pay for a trip of 15 km?
(A) Rs. 50 (B) Rs. 30 (C) Rs. 40 (D) Rs. 40

Q1292. [D] If fixed charge is Rs. 25 and per-km rate is Rs. 3, how much will a passenger pay for a trip of 15 km?
(A) Rs. 70 (B) Rs. 45 (C) Rs. 55 (D) Rs. 50

Q1293. [D] If fixed charge is Rs. 30 and per-km rate is Rs. 4, how much will a passenger pay for a trip of 15 km?
(A) Rs. 90 (B) Rs. 60 (C) Rs. 70 (D) Rs. 60

Q1294. [D] If fixed charge is Rs. 35 and per-km rate is Rs. 5, how much will a passenger pay for a trip of 15 km?
(A) Rs. 110 (B) Rs. 75 (C) Rs. 85 (D) Rs. 70

Q1295. [D] If fixed charge is Rs. 40 and per-km rate is Rs. 6, how much will a passenger pay for a trip of 15 km?
(A) Rs. 130 (B) Rs. 90 (C) Rs. 100 (D) Rs. 80

Q1296. [D] If fixed charge is Rs. 45 and per-km rate is Rs. 2, how much will a passenger pay for a trip of 15 km?
(A) Rs. 75 (B) Rs. 30 (C) Rs. 65 (D) Rs. 90

Q1297. [D] If fixed charge is Rs. 50 and per-km rate is Rs. 3, how much will a passenger pay for a trip of 15 km?
(A) Rs. 95 (B) Rs. 45 (C) Rs. 80 (D) Rs. 100

Q1298. [D] If fixed charge is Rs. 55 and per-km rate is Rs.

4, how much will a passenger pay for a trip of 15 km?
(A) Rs. 115 (B) Rs. 60 (C) Rs. 95 (D) Rs. 110

Q1299. [D] If fixed charge is Rs. 60 and per-km rate is Rs. 5, how much will a passenger pay for a trip of 15 km?
(A) Rs. 135 (B) Rs. 75 (C) Rs. 110 (D) Rs. 120

Q1300. [D] If fixed charge is Rs. 65 and per-km rate is Rs. 6, how much will a passenger pay for a trip of 15 km?
(A) Rs. 155 (B) Rs. 90 (C) Rs. 125 (D) Rs. 130

Q1301. [E] If the per-km rate y doubles while fixed charge x remains Rs. 20, calculate the cost for a 3 km journey.

Q1302. [E] If the per-km rate y doubles while fixed charge x remains Rs. 25, calculate the cost for a 4 km journey.

Q1303. [E] If the per-km rate y doubles while fixed charge x remains Rs. 30, calculate the cost for a 5 km journey.

Q1304. [E] If the per-km rate y doubles while fixed charge x remains Rs. 35, calculate the cost for a 6 km journey.

Q1305. [E] If the per-km rate y doubles while fixed charge x remains Rs. 40, calculate the cost for a 3 km journey.

Q1306. [E] If the per-km rate y doubles while fixed charge x remains Rs. 45, calculate the cost for a 4 km journey.

Q1307. [E] If the per-km rate y doubles while fixed charge x remains Rs. 50, calculate the cost for a 5 km journey.

Q1308. [E] If the per-km rate y doubles while fixed charge x remains Rs. 55, calculate the cost for a 6 km journey.

Q1309. [E] If the per-km rate y doubles while fixed charge x remains Rs. 60, calculate the cost for a 3 km journey.

Q1310. [E] If the per-km rate y doubles while fixed charge x remains Rs. 65, calculate the cost for a 4 km journey.

Q1311. [E] If the per-km rate y doubles while fixed charge x remains Rs. 20, calculate the cost for a 5 km journey.

Q1312. [E] If the per-km rate y doubles while fixed charge x remains Rs. 25, calculate the cost for a 6 km journey.

Q1313. [E] If the per-km rate y doubles while fixed charge x remains Rs. 30, calculate the cost for a 3 km journey.

Q1314. [E] If the per-km rate y doubles while fixed charge x remains Rs. 35, calculate the cost for a 4 km journey.

Q1315. [E] If the per-km rate y doubles while fixed charge x remains Rs. 40, calculate the cost for a 5 km journey.

Q1316. [E] If the per-km rate y doubles while fixed charge x remains Rs. 45, calculate the cost for a 6 km journey.

Q1317. [E] If the per-km rate y doubles while fixed charge x remains Rs. 50, calculate the cost for a 3 km journey.

Q1318. [E] If the per-km rate y doubles while fixed charge x remains Rs. 55, calculate the cost for a 4 km journey.

Q1319. [E] If the per-km rate y doubles while fixed charge x remains Rs. 60, calculate the cost for a 5 km journey.

Q1320. [E] If the per-km rate y doubles while fixed charge x remains Rs. 65, calculate the cost for a 6 km journey.

Q1321. [E] If the per-km rate y doubles while fixed charge x remains Rs. 20, calculate the cost for a 3 km journey.

Q1322. [E] If the per-km rate y doubles while fixed charge x remains Rs. 25, calculate the cost for a 4 km journey.

Q1323. [E] If the per-km rate y doubles while fixed charge x remains Rs. 30, calculate the cost for a 5 km journey.

Q1324. [E] If the per-km rate y doubles while fixed charge x remains Rs. 35, calculate the cost for a 6 km journey.

Q1325. [E] If the per-km rate y doubles while fixed charge x remains Rs. 40, calculate the cost for a 3 km journey.

Q1326. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1327. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1328. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1329. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1330. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1331. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1332. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1333. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1334. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1335. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1336. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1337. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1338. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1339. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1340. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1341. [F] A hotel charges a fixed room fee plus per-

person dining fee. Write the system of equations for two tour groups.

Q1342. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1343. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1344. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1345. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1346. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1347. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1348. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1349. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1350. [F] A hotel charges a fixed room fee plus per-person dining fee. Write the system of equations for two tour groups.

Q1351. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1352. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1353. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1354. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

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Q1356. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1357. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1358. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1359. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1360. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1361. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1362. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1363. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1364. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

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Q1366. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1367. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1368. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1369. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1370. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1371. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1372. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1373. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1374. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1375. [G] Explain why y (variable rate) must be positive in a realistic fixed-plus-variable pricing model.

Q1376. [H] Find the total charge for a trip of D km given fixed charge $x = 45$ and variable rate $y = 2$.

Q1377. [H] Find the total charge for a trip of D km given fixed charge $x = 50$ and variable rate $y = 3$.

Q1378. [H] Find the total charge for a trip of D km given fixed charge $x = 55$ and variable rate $y = 4$.

Q1379. [H] Find the total charge for a trip of D km given fixed charge $x = 60$ and variable rate $y = 5$.

Q1380. [H] Find the total charge for a trip of D km given fixed charge $x = 65$ and variable rate $y = 6$.

Q1381. [H] Find the total charge for a trip of D km given fixed charge $x = 20$ and variable rate $y = 2$.

Q1382. [H] Find the total charge for a trip of D km given fixed charge $x = 25$ and variable rate $y = 3$.

Q1383. [H] Find the total charge for a trip of D km given fixed charge $x = 30$ and variable rate $y = 4$.

Q1384. [H] Find the total charge for a trip of D km given fixed charge $x = 35$ and variable rate $y = 5$.

Q1385. [H] Find the total charge for a trip of D km given fixed charge $x = 40$ and variable rate $y = 6$.

Q1386. [H] Find the total charge for a trip of D km given fixed charge $x = 45$ and variable rate $y = 2$.

Q1387. [H] Find the total charge for a trip of D km given fixed charge $x = 50$ and variable rate $y = 3$.

Q1388. [H] Find the total charge for a trip of D km given fixed charge $x = 55$ and variable rate $y = 4$.

Q1389. [H] Find the total charge for a trip of D km given fixed charge $x = 60$ and variable rate $y = 5$.

Q1390. [H] Find the total charge for a trip of D km given fixed charge $x = 65$ and variable rate $y = 6$.

Q1391. [H] Find the total charge for a trip of D km given fixed charge $x = 20$ and variable rate $y = 2$.

Q1392. [H] Find the total charge for a trip of D km given fixed charge $x = 25$ and variable rate $y = 3$.

Q1393. [H] Find the total charge for a trip of D km given fixed charge $x = 30$ and variable rate $y = 4$.

Q1394. [H] Find the total charge for a trip of D km given fixed charge $x = 35$ and variable rate $y = 5$.

Q1395. [H] Find the total charge for a trip of D km given fixed charge $x = 40$ and variable rate $y = 6$.

Q1396. [H] Find the total charge for a trip of D km given fixed charge $x = 45$ and variable rate $y = 2$.

Q1397. [H] Find the total charge for a trip of D km given fixed charge $x = 50$ and variable rate $y = 3$.

Q1398. [H] Find the total charge for a trip of D km given fixed charge $x = 55$ and variable rate $y = 4$.

Q1399. [H] Find the total charge for a trip of D km given fixed charge $x = 60$ and variable rate $y = 5$.

Q1400. [H] Find the total charge for a trip of D km given fixed charge $x = 65$ and variable rate $y = 6$.

Pattern 8: Time & Work Reciprocal Rate Modeling

Q1401. [A] 2 men and 5 women can finish an embroidery job in 4 days, while 3 men and 4 women can finish it in 3 days. Find the time taken by 1 man alone and 1 woman alone.

Q1402. [A] 3 men and 6 women can finish an embroidery job in 5 days, while 4 men and 5 women can finish it in 4 days. Find the time taken by 1 man alone and 1 woman alone.

Q1403. [A] 4 men and 7 women can finish an embroidery job in 6 days, while 5 men and 6 women can finish it in 3 days. Find the time taken by 1 man alone and 1 woman alone.

Q1404. [A] 2 men and 8 women can finish an embroidery job in 4 days, while 3 men and 7 women can finish it in 4 days. Find the time taken by 1 man alone and 1 woman alone.

Q1405. [A] 3 men and 5 women can finish an embroidery job in 5 days, while 4 men and 4 women can finish it in 3 days. Find the time taken by 1 man alone and 1 woman alone.

Q1406. [A] 4 men and 6 women can finish an embroidery job in 6 days, while 5 men and 5 women can finish it in 4 days. Find the time taken by 1 man alone and 1 woman alone.

Q1407. [A] 2 men and 7 women can finish an embroidery job in 4 days, while 3 men and 6 women can finish it in 3 days. Find the time taken by 1 man alone and 1 woman alone.

Q1408. [A] 3 men and 8 women can finish an embroidery job in 5 days, while 4 men and 7 women can finish it in 4 days. Find the time taken by 1 man alone and 1 woman alone.

Q1409. [A] 4 men and 5 women can finish an embroidery job in 6 days, while 5 men and 4 women can finish it in 3 days. Find the time taken by 1 man alone and 1 woman alone.

Q1410. [A] 2 men and 6 women can finish an embroidery job in 4 days, while 3 men and 5 women can finish it in 4 days. Find the time taken by 1 man alone and 1 woman alone.

Q1411. [A] 3 men and 7 women can finish an embroidery job in 5 days, while 4 men and 6 women can finish it in 3 days. Find the time taken by 1 man alone and 1 woman alone.

Q1412. [A] 4 men and 8 women can finish an embroidery job in 6 days, while 5 men and 7 women can finish it in 4 days. Find the time taken by 1 man alone and 1 woman alone.

Q1413. [A] 2 men and 5 women can finish an embroidery job in 4 days, while 3 men and 4 women can finish it in 3 days. Find the time taken by 1 man alone and 1 woman alone.

Q1414. [A] 3 men and 6 women can finish an embroidery job in 5 days, while 4 men and 5 women can finish it in 4 days. Find the time taken by 1 man alone and 1 woman alone.

Q1415. [A] 4 men and 7 women can finish an embroidery job in 6 days, while 5 men and 6 women can finish it in 3 days. Find the time taken by 1 man alone and 1 woman alone.

Q1416. [A] 2 men and 8 women can finish an embroidery job in 4 days, while 3 men and 7 women can finish it in 4 days. Find the time taken by 1 man alone and 1 woman alone.

Q1417. [A] 3 men and 5 women can finish an embroidery job in 5 days, while 4 men and 4 women can finish it in 3 days. Find the time taken by 1 man alone and 1 woman alone.

Q1418. [A] 4 men and 6 women can finish an embroidery job in 6 days, while 5 men and 5 women can finish it in 4 days. Find the time taken by 1 man alone and 1 woman alone.

Q1419. [A] 2 men and 7 women can finish an embroidery job in 4 days, while 3 men and 6 women can finish it in 3 days. Find the time taken by 1 man alone and 1 woman

alone.

Q1420. [A] 3 men and 8 women can finish an embroidery job in 5 days, while 4 men and 7 women can finish it in 4 days. Find the time taken by 1 man alone and 1 woman alone.

Q1421. [A] 4 men and 5 women can finish an embroidery job in 6 days, while 5 men and 4 women can finish it in 3 days. Find the time taken by 1 man alone and 1 woman alone.

Q1422. [A] 2 men and 6 women can finish an embroidery job in 4 days, while 3 men and 5 women can finish it in 4 days. Find the time taken by 1 man alone and 1 woman alone.

Q1423. [A] 3 men and 7 women can finish an embroidery job in 5 days, while 4 men and 6 women can finish it in 3 days. Find the time taken by 1 man alone and 1 woman alone.

Q1424. [A] 4 men and 8 women can finish an embroidery job in 6 days, while 5 men and 7 women can finish it in 4 days. Find the time taken by 1 man alone and 1 woman alone.

Q1425. [A] 2 men and 5 women can finish an embroidery job in 4 days, while 3 men and 4 women can finish it in 3 days. Find the time taken by 1 man alone and 1 woman alone.

Q1426. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 3 workers and 6 assistants in 5 days.

Q1427. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 4 workers and 7 assistants in 6 days.

Q1428. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 2 workers and 8 assistants in 4 days.

Q1429. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 3 workers and 5 assistants in 5 days.

Q1430. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 4 workers and 6 assistants in 6 days.

Q1431. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 2 workers and 7 assistants in 4 days.

Q1432. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 3 workers and 8 assistants in 5 days.

Q1433. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 4 workers and 5 assistants in 6 days.

Q1434. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 2 workers and 6 assistants in 4 days.

Q1435. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 3 workers and 7 assistants in 5 days.

Q1436. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 4 workers and 8 assistants in 6 days.

Q1437. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 2 workers and 5 assistants in 4 days.

Q1438. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 3 workers and 6 assistants in 5 days.

Q1439. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 4 workers and 7 assistants in 6 days.

Q1440. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 2 workers and 8 assistants in 4 days.

Q1441. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 3 workers and 5 assistants in 5 days.

Q1442. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 4 workers and 6 assistants in 6 days.

Q1443. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 2 workers and 7 assistants in 4 days.

Q1444. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 3 workers and 8 assistants in 5 days.

Q1445. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 4 workers and 5 assistants in 6 days.

Q1446. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 2 workers and 6 assistants in 4 days.

Q1447. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 3 workers and 7 assistants in 5 days.

Q1448. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 4 workers and 8 assistants in 6 days.

Q1449. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 2 workers and 5 assistants in 4 days.

Q1450. [B] Set up the system of equations using reciprocal variables $u = 1/x$ and $v = 1/y$ for work done by 3 workers and 6 assistants in 5 days.

Q1451. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1452. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1453. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1454. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1455. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1456. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1457. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1458. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1459. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1460. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1461. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1462. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1463. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1464. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1465. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1466. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1467. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1468. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1469. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1470. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1471. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1472. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1473. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1474. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1475. [C] If 1 man takes 12 days and 1 woman takes 36 days to complete a task, find the fraction of work completed by 1 man and 1 woman together in 1 day.

Q1476. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1477. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1478. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1479. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1480. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1481. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1482. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1483. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1484. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1485. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1486. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1487. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1488. [D] If 2 workers complete a task in 10 days, what

is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1489. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1490. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1491. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1492. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1493. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1494. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1495. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1496. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1497. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1498. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1499. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1500. [D] If 2 workers complete a task in 10 days, what is the daily work rate u of 1 worker?
(A) 1/20 (B) 1/10 (C) 1/5 (D) 20

Q1501. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1502. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1503. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1504. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1505. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1506. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1507. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1508. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1509. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1510. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1511. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1512. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1513. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1514. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1515. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1516. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1517. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1518. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1519. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1520. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1521. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1522. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1523. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1524. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1525. [E] Verify if 1 man taking 18 days and 1 woman taking 36 days satisfies $2u + 5v = 1/4$.

Q1526. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1527. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1528. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1529. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1530. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1531. [F] An automated machine and a manual techni-

cian clean a water reservoir. Model their combined daily rate.

Q1532. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1533. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1534. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1535. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1536. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1537. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1538. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1539. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1540. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1541. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1542. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1543. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1544. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1545. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1546. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1547. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1548. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1549. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1550. [F] An automated machine and a manual technician clean a water reservoir. Model their combined daily rate.

Q1551. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1552. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1553. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1554. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1555. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1556. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1557. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1558. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1559. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1560. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1561. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1562. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1563. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1564. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1565. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1566. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1567. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1568. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1569. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1570. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1571. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1572. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1573. [G] Why do we convert time x into work rate $1/x$

when setting up linear equations for combined work?

Q1574. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1575. [G] Why do we convert time x into work rate $1/x$ when setting up linear equations for combined work?

Q1576. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1577. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1578. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1579. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1580. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1581. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1582. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1583. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1584. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1585. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1586. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1587. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1588. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1589. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1590. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1591. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1592. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1593. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1594. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1595. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1596. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1597. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1598. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1599. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Q1600. [H] Calculate total days needed if 3 men and 6 women work together, given 1 man takes 18 days and 1 woman takes 36 days.

Pattern 9: Two-Digit Numbers & Digit Reversal Patterns

Q1601. [A] The sum of the digits of a two-digit number is 3. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1602. [A] The sum of the digits of a two-digit number is 5. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1603. [A] The sum of the digits of a two-digit number is 7. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1604. [A] The sum of the digits of a two-digit number is 9. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1605. [A] The sum of the digits of a two-digit number is 11. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1606. [A] The sum of the digits of a two-digit number is 13. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1607. [A] The sum of the digits of a two-digit number is 9. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1608. [A] The sum of the digits of a two-digit number is 5. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1609. [A] The sum of the digits of a two-digit number is 7. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1610. [A] The sum of the digits of a two-digit number is 9. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1611. [A] The sum of the digits of a two-digit number is 11. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1612. [A] The sum of the digits of a two-digit number is 13. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1613. [A] The sum of the digits of a two-digit number is 8. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1614. [A] The sum of the digits of a two-digit number is 10. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1615. [A] The sum of the digits of a two-digit number is 5. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1616. [A] The sum of the digits of a two-digit number is 7. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1617. [A] The sum of the digits of a two-digit number is 9. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1618. [A] The sum of the digits of a two-digit number is 11. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1619. [A] The sum of the digits of a two-digit number is 7. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1620. [A] The sum of the digits of a two-digit number is 9. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1621. [A] The sum of the digits of a two-digit number is 11. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1622. [A] The sum of the digits of a two-digit number is 6. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1623. [A] The sum of the digits of a two-digit number is 8. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1624. [A] The sum of the digits of a two-digit number is 10. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1625. [A] The sum of the digits of a two-digit number is 6. Nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

Q1626. [B] The sum of a two-digit number and the number formed by reversing its digits is 88. If the digits differ by 4, find the original number.

Q1627. [B] The sum of a two-digit number and the number formed by reversing its digits is 110. If the digits differ by 4, find the original number.

Q1628. [B] The sum of a two-digit number and the number formed by reversing its digits is 132. If the digits differ by 4, find the original number.

Q1629. [B] The sum of a two-digit number and the number formed by reversing its digits is 77. If the digits differ by 3, find the original number.

Q1630. [B] The sum of a two-digit number and the number formed by reversing its digits is 99. If the digits differ by 3, find the original number.

Q1631. [B] The sum of a two-digit number and the number formed by reversing its digits is 55. If the digits differ by 3, find the original number.

Q1632. [B] The sum of a two-digit number and the number formed by reversing its digits is 77. If the digits differ by 3, find the original number.

Q1633. [B] The sum of a two-digit number and the number formed by reversing its digits is 99. If the digits differ by 3, find the original number.

Q1634. [B] The sum of a two-digit number and the number formed by reversing its digits is 121. If the digits differ by 3, find the original number.

Q1635. [B] The sum of a two-digit number and the number formed by reversing its digits is 143. If the digits differ by 3, find the original number.

Q1636. [B] The sum of a two-digit number and the number formed by reversing its digits is 88. If the digits differ by 4, find the original number.

Q1637. [B] The sum of a two-digit number and the number formed by reversing its digits is 44. If the digits differ by 2, find the original number.

Q1638. [B] The sum of a two-digit number and the number formed by reversing its digits is 66. If the digits differ by 2, find the original number.

Q1639. [B] The sum of a two-digit number and the number formed by reversing its digits is 88. If the digits differ by 2, find the original number.

Q1640. [B] The sum of a two-digit number and the number formed by reversing its digits is 110. If the digits differ by 2, find the original number.

Q1641. [B] The sum of a two-digit number and the number formed by reversing its digits is 132. If the digits differ by 2, find the original number.

Q1642. [B] The sum of a two-digit number and the number formed by reversing its digits is 154. If the digits differ by 2, find the original number.

Q1643. [B] The sum of a two-digit number and the number

formed by reversing its digits is 33. If the digits differ by 1, find the original number.

Q1644. [B] The sum of a two-digit number and the number formed by reversing its digits is 55. If the digits differ by 1, find the original number.

Q1645. [B] The sum of a two-digit number and the number formed by reversing its digits is 77. If the digits differ by 1, find the original number.

Q1646. [B] The sum of a two-digit number and the number formed by reversing its digits is 99. If the digits differ by 1, find the original number.

Q1647. [B] The sum of a two-digit number and the number formed by reversing its digits is 121. If the digits differ by 1, find the original number.

Q1648. [B] The sum of a two-digit number and the number formed by reversing its digits is 143. If the digits differ by 1, find the original number.

Q1649. [B] The sum of a two-digit number and the number formed by reversing its digits is 99. If the digits differ by 7, find the original number.

Q1650. [B] The sum of a two-digit number and the number formed by reversing its digits is 55. If the digits differ by 1, find the original number.

Q1651. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1652. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1653. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1654. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1655. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1656. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1657. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1658. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1659. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1660. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1661. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1662. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1663. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1664. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1665. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1666. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1667. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1668. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1669. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1670. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1671. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1672. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1673. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1674. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1675. [C] A two-digit number is 4 times the sum of its digits and 2 times the product of its digits. Find the number.

Q1676. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1677. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1678. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?

(A) 63 (B) 36 (C) 54 (D) 27

Q1679. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1680. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1681. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1682. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1683. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1684. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1685. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1686. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1687. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1688. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1689. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1690. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1691. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1692. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1693. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1694. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1695. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1696. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1697. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1698. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1699. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1700. [D] If the tens digit x is double the units digit y and their sum is 9, what is the number?
(A) 63 (B) 36 (C) 54 (D) 27

Q1701. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1702. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1703. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1704. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1705. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1706. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1707. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1708. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1709. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1710. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1711. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1712. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1713. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1714. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1715. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1716. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1717. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1718. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1719. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1720. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1721. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1722. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1723. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1724. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1725. [E] Prove that the sum of any two-digit number $10x + y$ and its reverse $10y + x$ is always divisible by 11.

Q1726. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1727. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1728. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1729. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1730. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1731. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1732. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1733. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1734. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1735. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1736. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1737. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1738. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1739. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1740. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1741. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1742. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1743. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1744. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1745. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1746. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1747. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1748. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1749. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1750. [F] A security passcode is a two-digit number. Adding 45 to the passcode swaps its digits. Find the passcode if the product of its digits is 14.

Q1751. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1752. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1753. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1754. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

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Q1756. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1757. [G] Express the difference between a two-digit

number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1758. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1759. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1760. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1761. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1762. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1763. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1764. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1765. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1766. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1767. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1768. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1769. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1770. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1771. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1772. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1773. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1774. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1775. [G] Express the difference between a two-digit number $10x + y$ and its reversed form $10y + x$ in simplified algebraic terms.

Q1776. [H] Find the two-digit number where the tens digit exceeds the units digit by 1 and the number is 7 times the sum of its digits.

Q1777. [H] Find the two-digit number where the tens digit exceeds the units digit by 1 and the number is 7 times the sum of its digits.

Q1778. [H] Find the two-digit number where the tens digit exceeds the units digit by 1 and the number is 7 times the sum of its digits.

Q1779. [H] Find the two-digit number where the tens digit exceeds the units digit by 1 and the number is 7 times the sum of its digits.

Q1780. [H] Find the two-digit number where the tens digit exceeds the units digit by 1 and the number is 7 times the sum of its digits.

Q1781. [H] Find the two-digit number where the tens digit exceeds the units digit by 6 and the number is 7 times the sum of its digits.

Q1782. [H] Find the two-digit number where the tens digit exceeds the units digit by 6 and the number is 7 times the sum of its digits.

Q1783. [H] Find the two-digit number where the tens digit exceeds the units digit by 1 and the number is 7 times the sum of its digits.

Q1784. [H] Find the two-digit number where the tens digit exceeds the units digit by 1 and the number is 7 times the sum of its digits.

Q1785. [H] Find the two-digit number where the tens digit exceeds the units digit by 1 and the number is 7 times the sum of its digits.

Q1786. [H] Find the two-digit number where the tens digit exceeds the units digit by 1 and the number is 7 times the sum of its digits.

Q1787. [H] Find the two-digit number where the tens digit exceeds the units digit by 5 and the number is 7 times the sum of its digits.

Q1788. [H] Find the two-digit number where the tens digit exceeds the units digit by 5 and the number is 7 times the sum of its digits.

Q1789. [H] Find the two-digit number where the tens digit exceeds the units digit by 5 and the number is 7 times the sum of its digits.

Q1790. [H] Find the two-digit number where the tens digit exceeds the units digit by 2 and the number is 7 times the sum of its digits.

Q1791. [H] Find the two-digit number where the tens digit exceeds the units digit by 2 and the number is 7 times the sum of its digits.

Q1792. [H] Find the two-digit number where the tens digit exceeds the units digit by 2 and the number is 7 times the

sum of its digits.

Q1793. [H] Find the two-digit number where the tens digit exceeds the units digit by 4 and the number is 7 times the sum of its digits.

Q1794. [H] Find the two-digit number where the tens digit exceeds the units digit by 4 and the number is 7 times the sum of its digits.

Q1795. [H] Find the two-digit number where the tens digit exceeds the units digit by 4 and the number is 7 times the sum of its digits.

Q1796. [H] Find the two-digit number where the tens digit exceeds the units digit by 4 and the number is 7 times the sum of its digits.

Q1797. [H] Find the two-digit number where the tens digit exceeds the units digit by 3 and the number is 7 times the sum of its digits.

Q1798. [H] Find the two-digit number where the tens digit exceeds the units digit by 3 and the number is 7 times the sum of its digits.

Q1799. [H] Find the two-digit number where the tens digit exceeds the units digit by 3 and the number is 7 times the sum of its digits.

Q1800. [H] Find the two-digit number where the tens digit exceeds the units digit by 3 and the number is 7 times the sum of its digits.

Pattern 10: Fraction Modification & Rational Systems

Q1801. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1802. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1803. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1804. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1805. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1806. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1807. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1808. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1809. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1810. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1811. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1812. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1813. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1814. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1815. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1816. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1817. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1818. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1819. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1820. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1821. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1822. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1823. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1824. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1825. [A] A fraction becomes 1/3 when 1 is subtracted from the numerator and 1/4 when 8 is added to its denominator. Find the fraction.

Q1826. [B] A fraction becomes 9/11 if 2 is added to both

numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1827. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1828. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1829. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1830. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1831. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1832. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1833. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1834. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1835. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1836. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1837. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1838. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1839. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1840. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1841. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1842. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1843. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1844. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1845. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1846. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1847. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1848. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1849. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1850. [B] A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.

Q1851. [C] The denominator of a fraction exceeds its numerator by 4. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1852. [C] The denominator of a fraction exceeds its numerator by 4. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1853. [C] The denominator of a fraction exceeds its numerator by 4. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1854. [C] The denominator of a fraction exceeds its numerator by 4. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1855. [C] The denominator of a fraction exceeds its numerator by 2. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1856. [C] The denominator of a fraction exceeds its numerator by 3. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1857. [C] The denominator of a fraction exceeds its numerator by 3. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1858. [C] The denominator of a fraction exceeds its numerator by 3. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1859. [C] The denominator of a fraction exceeds its numerator by 3. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1860. [C] The denominator of a fraction exceeds its numerator by 3. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1861. [C] The denominator of a fraction exceeds its numerator by 2. If 1 is added to the numerator, the fraction

equals 1/2. Find the fraction.

Q1862. [C] The denominator of a fraction exceeds its numerator by 2. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1863. [C] The denominator of a fraction exceeds its numerator by 2. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1864. [C] The denominator of a fraction exceeds its numerator by 2. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1865. [C] The denominator of a fraction exceeds its numerator by 2. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1866. [C] The denominator of a fraction exceeds its numerator by 7. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1867. [C] The denominator of a fraction exceeds its numerator by 1. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1868. [C] The denominator of a fraction exceeds its numerator by 1. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1869. [C] The denominator of a fraction exceeds its numerator by 1. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1870. [C] The denominator of a fraction exceeds its numerator by 1. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1871. [C] The denominator of a fraction exceeds its numerator by 6. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1872. [C] The denominator of a fraction exceeds its numerator by 6. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1873. [C] The denominator of a fraction exceeds its numerator by 2. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1874. [C] The denominator of a fraction exceeds its numerator by 2. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1875. [C] The denominator of a fraction exceeds its numerator by 2. If 1 is added to the numerator, the fraction equals 1/2. Find the fraction.

Q1876. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1877. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1878. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1879. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1880. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1881. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1882. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1883. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1884. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1885. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1886. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1887. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1888. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1889. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1890. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1891. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1892. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1893. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1894. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1895. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1896. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1897. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1898. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1899. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1900. [D] If $x/y = 3/4$, what does the fraction become when 3 is added to numerator and 2 is subtracted from denominator?
(A) 3/5 (B) 5/6 (C) 7/8 (D) 1/2

Q1901. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1902. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1903. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1904. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1905. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1906. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1907. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1908. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1909. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1910. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1911. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1912. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1913. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1914. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1915. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1916. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1917. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1918. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1919. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1920. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1921. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1922. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1923. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y $(x < y)$ increases its value.

Q1924. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction

x/y ($x < y$) increases its value.

Q1925. [E] Show that adding the same positive constant c to both numerator and denominator of a proper fraction x/y ($x < y$) increases its value.

Q1926. [F] An alloy mixture fraction x/y is modified by adding 1 units to copper content. Set up the rational equation.

Q1927. [F] An alloy mixture fraction x/y is modified by adding 2 units to copper content. Set up the rational equation.

Q1928. [F] An alloy mixture fraction x/y is modified by adding 3 units to copper content. Set up the rational equation.

Q1929. [F] An alloy mixture fraction x/y is modified by adding 4 units to copper content. Set up the rational equation.

Q1930. [F] An alloy mixture fraction x/y is modified by adding 5 units to copper content. Set up the rational equation.

Q1931. [F] An alloy mixture fraction x/y is modified by adding 1 units to copper content. Set up the rational equation.

Q1932. [F] An alloy mixture fraction x/y is modified by adding 2 units to copper content. Set up the rational equation.

Q1933. [F] An alloy mixture fraction x/y is modified by adding 3 units to copper content. Set up the rational equation.

Q1934. [F] An alloy mixture fraction x/y is modified by adding 4 units to copper content. Set up the rational equation.

Q1935. [F] An alloy mixture fraction x/y is modified by adding 5 units to copper content. Set up the rational equation.

Q1936. [F] An alloy mixture fraction x/y is modified by adding 1 units to copper content. Set up the rational equation.

Q1937. [F] An alloy mixture fraction x/y is modified by adding 2 units to copper content. Set up the rational equation.

Q1938. [F] An alloy mixture fraction x/y is modified by adding 3 units to copper content. Set up the rational equation.

Q1939. [F] An alloy mixture fraction x/y is modified by adding 4 units to copper content. Set up the rational equation.

Q1940. [F] An alloy mixture fraction x/y is modified by adding 5 units to copper content. Set up the rational equation.

Q1941. [F] An alloy mixture fraction x/y is modified by adding 1 units to copper content. Set up the rational equation.

Q1942. [F] An alloy mixture fraction x/y is modified by adding 2 units to copper content. Set up the rational equation.

Q1943. [F] An alloy mixture fraction x/y is modified by adding 3 units to copper content. Set up the rational equation.

Q1944. [F] An alloy mixture fraction x/y is modified by adding 4 units to copper content. Set up the rational equation.

Q1945. [F] An alloy mixture fraction x/y is modified by adding 5 units to copper content. Set up the rational equation.

Q1946. [F] An alloy mixture fraction x/y is modified by adding 1 units to copper content. Set up the rational equation.

Q1947. [F] An alloy mixture fraction x/y is modified by adding 2 units to copper content. Set up the rational equation.

Q1948. [F] An alloy mixture fraction x/y is modified by adding 3 units to copper content. Set up the rational equation.

Q1949. [F] An alloy mixture fraction x/y is modified by adding 4 units to copper content. Set up the rational equation.

Q1950. [F] An alloy mixture fraction x/y is modified by adding 5 units to copper content. Set up the rational equation.

Q1951. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1952. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1953. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1954. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1955. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1956. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1957. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1958. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1959. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1960. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1961. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1962. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1963. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1964. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1965. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1966. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1967. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1968. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1969. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1970. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1971. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1972. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1973. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1974. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1975. [G] Solve for x and y: $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$.

Q1976. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1977. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1978. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1979. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1980. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1981. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1982. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1983. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1984. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1985. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1986. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1987. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1988. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1989. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1990. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1991. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1992. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1993. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1994. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1995. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1996. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1997. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1998. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q1999. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Q2000. [H] Find the sum of numerator and denominator of a fraction x/y if $x + y = 12$ and $\frac{x+3}{y-2} = 1$.

Pattern 11: Ratio Conditions for System Consistency

Q2001. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2002. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2003. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2004. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2005. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2006. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2007. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2008. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2009. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2010. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2011. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2012. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2013. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2014. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2015. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2016. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2017. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2018. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2019. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2020. [A] For what value of k will the system $x + 2y = 3$

and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2021. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2022. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2023. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2024. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2025. [A] For what value of k will the system $x + 2y = 3$ and $5x + ky + 7 = 0$ represent inconsistent (parallel) lines?

Q2026. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2027. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2028. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2029. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2030. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2031. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2032. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2033. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2034. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2035. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2036. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2037. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2038. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2039. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2040. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2041. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2042. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2043. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2044. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2045. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2046. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2047. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2048. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2049. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2050. [B] Find k for which $2x + 3y = 5$ and $4x + ky = 10$ represent coincident (infinitely many solutions) lines.

Q2051. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2052. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2053. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2054. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2055. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2056. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2057. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2058. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2059. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2060. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2061. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2062. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2063. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2064. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ have infinitely many solutions.

Q2065. [C] Determine the values of a and b for which

$2x - 3y = 7$ and $(a+b)x - (a+b-3)y = 4a + b$ have infinitely many solutions.

Q2066. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a+b)x - (a+b-3)y = 4a + b$ have infinitely many solutions.

Q2067. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a+b)x - (a+b-3)y = 4a + b$ have infinitely many solutions.

Q2068. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a+b)x - (a+b-3)y = 4a + b$ have infinitely many solutions.

Q2069. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a+b)x - (a+b-3)y = 4a + b$ have infinitely many solutions.

Q2070. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a+b)x - (a+b-3)y = 4a + b$ have infinitely many solutions.

Q2071. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a+b)x - (a+b-3)y = 4a + b$ have infinitely many solutions.

Q2072. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a+b)x - (a+b-3)y = 4a + b$ have infinitely many solutions.

Q2073. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a+b)x - (a+b-3)y = 4a + b$ have infinitely many solutions.

Q2074. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a+b)x - (a+b-3)y = 4a + b$ have infinitely many solutions.

Q2075. [C] Determine the values of a and b for which $2x - 3y = 7$ and $(a+b)x - (a+b-3)y = 4a + b$ have infinitely many solutions.

Q2076. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2077. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2078. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2079. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2080. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2081. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2082. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2083. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2084. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2085. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2086. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2087. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2088. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2089. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2090. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2091. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2092. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2093. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2094. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?

(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2095. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2096. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2097. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2098. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2099. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2100. [D] For what value of p does the system $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a UNIQUE solution?
(A) $p \neq 4$ (B) $p = 4$ (C) $p = 2$ (D) $p \neq 2$

Q2101. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2102. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2103. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2104. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2105. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2106. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2107. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2108. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2109. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2110. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2111. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2112. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2113. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2114. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2115. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2116. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2117. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2118. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2119. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2120. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2121. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2122. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2123. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2124. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2125. [E] Verify whether $k = 3$ gives a unique solution for $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$.

Q2126. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2127. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2128. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2129. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2130. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2131. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2132. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2133. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2134. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2135. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2136. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2137. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2138. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2139. [F] Two supply lines are represented by $x + 2y = 5$ and $3x + 6y = c$. Find c for coincident operation.

Q2140. [F] Two supply lines are represented by $x+2y=5$ and $3x+6y=c$. Find c for coincident operation.

Q2141. [F] Two supply lines are represented by $x+2y=5$ and $3x+6y=c$. Find c for coincident operation.

Q2142. [F] Two supply lines are represented by $x+2y=5$ and $3x+6y=c$. Find c for coincident operation.

Q2143. [F] Two supply lines are represented by $x+2y=5$ and $3x+6y=c$. Find c for coincident operation.

Q2144. [F] Two supply lines are represented by $x+2y=5$ and $3x+6y=c$. Find c for coincident operation.

Q2145. [F] Two supply lines are represented by $x+2y=5$ and $3x+6y=c$. Find c for coincident operation.

Q2146. [F] Two supply lines are represented by $x+2y=5$ and $3x+6y=c$. Find c for coincident operation.

Q2147. [F] Two supply lines are represented by $x+2y=5$ and $3x+6y=c$. Find c for coincident operation.

Q2148. [F] Two supply lines are represented by $x+2y=5$ and $3x+6y=c$. Find c for coincident operation.

Q2149. [F] Two supply lines are represented by $x+2y=5$ and $3x+6y=c$. Find c for coincident operation.

Q2150. [F] Two supply lines are represented by $x+2y=5$ and $3x+6y=c$. Find c for coincident operation.

Q2151. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2152. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2153. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2154. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2155. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2156. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2157. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2158. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2159. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2160. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2161. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2162. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2163. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2164. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2165. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2166. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2167. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2168. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2169. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2170. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2171. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2172. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2173. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2174. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2175. [G] State the geometric interpretation of a system of equations with $a_1/a_2=b_1/b_2\neq c_1/c_2$.

Q2176. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2177. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2178. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2179. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2180. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2181. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2182. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2183. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2184. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2185. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2186. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2187. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2188. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2189. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2190. [H] Find m such that $(2m-1)x+(m-1)y=2m+1$ and $3x+y=1$ have no solution.

Q2191. [H] Find m such that $(2m-1)x+(m-1)y = 2m+1$ and $3x+y = 1$ have no solution.

Q2192. [H] Find m such that $(2m-1)x+(m-1)y = 2m+1$ and $3x+y = 1$ have no solution.

Q2193. [H] Find m such that $(2m-1)x+(m-1)y = 2m+1$ and $3x+y = 1$ have no solution.

Q2194. [H] Find m such that $(2m-1)x+(m-1)y = 2m+1$ and $3x+y = 1$ have no solution.

Q2195. [H] Find m such that $(2m-1)x+(m-1)y = 2m+1$ and $3x+y = 1$ have no solution.

Q2196. [H] Find m such that $(2m-1)x+(m-1)y = 2m+1$ and $3x+y = 1$ have no solution.

Q2197. [H] Find m such that $(2m-1)x+(m-1)y = 2m+1$ and $3x+y = 1$ have no solution.

Q2198. [H] Find m such that $(2m-1)x+(m-1)y = 2m+1$ and $3x+y = 1$ have no solution.

Q2199. [H] Find m such that $(2m-1)x+(m-1)y = 2m+1$ and $3x+y = 1$ have no solution.

Q2200. [H] Find m such that $(2m-1)x+(m-1)y = 2m+1$ and $3x+y = 1$ have no solution.

Pattern 12: Upstream / Downstream Rate Modeling

Q2201. [A] A motorboat whose speed in still water is 15 km/h takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2202. [A] A motorboat whose speed in still water is 16 km/h takes 1 hour more to go 30 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2203. [A] A motorboat whose speed in still water is 17 km/h takes 1 hour more to go 36 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2204. [A] A motorboat whose speed in still water is 18 km/h takes 1 hour more to go 42 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2205. [A] A motorboat whose speed in still water is 19 km/h takes 1 hour more to go 48 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2206. [A] A motorboat whose speed in still water is 20 km/h takes 1 hour more to go 54 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2207. [A] A motorboat whose speed in still water is 21 km/h takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2208. [A] A motorboat whose speed in still water is 22 km/h takes 1 hour more to go 30 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2209. [A] A motorboat whose speed in still water is 23 km/h takes 1 hour more to go 36 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2210. [A] A motorboat whose speed in still water is 24 km/h takes 1 hour more to go 42 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2211. [A] A motorboat whose speed in still water is 15 km/h takes 1 hour more to go 48 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2212. [A] A motorboat whose speed in still water is 16 km/h takes 1 hour more to go 54 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2213. [A] A motorboat whose speed in still water is 17 km/h takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2214. [A] A motorboat whose speed in still water is 18 km/h takes 1 hour more to go 30 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2215. [A] A motorboat whose speed in still water is 19 km/h takes 1 hour more to go 36 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2216. [A] A motorboat whose speed in still water is 20 km/h takes 1 hour more to go 42 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2217. [A] A motorboat whose speed in still water is 21 km/h takes 1 hour more to go 48 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2218. [A] A motorboat whose speed in still water is 22 km/h takes 1 hour more to go 54 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2219. [A] A motorboat whose speed in still water is 23 km/h takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2220. [A] A motorboat whose speed in still water is 24 km/h takes 1 hour more to go 30 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2221. [A] A motorboat whose speed in still water is 15 km/h takes 1 hour more to go 36 km upstream than to

return downstream to the same spot. Find the speed of the stream y.

Q2222. [A] A motorboat whose speed in still water is 16 km/h takes 1 hour more to go 42 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2223. [A] A motorboat whose speed in still water is 17 km/h takes 1 hour more to go 48 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2224. [A] A motorboat whose speed in still water is 18 km/h takes 1 hour more to go 54 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2225. [A] A motorboat whose speed in still water is 19 km/h takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream y.

Q2226. [B] A crew rows 30 km downstream and 15.0 km upstream in 2.2 hours. Form the quadratic rate equation.

Q2227. [B] A crew rows 36 km downstream and 18.0 km upstream in 2.5 hours. Form the quadratic rate equation.

Q2228. [B] A crew rows 42 km downstream and 21.0 km upstream in 2.8 hours. Form the quadratic rate equation.

Q2229. [B] A crew rows 48 km downstream and 24.0 km upstream in 3.0 hours. Form the quadratic rate equation.

Q2230. [B] A crew rows 54 km downstream and 27.0 km upstream in 3.3 hours. Form the quadratic rate equation.

Q2231. [B] A crew rows 24 km downstream and 12.0 km upstream in 2.3 hours. Form the quadratic rate equation.

Q2232. [B] A crew rows 30 km downstream and 15.0 km upstream in 2.7 hours. Form the quadratic rate equation.

Q2233. [B] A crew rows 36 km downstream and 18.0 km upstream in 3.1 hours. Form the quadratic rate equation.

Q2234. [B] A crew rows 42 km downstream and 21.0 km upstream in 3.4 hours. Form the quadratic rate equation.

Q2235. [B] A crew rows 48 km downstream and 24.0 km upstream in 3.8 hours. Form the quadratic rate equation.

Q2236. [B] A crew rows 54 km downstream and 27.0 km upstream in 4.0 hours. Form the quadratic rate equation.

Q2237. [B] A crew rows 24 km downstream and 12.0 km upstream in 1.7 hours. Form the quadratic rate equation.

Q2238. [B] A crew rows 30 km downstream and 15.0 km upstream in 2.0 hours. Form the quadratic rate equation.

Q2239. [B] A crew rows 36 km downstream and 18.0 km upstream in 2.3 hours. Form the quadratic rate equation.

Q2240. [B] A crew rows 42 km downstream and 21.0 km upstream in 2.6 hours. Form the quadratic rate equation.

Q2241. [B] A crew rows 48 km downstream and 24.0 km upstream in 4.7 hours. Form the quadratic rate equation.

Q2242. [B] A crew rows 54 km downstream and 27.0 km upstream in 4.9 hours. Form the quadratic rate equation.

Q2243. [B] A crew rows 24 km downstream and 12.0 km upstream in 2.1 hours. Form the quadratic rate equation.

Q2244. [B] A crew rows 30 km downstream and 15.0 km upstream in 2.5 hours. Form the quadratic rate equation.

Q2245. [B] A crew rows 36 km downstream and 18.0 km upstream in 2.8 hours. Form the quadratic rate equation.

Q2246. [B] A crew rows 42 km downstream and 21.0 km upstream in 3.1 hours. Form the quadratic rate equation.

Q2247. [B] A crew rows 48 km downstream and 24.0 km upstream in 3.3 hours. Form the quadratic rate equation.

Q2248. [B] A crew rows 54 km downstream and 27.0 km upstream in 3.6 hours. Form the quadratic rate equation.

Q2249. [B] A crew rows 24 km downstream and 12.0 km upstream in 1.5 hours. Form the quadratic rate equation.

Q2250. [B] A crew rows 30 km downstream and 15.0 km upstream in 1.8 hours. Form the quadratic rate equation.

Q2251. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2252. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2253. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2254. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2255. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2256. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2257. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2258. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2259. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2260. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2261. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2262. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2263. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2264. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2265. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2266. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2267. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2268. [C] If boat speed is x and stream speed is y, write

the expressions for upstream speed and downstream speed.

Q2269. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2270. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2271. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2272. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2273. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2274. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2275. [C] If boat speed is x and stream speed is y, write the expressions for upstream speed and downstream speed.

Q2276. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2277. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2278. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2279. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2280. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2281. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2282. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2283. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2284. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2285. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2286. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2287. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2288. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2289. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2290. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2291. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2292. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2293. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2294. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2295. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2296. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2297. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2298. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2299. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2300. [D] A boat travels downstream at 20 km/h and upstream at 12 km/h. What is the speed of the boat in still water?
(A) 16 km/h (B) 14 km/h (C) 18 km/h (D) 4 km/h

Q2301. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2302. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2303. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2304. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2305. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2306. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2307. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2308. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2309. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2310. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2311. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2312. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2313. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2314. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2315. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2316. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2317. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2318. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2319. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2320. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2321. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2322. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2323. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2324. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2325. [E] Why must the boat speed x always be strictly greater than the stream speed y ($x > y$)?

Q2326. [F] A river ferry route covers 54 km. Calculate the round-trip time if $x = 20$ km/h and $y = 2$ km/h.

Q2327. [F] A river ferry route covers 24 km. Calculate the round-trip time if $x = 21$ km/h and $y = 3$ km/h.

Q2328. [F] A river ferry route covers 30 km. Calculate the round-trip time if $x = 22$ km/h and $y = 4$ km/h.

Q2329. [F] A river ferry route covers 36 km. Calculate the round-trip time if $x = 23$ km/h and $y = 5$ km/h.

Q2330. [F] A river ferry route covers 42 km. Calculate the round-trip time if $x = 24$ km/h and $y = 6$ km/h.

Q2331. [F] A river ferry route covers 48 km. Calculate the round-trip time if $x = 15$ km/h and $y = 2$ km/h.

Q2332. [F] A river ferry route covers 54 km. Calculate the round-trip time if $x = 16$ km/h and $y = 3$ km/h.

Q2333. [F] A river ferry route covers 24 km. Calculate the round-trip time if $x = 17$ km/h and $y = 4$ km/h.

Q2334. [F] A river ferry route covers 30 km. Calculate the round-trip time if $x = 18$ km/h and $y = 5$ km/h.

Q2335. [F] A river ferry route covers 36 km. Calculate the round-trip time if $x = 19$ km/h and $y = 6$ km/h.

Q2336. [F] A river ferry route covers 42 km. Calculate the round-trip time if $x = 20$ km/h and $y = 2$ km/h.

Q2337. [F] A river ferry route covers 48 km. Calculate the round-trip time if $x = 21$ km/h and $y = 3$ km/h.

Q2338. [F] A river ferry route covers 54 km. Calculate the round-trip time if $x = 22$ km/h and $y = 4$ km/h.

Q2339. [F] A river ferry route covers 24 km. Calculate the round-trip time if $x = 23$ km/h and $y = 5$ km/h.

Q2340. [F] A river ferry route covers 30 km. Calculate the round-trip time if $x = 24$ km/h and $y = 6$ km/h.

Q2341. [F] A river ferry route covers 36 km. Calculate the round-trip time if $x = 15$ km/h and $y = 2$ km/h.

Q2342. [F] A river ferry route covers 42 km. Calculate the round-trip time if $x = 16$ km/h and $y = 3$ km/h.

Q2343. [F] A river ferry route covers 48 km. Calculate the round-trip time if $x = 17$ km/h and $y = 4$ km/h.

Q2344. [F] A river ferry route covers 54 km. Calculate the round-trip time if $x = 18$ km/h and $y = 5$ km/h.

Q2345. [F] A river ferry route covers 24 km. Calculate the round-trip time if $x = 19$ km/h and $y = 6$ km/h.

Q2346. [F] A river ferry route covers 30 km. Calculate the round-trip time if $x = 20$ km/h and $y = 2$ km/h.

Q2347. [F] A river ferry route covers 36 km. Calculate the round-trip time if $x = 21$ km/h and $y = 3$ km/h.

Q2348. [F] A river ferry route covers 42 km. Calculate the

round-trip time if $x = 22$ km/h and $y = 4$ km/h.

Q2349. [F] A river ferry route covers 48 km. Calculate the round-trip time if $x = 23$ km/h and $y = 5$ km/h.

Q2350. [F] A river ferry route covers 54 km. Calculate the round-trip time if $x = 24$ km/h and $y = 6$ km/h.

Q2351. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2352. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2353. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2354. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2355. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2356. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2357. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2358. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2359. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2360. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2361. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2362. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2363. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2364. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2365. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2366. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2367. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2368. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2369. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2370. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2371. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2372. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2373. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2374. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2375. [G] Express the stream speed y in terms of downstream speed D and upstream speed U.

Q2376. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2377. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2378. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2379. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2380. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2381. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2382. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2383. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2384. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2385. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2386. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2387. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2388. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2389. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2390. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2391. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2392. [H] A river boat rows 30 km downstream in 2

hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2393. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2394. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2395. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2396. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2397. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2398. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2399. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Q2400. [H] A river boat rows 30 km downstream in 2 hours and 18 km upstream in 3 hours. Calculate boat speed and stream speed.

Pattern 13: Equal Roots & Discriminant Proof Identities

Q2401. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2402. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2403. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2404. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2405. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2406. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2407. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2408. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2409. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2410. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2411. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2412. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2413. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2414. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2415. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2416. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2417. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2418. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2419. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2420. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2421. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2422. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2423. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2424. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2425. [A] If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q2426. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2427. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2428. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2429. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2430. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2431. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2432. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2433. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2434. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2435. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2436. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2437. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2438. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2439. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2440. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2441. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2442. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2443. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2444. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2445. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2446. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2447. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2448. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2449. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2450. [B] Prove that $x^2(a^2+b^2)+2x(ac+bd)+(c^2+d^2) = 0$ has no real roots if $ad \neq bc$.

Q2451. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2452. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2453. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2454. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2455. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2456. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2457. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2458. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2459. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2460. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2461. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2462. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2463. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2464. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2465. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2466. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2467. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2468. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2469. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2470. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2471. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2472. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2473. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2474. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2475. [C] Find the value of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots.

Q2476. [D] What is the discriminant D of $ax^2 + bx + c = 0$ when roots are real and equal?

(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2477. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2478. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2479. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2480. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2481. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2482. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2483. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2484. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2485. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2486. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2487. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2488. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2489. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2490. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2491. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2492. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2493. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2494. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2495. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2496. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2497. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2498. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2499. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2500. [D] What is the discriminant D of $ax^2+bx+c=0$ when roots are real and equal?
(A) $D=0$ (B) $D>0$ (C) $D<0$ (D) $D=1$

Q2501. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2502. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2503. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2504. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2505. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2506. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2507. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2508. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2509. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2510. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2511. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2512. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2513. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2514. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2515. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2516. [E] If roots of $(p+1)x^2-6(p+1)x+3(p+9)=0$ $(p\neq -1)$ are equal, find p.

Q2517. [E] If roots of $(p+1)x^2 - 6(p+1)x + 3(p+9) = 0$ $(p \neq -1)$ are equal, find p.

Q2518. [E] If roots of $(p+1)x^2 - 6(p+1)x + 3(p+9) = 0$ $(p \neq -1)$ are equal, find p.

Q2519. [E] If roots of $(p+1)x^2 - 6(p+1)x + 3(p+9) = 0$ $(p \neq -1)$ are equal, find p.

Q2520. [E] If roots of $(p+1)x^2 - 6(p+1)x + 3(p+9) = 0$ $(p \neq -1)$ are equal, find p.

Q2521. [E] If roots of $(p+1)x^2 - 6(p+1)x + 3(p+9) = 0$ $(p \neq -1)$ are equal, find p.

Q2522. [E] If roots of $(p+1)x^2 - 6(p+1)x + 3(p+9) = 0$ $(p \neq -1)$ are equal, find p.

Q2523. [E] If roots of $(p+1)x^2 - 6(p+1)x + 3(p+9) = 0$ $(p \neq -1)$ are equal, find p.

Q2524. [E] If roots of $(p+1)x^2 - 6(p+1)x + 3(p+9) = 0$ $(p \neq -1)$ are equal, find p.

Q2525. [E] If roots of $(p+1)x^2 - 6(p+1)x + 3(p+9) = 0$ $(p \neq -1)$ are equal, find p.

Q2526. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2527. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2528. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2529. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2530. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2531. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2532. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2533. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

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Q2536. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2537. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2538. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2539. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2540. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2541. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2542. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2543. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2544. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2545. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2546. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2547. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2548. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2549. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2550. [F] A stress load formula gives quadratic equation $x^2 + 2kx + k^2 = 0$. Show that it always has equal roots for any real k.

Q2551. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2552. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2553. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

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Q2556. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2557. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2558. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2559. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2560. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2561. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2562. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2563. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2564. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2565. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2566. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2567. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2568. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2569. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2570. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2571. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2572. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2573. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2574. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2575. [G] Show that the roots of $x^2 - 2px + (p^2 - q^2) = 0$ are rational if p, q are rational.

Q2576. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2577. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2578. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2579. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2580. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2581. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2582. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2583. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2584. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2585. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2586. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2587. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2588. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2589. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2590. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2591. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2592. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2593. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2594. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2595. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2596. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2597. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2598. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2599. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Q2600. [H] Find k if $2x^2 + kx + 3 = 0$ has two equal real roots.

Pattern 14: Age & Multiplier Ratio Word Problems

Q2601. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2602. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2603. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2604. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2605. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2606. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2607. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2608. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2609. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2610. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2611. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2612. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2613. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2614. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2615. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2616. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2617. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2618. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2619. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2620. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2621. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2622. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2623. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2624. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2625. [A] Five years hence, Jacob's age will be 3 times that of his son. Five years ago, Jacob's age was 7 times that of his son. Find their present ages.

Q2626. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2627. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2628. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2629. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2630. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2631. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2632. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2633. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2634. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2635. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2636. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2637. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2638. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2639. [B] Ten years ago, a father was 12 times as old as

his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2640. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2641. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2642. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2643. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2644. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2645. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2646. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2647. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2648. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2649. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2650. [B] Ten years ago, a father was 12 times as old as his son. Ten years hence, he will be twice as old as his son. Find their present ages.

Q2651. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2652. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2653. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2654. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2655. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2656. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2657. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2658. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2659. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2660. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2661. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2662. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2663. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2664. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2665. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2666. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2667. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2668. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2669. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2670. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2671. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2672. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2673. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2674. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2675. [C] The present age of a father is 3 years more than 3 times the age of his son. Three years hence, father's age will be 10 years more than twice son's age. Find present ages.

Q2676. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2677. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2678. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2679. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2680. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2681. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2682. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2683. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2684. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2685. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2686. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2687. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2688. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2689. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2690. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2691. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2692. [D] If Aftab is 42 years old and his daughter is 12

years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2693. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2694. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2695. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2696. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2697. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2698. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2699. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2700. [D] If Aftab is 42 years old and his daughter is 12 years old, how many years ago was Aftab 7 times as old as his daughter?
(A) 7 years ago (B) 5 years ago (C) 10 years ago (D) 3 years ago

Q2701. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2702. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2703. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2704. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2705. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2706. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2707. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2708. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2709. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2710. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2711. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2712. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2713. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2714. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2715. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2716. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2717. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2718. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2719. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2720. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2721. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2722. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2723. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2724. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2725. [E] A man's age is equal to the square of his son's age y. In 8 years, his age will be 4 more than 3 times son's age. Find their present ages.

Q2726. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2727. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2728. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2729. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2730. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2731. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2732. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2733. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2734. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2735. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2736. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2737. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2738. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2739. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2740. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2741. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2742. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2743. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2744. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2745. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2746. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2747. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2748. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2749. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2750. [F] A mother is 4 times as old as her daughter. In 16 years, she will be twice as old as her daughter. Find their present ages.

Q2751. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2752. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2753. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2754. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2755. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2756. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2757. [G] Form a system of equations for two friends

whose ages differ by 2 years and whose father's age is twice the older friend.

Q2758. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2759. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2760. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2761. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2762. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2763. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2764. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2765. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2766. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2767. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2768. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2769. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2770. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2771. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2772. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2773. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2774. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2775. [G] Form a system of equations for two friends whose ages differ by 2 years and whose father's age is twice the older friend.

Q2776. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2777. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2778. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2779. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2780. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2781. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2782. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2783. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2784. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2785. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2786. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2787. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2788. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2789. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2790. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2791. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2792. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their

ages was 124. Find their present ages.

Q2793. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2794. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2795. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2796. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2797. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2798. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2799. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

Q2800. [H] The sum of present ages of a mother and daughter is 45 years. Five years ago, the product of their ages was 124. Find their present ages.

PART 2: DEDICATED ANSWER KEY (Q1 – Q2800)

Pattern 1: Middle-Term Splitting with Radical Coefficients — Answer Key

Q1. Zeroes are $x = -\frac{1\sqrt{2}}{2}$ and $x = -\frac{1\sqrt{2}}{2}$.
Q2. Zeroes are $x = -\frac{2\sqrt{3}}{3}$ and $x = -\frac{2\sqrt{3}}{6}$.
Q3. Zeroes are $x = -\frac{3\sqrt{5}}{5}$ and $x = -\frac{3\sqrt{5}}{15}$.
Q4. Zeroes are $x = -\frac{1\sqrt{7}}{7}$ and $x = -\frac{4\sqrt{7}}{28}$.
Q5. Zeroes are $x = -1\sqrt{2}$ and $x = -\frac{5\sqrt{2}}{10}$.
Q6. Zeroes are $x = -1\sqrt{3}$ and $x = -\frac{6\sqrt{3}}{3}$.
Q7. Zeroes are $x = -\frac{1\sqrt{5}}{5}$ and $x = -\frac{7\sqrt{5}}{10}$.
Q8. Zeroes are $x = -\frac{2\sqrt{7}}{7}$ and $x = -\frac{1\sqrt{7}}{21}$.
Q9. Zeroes are $x = -\frac{3\sqrt{2}}{2}$ and $x = -\frac{2\sqrt{2}}{8}$.
Q10. Zeroes are $x = -\frac{1\sqrt{3}}{3}$ and $x = -\frac{3\sqrt{3}}{15}$.
Q11. Zeroes are $x = -\frac{2\sqrt{5}}{5}$ and $x = -\frac{4\sqrt{5}}{5}$.
Q12. Zeroes are $x = -\frac{3\sqrt{7}}{7}$ and $x = -\frac{5\sqrt{7}}{14}$.
Q13. Zeroes are $x = -\frac{1\sqrt{2}}{2}$ and $x = -\frac{6\sqrt{2}}{6}$.
Q14. Zeroes are $x = -\frac{2\sqrt{3}}{3}$ and $x = -\frac{7\sqrt{3}}{12}$.
Q15. Zeroes are $x = -\frac{3\sqrt{5}}{5}$ and $x = -\frac{1\sqrt{5}}{25}$.
Q16. Zeroes are $x = -\frac{1\sqrt{7}}{7}$ and $x = -\frac{2\sqrt{7}}{7}$.
Q17. Zeroes are $x = -1\sqrt{2}$ and $x = -\frac{3\sqrt{2}}{4}$.
Q18. Zeroes are $x = -1\sqrt{3}$ and $x = -\frac{4\sqrt{3}}{9}$.
Q19. Zeroes are $x = -\frac{1\sqrt{5}}{5}$ and $x = -\frac{5\sqrt{5}}{20}$.
Q20. Zeroes are $x = -\frac{2\sqrt{7}}{7}$ and $x = -\frac{6\sqrt{7}}{35}$.
Q21. Zeroes are $x = -\frac{3\sqrt{2}}{2}$ and $x = -\frac{7\sqrt{2}}{2}$.
Q22. Zeroes are $x = -\frac{1\sqrt{3}}{3}$ and $x = -\frac{1\sqrt{3}}{6}$.
Q23. Zeroes are $x = -\frac{2\sqrt{5}}{5}$ and $x = -\frac{2\sqrt{5}}{15}$.
Q24. Zeroes are $x = -\frac{3\sqrt{7}}{7}$ and $x = -\frac{3\sqrt{7}}{28}$.
Q25. Zeroes are $x = -\frac{1\sqrt{2}}{2}$ and $x = -\frac{4\sqrt{2}}{10}$.
Q26. $p(x) = \sqrt{3}x^2 + 11x + 5\sqrt{3}$.
Q27. $p(x) = 2\sqrt{5}x^2 + 36x + 6\sqrt{5}$.
Q28. $p(x) = 3\sqrt{7}x^2 + 28x + 7\sqrt{7}$.
Q29. $p(x) = 4\sqrt{2}x^2 + 17x + \sqrt{2}$.
Q30. $p(x) = 5\sqrt{3}x^2 + 47x + 2\sqrt{3}$.
Q31. $p(x) = \sqrt{5}x^2 + 8x + 3\sqrt{5}$.
Q32. $p(x) = 2\sqrt{7}x^2 + 32x + 4\sqrt{7}$.
Q33. $p(x) = 3\sqrt{2}x^2 + 23x + 5\sqrt{2}$.
Q34. $p(x) = 4\sqrt{3}x^2 + 18x + 6\sqrt{3}$.
Q35. $p(x) = 5\sqrt{5}x^2 + 57x + 7\sqrt{5}$.
Q36. $p(x) = \sqrt{7}x^2 + 22x + \sqrt{7}$.
Q37. $p(x) = 2\sqrt{2}x^2 + 6x + 2\sqrt{2}$.
Q38. $p(x) = 3\sqrt{3}x^2 + 21x + 3\sqrt{3}$.
Q39. $p(x) = 4\sqrt{5}x^2 + 64x + 4\sqrt{5}$.

Q40. $p(x) = 5\sqrt{7}x^2 + 40x + 5\sqrt{7}$.
Q41. $p(x) = \sqrt{2}x^2 + 10x + 6\sqrt{2}$.
Q42. $p(x) = 2\sqrt{3}x^2 + 25x + 7\sqrt{3}$.
Q43. $p(x) = 3\sqrt{5}x^2 + 16x + \sqrt{5}$.
Q44. $p(x) = 4\sqrt{7}x^2 + 58x + 2\sqrt{7}$.
Q45. $p(x) = 5\sqrt{2}x^2 + 33x + 3\sqrt{2}$.
Q46. $p(x) = \sqrt{3}x^2 + 7x + 4\sqrt{3}$.
Q47. $p(x) = 2\sqrt{5}x^2 + 25x + 5\sqrt{5}$.
Q48. $p(x) = 3\sqrt{7}x^2 + 69x + 6\sqrt{7}$.
Q49. $p(x) = 4\sqrt{2}x^2 + 15x + 7\sqrt{2}$.
Q50. $p(x) = 5\sqrt{3}x^2 + 31x + \sqrt{3}$.
Q51. $k = 17$, second zero is $x = -\frac{2\sqrt{5}}{5}$.
Q52. $k = 17$, second zero is $x = -\frac{3\sqrt{7}}{14}$.
Q53. $k = 16$, second zero is $x = -\frac{4\sqrt{2}}{6}$.
Q54. $k = 41$, second zero is $x = -\frac{5\sqrt{3}}{12}$.
Q55. $k = 31$, second zero is $x = -\frac{6\sqrt{5}}{25}$.
Q56. $k = 21$, second zero is $x = -\frac{7\sqrt{7}}{7}$.
Q57. $k = 13$, second zero is $x = -\frac{1\sqrt{2}}{4}$.
Q58. $k = 11$, second zero is $x = -\frac{2\sqrt{3}}{9}$.
Q59. $k = 43$, second zero is $x = -\frac{3\sqrt{5}}{20}$.
Q60. $k = 109$, second zero is $x = -\frac{4\sqrt{7}}{35}$.
Q61. $k = 7$, second zero is $x = -\frac{5\sqrt{2}}{2}$.
Q62. $k = 18$, second zero is $x = -\frac{6\sqrt{3}}{6}$.
Q63. $k = 52$, second zero is $x = -\frac{7\sqrt{5}}{15}$.
Q64. $k = 29$, second zero is $x = -\frac{1\sqrt{7}}{28}$.
Q65. $k = 22$, second zero is $x = -\frac{2\sqrt{2}}{10}$.
Q66. $k = 12$, second zero is $x = -\frac{3\sqrt{3}}{3}$.
Q67. $k = 14$, second zero is $x = -\frac{4\sqrt{5}}{10}$.
Q68. $k = 47$, second zero is $x = -\frac{5\sqrt{7}}{21}$.
Q69. $k = 30$, second zero is $x = -\frac{6\sqrt{2}}{8}$.
Q70. $k = 22$, second zero is $x = -\frac{7\sqrt{3}}{15}$.
Q71. $k = 11$, second zero is $x = -\frac{1\sqrt{5}}{5}$.
Q72. $k = 44$, second zero is $x = -\frac{2\sqrt{7}}{14}$.
Q73. $k = 9$, second zero is $x = -\frac{3\sqrt{2}}{6}$.
Q74. $k = 28$, second zero is $x = -\frac{4\sqrt{3}}{12}$.
Q75. $k = 80$, second zero is $x = -\frac{5\sqrt{5}}{25}$.
Q76. Correct Option: (A) $x = -\frac{1\sqrt{7}}{7}, -\frac{6\sqrt{7}}{7}$.
Q77. Correct Option: (A) $x = -1\sqrt{2}, -\frac{7\sqrt{2}}{4}$.
Q78. Correct Option: (A) $x = -1\sqrt{3}, -\frac{1\sqrt{3}}{9}$.
Q79. Correct Option: (A) $x = -\frac{1\sqrt{5}}{5}, -\frac{2\sqrt{5}}{20}$.
Q80. Correct Option: (A) $x = -\frac{2\sqrt{7}}{7}, -\frac{3\sqrt{7}}{35}$.
Q81. Correct Option: (A) $x = -\frac{3\sqrt{2}}{2}, -\frac{4\sqrt{2}}{2}$.
Q82. Correct Option: (A) $x = -\frac{1\sqrt{3}}{3}, -\frac{5\sqrt{3}}{6}$.
Q83. Correct Option: (A) $x = -\frac{2\sqrt{5}}{5}, -\frac{6\sqrt{5}}{15}$.
Q84. Correct Option: (A) $x = -\frac{3\sqrt{7}}{7}, -\frac{7\sqrt{7}}{28}$.
Q85. Correct Option: (A) $x = -\frac{1\sqrt{2}}{2}, -\frac{1\sqrt{2}}{10}$.
Q86. Correct Option: (A) $x = -\frac{2\sqrt{3}}{3}, -\frac{2\sqrt{3}}{3}$.
Q87. Correct Option: (A) $x = -\frac{3\sqrt{5}}{5}, -\frac{3\sqrt{5}}{10}$.
Q88. Correct Option: (A) $x = -\frac{1\sqrt{7}}{7}, -\frac{4\sqrt{7}}{21}$.
Q89. Correct Option: (A) $x = -1\sqrt{2}, -\frac{5\sqrt{2}}{8}$.
Q90. Correct Option: (A) $x = -1\sqrt{3}, -\frac{6\sqrt{3}}{15}$.
Q91. Correct Option: (A) $x = -\frac{1\sqrt{5}}{5}, -\frac{7\sqrt{5}}{5}$.
Q92. Correct Option: (A) $x = -\frac{2\sqrt{7}}{7}, -\frac{1\sqrt{7}}{14}$.
Q93. Correct Option: (A) $x = -\frac{3\sqrt{2}}{2}, -\frac{2\sqrt{2}}{6}$.
Q94. Correct Option: (A) $x = -\frac{1\sqrt{3}}{3}, -\frac{3\sqrt{3}}{12}$.
Q95. Correct Option: (A) $x = -\frac{2\sqrt{5}}{5}, -\frac{4\sqrt{5}}{25}$.
Q96. Correct Option: (A) $x = -\frac{3\sqrt{7}}{7}, -\frac{5\sqrt{7}}{7}$.
Q97. Correct Option: (A) $x = -\frac{1\sqrt{2}}{2}, -\frac{6\sqrt{2}}{4}$.
Q98. Correct Option: (A) $x = -\frac{2\sqrt{3}}{3}, -\frac{7\sqrt{3}}{9}$.
Q99. Correct Option: (A) $x = -\frac{3\sqrt{5}}{5}, -\frac{1\sqrt{5}}{20}$.
Q100. Correct Option: (A) $x = -\frac{1\sqrt{7}}{7}, -\frac{2\sqrt{7}}{35}$.
Q101. Zeroes: $-1\sqrt{2}, -\frac{3\sqrt{2}}{2}$. Sum $= -7/(\sqrt{2})$, Product $= (3\sqrt{2})/(\sqrt{2}) = 3/1$.
Q102. Zeroes: $-1\sqrt{3}, -\frac{4\sqrt{3}}{6}$. Sum $= -22/(2\sqrt{3})$, Product $= (4\sqrt{3})/(2\sqrt{3}) = 4/2$.
Q103. Zeroes: $-\frac{1\sqrt{5}}{5}, -\frac{5\sqrt{5}}{15}$. Sum $= -20/(3\sqrt{5})$, Product $= (5\sqrt{5})/(3\sqrt{5}) = 5/3$.
Q104. Zeroes: $-\frac{2\sqrt{7}}{7}, -\frac{6\sqrt{7}}{28}$. Sum $= -62/(4\sqrt{7})$, Product $= (6\sqrt{7})/(4\sqrt{7}) = 6/4$.
Q105. Zeroes: $-\frac{3\sqrt{2}}{2}, -\frac{7\sqrt{2}}{10}$. Sum $= -37/(5\sqrt{2})$, Product $= (7\sqrt{2})/(5\sqrt{2}) = 7/5$.
Q106. Zeroes: $-\frac{1\sqrt{3}}{3}, -\frac{1\sqrt{3}}{3}$. Sum $= -4/(\sqrt{3})$, Product $= (\sqrt{3})/(\sqrt{3}) = 1/1$.
Q107. Zeroes: $-\frac{2\sqrt{5}}{5}, -\frac{2\sqrt{5}}{10}$. Sum $= -22/(2\sqrt{5})$, Product $= (2\sqrt{5})/(2\sqrt{5}) = 2/2$.
Q108. Zeroes: $-\frac{3\sqrt{7}}{7}, -\frac{3\sqrt{7}}{21}$. Sum $= -66/(3\sqrt{7})$, Product $= (3\sqrt{7})/(3\sqrt{7}) = 3/3$.
Q109. Zeroes: $-\frac{1\sqrt{2}}{2}, -\frac{4\sqrt{2}}{8}$. Sum $= -12/(4\sqrt{2})$, Product $= (4\sqrt{2})/(4\sqrt{2}) = 4/4$.
Q110. Zeroes: $-\frac{2\sqrt{3}}{3}, -\frac{5\sqrt{3}}{15}$. Sum $= -35/(5\sqrt{3})$, Product $= (5\sqrt{3})/(5\sqrt{3}) = 5/5$.
Q111. Zeroes: $-\frac{3\sqrt{5}}{5}, -\frac{6\sqrt{5}}{5}$. Sum $= -21/(\sqrt{5})$, Product $= (6\sqrt{5})/(\sqrt{5}) = 6/1$.
Q112. Zeroes: $-\frac{1\sqrt{7}}{7}, -\frac{7\sqrt{7}}{14}$. Sum $= -21/(2\sqrt{7})$, Product $= (7\sqrt{7})/(2\sqrt{7}) = 7/2$.
Q113. Zeroes: $-1\sqrt{2}, -\frac{1\sqrt{2}}{6}$. Sum $= -13/(3\sqrt{2})$, Product $= (\sqrt{2})/(3\sqrt{2}) = 1/3$.

Q114. Zeroes: $-1\sqrt{3}, -\frac{2\sqrt{3}}{12}$. Sum $= -38/(4\sqrt{3})$, Product $= (2\sqrt{3})/(4\sqrt{3}) = 2/4$.

Q115. Zeroes: $-\frac{1\sqrt{5}}{5}, -\frac{3\sqrt{5}}{25}$. Sum $= -28/(5\sqrt{5})$, Product $= (3\sqrt{5})/(5\sqrt{5}) = 3/5$.

Q116. Zeroes: $-\frac{2\sqrt{7}}{7}, -\frac{4\sqrt{7}}{7}$. Sum $= -18/(\sqrt{7})$, Product $= (4\sqrt{7})/(\sqrt{7}) = 4/1$.

Q117. Zeroes: $-\frac{3\sqrt{2}}{2}, -\frac{5\sqrt{2}}{4}$. Sum $= -17/(2\sqrt{2})$, Product $= (5\sqrt{2})/(2\sqrt{2}) = 5/2$.

Q118. Zeroes: $-\frac{1\sqrt{3}}{3}, -\frac{6\sqrt{3}}{9}$. Sum $= -15/(3\sqrt{3})$, Product $= (6\sqrt{3})/(3\sqrt{3}) = 6/3$.

Q119. Zeroes: $-\frac{2\sqrt{5}}{5}, -\frac{7\sqrt{5}}{20}$. Sum $= -47/(4\sqrt{5})$, Product $= (7\sqrt{5})/(4\sqrt{5}) = 7/4$.

Q120. Zeroes: $-\frac{3\sqrt{7}}{7}, -\frac{1\sqrt{7}}{35}$. Sum $= -106/(5\sqrt{7})$, Product $= (\sqrt{7})/(5\sqrt{7}) = 1/5$.

Q121. Zeroes: $-\frac{1\sqrt{2}}{2}, -\frac{2\sqrt{2}}{2}$. Sum $= -4/(\sqrt{2})$, Product $= (2\sqrt{2})/(\sqrt{2}) = 2/1$.

Q122. Zeroes: $-\frac{2\sqrt{3}}{3}, -\frac{3\sqrt{3}}{6}$. Sum $= -15/(2\sqrt{3})$, Product $= (3\sqrt{3})/(2\sqrt{3}) = 3/2$.

Q123. Zeroes: $-\frac{3\sqrt{5}}{5}, -\frac{4\sqrt{5}}{15}$. Sum $= -49/(3\sqrt{5})$, Product $= (4\sqrt{5})/(3\sqrt{5}) = 4/3$.

Q124. Zeroes: $-\frac{1\sqrt{7}}{7}, -\frac{5\sqrt{7}}{28}$. Sum $= -33/(4\sqrt{7})$, Product $= (5\sqrt{7})/(4\sqrt{7}) = 5/4$.

Q125. Zeroes: $-1\sqrt{2}, -\frac{6\sqrt{2}}{10}$. Sum $= -26/(5\sqrt{2})$, Product $= (6\sqrt{2})/(5\sqrt{2}) = 6/5$.

Q126. $x = -1\sqrt{3}$ and $x = -\frac{7\sqrt{3}}{3}$ (algebraic roots).

Q127. $x = -\frac{1\sqrt{5}}{5}$ and $x = -\frac{1\sqrt{5}}{10}$ (algebraic roots).

Q128. $x = -\frac{2\sqrt{7}}{7}$ and $x = -\frac{2\sqrt{7}}{21}$ (algebraic roots).

Q129. $x = -\frac{3\sqrt{2}}{2}$ and $x = -\frac{3\sqrt{2}}{8}$ (algebraic roots).

Q130. $x = -\frac{1\sqrt{3}}{3}$ and $x = -\frac{4\sqrt{3}}{15}$ (algebraic roots).

Q131. $x = -\frac{2\sqrt{5}}{5}$ and $x = -\frac{5\sqrt{5}}{5}$ (algebraic roots).

Q132. $x = -\frac{3\sqrt{7}}{7}$ and $x = -\frac{6\sqrt{7}}{14}$ (algebraic roots).

Q133. $x = -\frac{1\sqrt{2}}{2}$ and $x = -\frac{7\sqrt{2}}{6}$ (algebraic roots).

Q134. $x = -\frac{2\sqrt{3}}{3}$ and $x = -\frac{1\sqrt{3}}{12}$ (algebraic roots).

Q135. $x = -\frac{3\sqrt{5}}{5}$ and $x = -\frac{2\sqrt{5}}{25}$ (algebraic roots).

Q136. $x = -\frac{1\sqrt{7}}{7}$ and $x = -\frac{3\sqrt{7}}{7}$ (algebraic roots).

Q137. $x = -1\sqrt{2}$ and $x = -\frac{4\sqrt{2}}{4}$ (algebraic roots).

Q138. $x = -1\sqrt{3}$ and $x = -\frac{5\sqrt{3}}{9}$ (algebraic roots).

Q139. $x = -\frac{1\sqrt{5}}{5}$ and $x = -\frac{6\sqrt{5}}{20}$ (algebraic roots).

Q140. $x = -\frac{2\sqrt{7}}{7}$ and $x = -\frac{7\sqrt{7}}{35}$ (algebraic roots).

Q141. $x = -\frac{3\sqrt{2}}{2}$ and $x = -\frac{1\sqrt{2}}{2}$ (algebraic roots).

Q142. $x = -\frac{1\sqrt{3}}{3}$ and $x = -\frac{2\sqrt{3}}{6}$ (algebraic roots).

Q143. $x = -\frac{2\sqrt{5}}{5}$ and $x = -\frac{3\sqrt{5}}{15}$ (algebraic roots).

Q144. $x = -\frac{3\sqrt{7}}{7}$ and $x = -\frac{4\sqrt{7}}{28}$ (algebraic roots).

Q145. $x = -\frac{1\sqrt{2}}{2}$ and $x = -\frac{5\sqrt{2}}{10}$ (algebraic roots).

Q146. $x = -\frac{2\sqrt{3}}{3}$ and $x = -\frac{6\sqrt{3}}{3}$ (algebraic roots).

Q147. $x = -\frac{3\sqrt{5}}{5}$ and $x = -\frac{7\sqrt{5}}{10}$ (algebraic roots).

Q148. $x = -\frac{1\sqrt{7}}{7}$ and $x = -\frac{1\sqrt{7}}{21}$ (algebraic roots).

Q149. $x = -1\sqrt{2}$ and $x = -\frac{2\sqrt{2}}{8}$ (algebraic roots).

Q150. $x = -1\sqrt{3}$ and $x = -\frac{3\sqrt{3}}{15}$ (algebraic roots).

Q151. $D = b^2 - 4ac = 9^2 - 4(5)(4) = 1 > 0$. Real and distinct roots.

Q152. $D = b^2 - 4ac = 33^2 - 4(14)(5) = 809 > 0$. Real and distinct roots.

Q153. $D = b^2 - 4ac = 24^2 - 4(6)(6) = 432 > 0$. Real and distinct roots.

Q154. $D = b^2 - 4ac = 19^2 - 4(12)(7) = 25 > 0$. Real and distinct roots.

Q155. $D = b^2 - 4ac = 51^2 - 4(25)(1) = 2501 > 0$. Real and distinct roots.

Q156. $D = b^2 - 4ac = 23^2 - 4(7)(2) = 473 > 0$. Real and distinct roots.

Q157. $D = b^2 - 4ac = 7^2 - 4(4)(3) = 1 > 0$. Real and distinct roots.

Q158. $D = b^2 - 4ac = 22^2 - 4(9)(4) = 340 > 0$. Real and distinct roots.

Q159. $D = b^2 - 4ac = 65^2 - 4(20)(5) = 3825 > 0$. Real and distinct roots.

Q160. $D = b^2 - 4ac = 41^2 - 4(35)(6) = 841 > 0$. Real and distinct roots.

Q161. $D = b^2 - 4ac = 11^2 - 4(2)(7) = 65 > 0$. Real and distinct roots.

Q162. $D = b^2 - 4ac = 19^2 - 4(6)(1) = 337 > 0$. Real and distinct roots.

Q163. $D = b^2 - 4ac = 17^2 - 4(15)(2) = 169 > 0$. Real and distinct roots.

Q164. $D = b^2 - 4ac = 59^2 - 4(28)(3) = 3145 > 0$. Real and distinct roots.

Q165. $D = b^2 - 4ac = 34^2 - 4(10)(4) = 996 > 0$. Real and distinct roots.

Q166. $D = b^2 - 4ac = 8^2 - 4(3)(5) = 4 > 0$. Real and distinct roots.

Q167. $D = b^2 - 4ac = 26^2 - 4(10)(6) = 436 > 0$. Real and distinct roots.

Q168. $D = b^2 - 4ac = 70^2 - 4(21)(7) = 4312 > 0$. Real and distinct roots.

Q169. $D = b^2 - 4ac = 9^2 - 4(8)(1) = 49 > 0$. Real and distinct roots.

Q170. $D = b^2 - 4ac = 32^2 - 4(15)(2) = 904 > 0$. Real and distinct roots.

Q171. $D = b^2 - 4ac = 18^2 - 4(5)(3) = 264 > 0$. Real and distinct roots.

Q172. $D = b^2 - 4ac = 18^2 - 4(14)(4) = 100 > 0$. Real and distinct roots.

Q173. $D = b^2 - 4ac = 17^2 - 4(6)(5) = 169 > 0$. Real and distinct roots.

Q174. $D = b^2 - 4ac = 42^2 - 4(12)(6) = 1476 > 0$. Real

and distinct roots.

Q175. $D = b^2 - 4ac = 32^2 - 4(25)(7) = 324 > 0$. Real and distinct roots.

Q176. $\alpha = -\frac{2\sqrt{7}}{7}, \beta = -\frac{1\sqrt{7}}{7} \implies \alpha/\beta = (2 \cdot 1)/1 = 2/1.$

Q177. $\alpha = -\frac{3\sqrt{2}}{2}, \beta = -\frac{2\sqrt{2}}{4} \implies \alpha/\beta = (3 \cdot 2)/2 = 6/2.$

Q178. $\alpha = -\frac{1\sqrt{3}}{3}, \beta = -\frac{3\sqrt{3}}{9} \implies \alpha/\beta = (1 \cdot 3)/3 = 3/3.$

Q179. $\alpha = -\frac{2\sqrt{5}}{5}, \beta = -\frac{4\sqrt{5}}{20} \implies \alpha/\beta = (2 \cdot 4)/4 = 8/4.$

Q180. $\alpha = -\frac{3\sqrt{7}}{7}, \beta = -\frac{5\sqrt{7}}{35} \implies \alpha/\beta = (3 \cdot 5)/5 = 15/5.$

Q181. $\alpha = -\frac{1\sqrt{2}}{2}, \beta = -\frac{6\sqrt{2}}{2} \implies \alpha/\beta = (1 \cdot 1)/6 = 1/6.$

Q182. $\alpha = -\frac{2\sqrt{3}}{3}, \beta = -\frac{7\sqrt{3}}{6} \implies \alpha/\beta = (2 \cdot 2)/7 = 4/7.$

Q183. $\alpha = -\frac{3\sqrt{5}}{5}, \beta = -\frac{1\sqrt{5}}{15} \implies \alpha/\beta = (3 \cdot 3)/1 = 9/1.$

Q184. $\alpha = -\frac{1\sqrt{7}}{7}, \beta = -\frac{2\sqrt{7}}{28} \implies \alpha/\beta = (1 \cdot 4)/2 = 4/2.$

Q185. $\alpha = -1\sqrt{2}, \beta = -\frac{3\sqrt{2}}{10} \implies \alpha/\beta = (2 \cdot 5)/3 = 10/3.$

Q186. $\alpha = -1\sqrt{3}, \beta = -\frac{4\sqrt{3}}{3} \implies \alpha/\beta = (3 \cdot 1)/4 = 3/4.$

Q187. $\alpha = -\frac{1\sqrt{5}}{5}, \beta = -\frac{5\sqrt{5}}{10} \implies \alpha/\beta = (1 \cdot 2)/5 = 2/5.$

Q188. $\alpha = -\frac{2\sqrt{7}}{7}, \beta = -\frac{6\sqrt{7}}{21} \implies \alpha/\beta = (2 \cdot 3)/6 = 6/6.$

Q189. $\alpha = -\frac{3\sqrt{2}}{2}, \beta = -\frac{7\sqrt{2}}{8} \implies \alpha/\beta = (3 \cdot 4)/7 = 12/7.$

Q190. $\alpha = -\frac{1\sqrt{3}}{3}, \beta = -\frac{1\sqrt{3}}{15} \implies \alpha/\beta = (1 \cdot 5)/1 = 5/1.$

Q191. $\alpha = -\frac{2\sqrt{5}}{5}, \beta = -\frac{2\sqrt{5}}{5} \implies \alpha/\beta = (2 \cdot 1)/2 = 2/2.$

Q192. $\alpha = -\frac{3\sqrt{7}}{7}, \beta = -\frac{3\sqrt{7}}{14} \implies \alpha/\beta = (3 \cdot 2)/3 = 6/3.$

Q193. $\alpha = -\frac{1\sqrt{2}}{2}, \beta = -\frac{4\sqrt{2}}{6} \implies \alpha/\beta = (1 \cdot 3)/4 = 3/4.$

Q194. $\alpha = -\frac{2\sqrt{3}}{3}, \beta = -\frac{5\sqrt{3}}{12} \implies \alpha/\beta = (2 \cdot 4)/5 = 8/5.$

Q195. $\alpha = -\frac{3\sqrt{5}}{5}, \beta = -\frac{6\sqrt{5}}{25} \implies \alpha/\beta = (3 \cdot 5)/6 = 15/6.$

Q196. $\alpha = -\frac{1\sqrt{7}}{7}, \beta = -\frac{7\sqrt{7}}{7} \implies \alpha/\beta = (1 \cdot 1)/7 = 1/7.$

Q197. $\alpha = -1\sqrt{2}, \beta = -\frac{1\sqrt{2}}{4} \implies \alpha/\beta = (2 \cdot 2)/1 = 4/1.$

Q198. $\alpha = -1\sqrt{3}, \beta = -\frac{2\sqrt{3}}{9} \implies \alpha/\beta = (3 \cdot 3)/2 = 9/2.$

Q199. $\alpha = -\frac{1\sqrt{5}}{5}, \beta = -\frac{3\sqrt{5}}{20} \implies \alpha/\beta = (1 \cdot 4)/3 = 4/3.$

Q200. $\alpha = -\frac{2\sqrt{7}}{7}, \beta = -\frac{4\sqrt{7}}{35} \implies \alpha/\beta = (2 \cdot 5)/4 = 10/4.$

Pattern 2: Condition-Based Reciprocal Relationships of Zeroes — Answer Key

Q201. Product of zeroes $= 1 \implies \frac{2mk}{k^2+4} = 1 \implies k^2 - 4k + 4 = 0 \implies k = 2.$

Q202. Product of zeroes $= 1 \implies \frac{2mk}{k^2+9} = 1 \implies k^2 - 6k + 9 = 0 \implies k = 3.$

Q203. Product of zeroes $= 1 \implies \frac{2mk}{k^2+16} = 1 \implies k^2 - 8k + 16 = 0 \implies k = 4.$

Q204. Product of zeroes $= 1 \implies \frac{2mk}{k^2+25} = 1 \implies k^2 - 10k + 25 = 0 \implies k = 5.$

Q205. Product of zeroes $= 1 \implies \frac{2mk}{k^2+36} = 1 \implies k^2 - 12k + 36 = 0 \implies k = 6.$

Q206. Product of zeroes $= 1 \implies \frac{2mk}{k^2+49} = 1 \implies k^2 - 14k + 49 = 0 \implies k = 7.$

Q207. Product of zeroes $= 1 \implies \frac{2mk}{k^2+4} = 1 \implies k^2 - 4k + 4 = 0 \implies k = 2.$

Q208. Product of zeroes $= 1 \implies \frac{2mk}{k^2+9} = 1 \implies k^2 - 6k + 9 = 0 \implies k = 3.$

Q209. Product of zeroes $= 1 \implies \frac{2mk}{k^2+16} = 1 \implies k^2 - 8k + 16 = 0 \implies k = 4.$

Q210. Product of zeroes $= 1 \implies \frac{2mk}{k^2+25} = 1 \implies k^2 - 10k + 25 = 0 \implies k = 5.$

Q211. Product of zeroes $= 1 \implies \frac{2mk}{k^2+36} = 1 \implies k^2 - 12k + 36 = 0 \implies k = 6.$

Q212. Product of zeroes $= 1 \implies \frac{2mk}{k^2+49} = 1 \implies k^2 - 14k + 49 = 0 \implies k = 7.$

Q213. Product of zeroes $= 1 \implies \frac{2mk}{k^2+4} = 1 \implies k^2 - 4k + 4 = 0 \implies k = 2.$

Q214. Product of zeroes $= 1 \implies \frac{2mk}{k^2+9} = 1 \implies k^2 - 6k + 9 = 0 \implies k = 3.$

Q215. Product of zeroes $= 1 \implies \frac{2mk}{k^2+16} = 1 \implies k^2 - 8k + 16 = 0 \implies k = 4.$

Q216. Product of zeroes $= 1 \implies \frac{2mk}{k^2+25} = 1 \implies k^2 - 10k + 25 = 0 \implies k = 5.$

Q217. Product of zeroes $= 1 \implies \frac{2mk}{k^2+36} = 1 \implies k^2 - 12k + 36 = 0 \implies k = 6.$

Q218. Product of zeroes $= 1 \implies \frac{2mk}{k^2+49} = 1 \implies k^2 - 14k + 49 = 0 \implies k = 7.$

Q219. Product of zeroes $= 1 \implies \frac{2mk}{k^2+4} = 1 \implies k^2 - 4k + 4 = 0 \implies k = 2.$

Q220. Product of zeroes $= 1 \implies \frac{2mk}{k^2+9} = 1 \implies k^2 - 6k + 9 = 0 \implies k = 3.$

Q221. Product of zeroes $= 1 \implies \frac{2mk}{k^2+16} = 1 \implies k^2 - 8k + 16 = 0 \implies k = 4.$

Q222. Product of zeroes $= 1 \implies \frac{2mk}{k^2+25} = 1 \implies k^2 - 10k + 25 = 0 \implies k = 5.$

Q223. Product of zeroes $= 1 \implies \frac{2mk}{k^2+36} = 1 \implies k^2 - 12k + 36 = 0 \implies k = 6.$

Q224. Product of zeroes $= 1 \implies \frac{2mk}{k^2+49} = 1 \implies k^2 - 14k + 49 = 0 \implies k = 7.$

Q225. Product of zeroes $= 1 \implies \frac{2mk}{k^2+4} = 1 \implies k^2 - 4k + 4 = 0 \implies k = 2.$

Q226. Product $= 1 \implies a^2 - 6a + 6 = 0 \implies a = 3.$

Q227. Product $= 1 \implies a^2 - 8a + 8 = 0 \implies a = 4.$

Q228. Product $= 1 \implies a^2 - 10a + 10 = 0 \implies a = 5.$

Q229. Product $= 1 \implies a^2 - 12a + 12 = 0 \implies a = 6.$

Q230. Product $= 1 \implies a^2 - 14a + 14 = 0 \implies a = 7.$

Q231. Product $= 1 \implies a^2 - 4a + 4 = 0 \implies a = 2.$

Q232. Product $= 1 \implies a^2 - 6a + 6 = 0 \implies a = 3.$

Q233. Product $= 1 \implies a^2 - 8a + 8 = 0 \implies a = 4.$

Q234. Product $= 1 \implies a^2 - 10a + 10 = 0 \implies a = 5.$

Q235. Product $= 1 \implies a^2 - 12a + 12 = 0 \implies a = 6.$

Q236. Product $= 1 \implies a^2 - 14a + 14 = 0 \implies a = 7.$

Q237. Product $= 1 \implies a^2 - 4a + 4 = 0 \implies a = 2.$

Q238. Product $= 1 \implies a^2 - 6a + 6 = 0 \implies a = 3$.
Q239. Product $= 1 \implies a^2 - 8a + 8 = 0 \implies a = 4$.
Q240. Product $= 1 \implies a^2 - 10a + 10 = 0 \implies a = 5$.
Q241. Product $= 1 \implies a^2 - 12a + 12 = 0 \implies a = 6$.
Q242. Product $= 1 \implies a^2 - 14a + 14 = 0 \implies a = 7$.
Q243. Product $= 1 \implies a^2 - 4a + 4 = 0 \implies a = 2$.
Q244. Product $= 1 \implies a^2 - 6a + 6 = 0 \implies a = 3$.
Q245. Product $= 1 \implies a^2 - 8a + 8 = 0 \implies a = 4$.
Q246. Product $= 1 \implies a^2 - 10a + 10 = 0 \implies a = 5$.
Q247. Product $= 1 \implies a^2 - 12a + 12 = 0 \implies a = 6$.
Q248. Product $= 1 \implies a^2 - 14a + 14 = 0 \implies a = 7$.
Q249. Product $= 1 \implies a^2 - 4a + 4 = 0 \implies a = 2$.
Q250. Product $= 1 \implies a^2 - 6a + 6 = 0 \implies a = 3$.
Q251. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q252. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q253. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q254. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q255. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q256. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q257. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q258. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q259. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q260. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q261. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q262. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q263. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q264. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q265. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q266. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q267. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q268. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q269. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q270. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q271. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q272. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q273. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q274. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q275. $\frac{k^2-1}{2k+1} = 1 \implies k^2 - 2k - 2 = 0 \implies k = 1 \pm \sqrt{3}$.
Q276. Correct Option: (A) $k = 5$.
Q277. Correct Option: (A) $k = 6$.
Q278. Correct Option: (A) $k = 7$.
Q279. Correct Option: (A) $k = 2$.
Q280. Correct Option: (A) $k = 3$.
Q281. Correct Option: (A) $k = 4$.
Q282. Correct Option: (A) $k = 5$.
Q283. Correct Option: (A) $k = 6$.
Q284. Correct Option: (A) $k = 7$.
Q285. Correct Option: (A) $k = 2$.
Q286. Correct Option: (A) $k = 3$.
Q287. Correct Option: (A) $k = 4$.
Q288. Correct Option: (A) $k = 5$.
Q289. Correct Option: (A) $k = 6$.
Q290. Correct Option: (A) $k = 7$.
Q291. Correct Option: (A) $k = 2$.
Q292. Correct Option: (A) $k = 3$.
Q293. Correct Option: (A) $k = 4$.
Q294. Correct Option: (A) $k = 5$.
Q295. Correct Option: (A) $k = 6$.
Q296. Correct Option: (A) $k = 7$.
Q297. Correct Option: (A) $k = 2$.
Q298. Correct Option: (A) $k = 3$.
Q299. Correct Option: (A) $k = 4$.
Q300. Correct Option: (A) $k = 5$.
Q301. No, because Product of zeroes $= 7/3 \neq 1$.
Q302. No, because Product of zeroes $= 7/3 \neq 1$.
Q303. No, because Product of zeroes $= 7/3 \neq 1$.
Q304. No, because Product of zeroes $= 7/3 \neq 1$.
Q305. No, because Product of zeroes $= 7/3 \neq 1$.
Q306. No, because Product of zeroes $= 7/3 \neq 1$.
Q307. No, because Product of zeroes $= 7/3 \neq 1$.
Q308. No, because Product of zeroes $= 7/3 \neq 1$.
Q309. No, because Product of zeroes $= 7/3 \neq 1$.
Q310. No, because Product of zeroes $= 7/3 \neq 1$.
Q311. No, because Product of zeroes $= 7/3 \neq 1$.
Q312. No, because Product of zeroes $= 7/3 \neq 1$.
Q313. No, because Product of zeroes $= 7/3 \neq 1$.
Q314. No, because Product of zeroes $= 7/3 \neq 1$.
Q315. No, because Product of zeroes $= 7/3 \neq 1$.
Q316. No, because Product of zeroes $= 7/3 \neq 1$.
Q317. No, because Product of zeroes $= 7/3 \neq 1$.
Q318. No, because Product of zeroes $= 7/3 \neq 1$.
Q319. No, because Product of zeroes $= 7/3 \neq 1$.
Q320. No, because Product of zeroes $= 7/3 \neq 1$.
Q321. No, because Product of zeroes $= 7/3 \neq 1$.
Q322. No, because Product of zeroes $= 7/3 \neq 1$.
Q323. No, because Product of zeroes $= 7/3 \neq 1$.
Q324. No, because Product of zeroes $= 7/3 \neq 1$.
Q325. No, because Product of zeroes $= 7/3 \neq 1$.
Q326. Product $= 1 \implies p = 7$.
Q327. Product $= 1 \implies p = 2$.
Q328. Product $= 1 \implies p = 3$.
Q329. Product $= 1 \implies p = 4$.

Q330. Product $= 1 \implies p = 5$.
Q331. Product $= 1 \implies p = 6$.
Q332. Product $= 1 \implies p = 7$.
Q333. Product $= 1 \implies p = 2$.
Q334. Product $= 1 \implies p = 3$.
Q335. Product $= 1 \implies p = 4$.
Q336. Product $= 1 \implies p = 5$.
Q337. Product $= 1 \implies p = 6$.
Q338. Product $= 1 \implies p = 7$.
Q339. Product $= 1 \implies p = 2$.
Q340. Product $= 1 \implies p = 3$.
Q341. Product $= 1 \implies p = 4$.
Q342. Product $= 1 \implies p = 5$.
Q343. Product $= 1 \implies p = 6$.
Q344. Product $= 1 \implies p = 7$.
Q345. Product $= 1 \implies p = 2$.
Q346. Product $= 1 \implies p = 3$.
Q347. Product $= 1 \implies p = 4$.
Q348. Product $= 1 \implies p = 5$.
Q349. Product $= 1 \implies p = 6$.
Q350. Product $= 1 \implies p = 7$.
Q351. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 2$ satisfies the quadratic condition.
Q352. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 3$ satisfies the quadratic condition.
Q353. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 4$ satisfies the quadratic condition.
Q354. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 5$ satisfies the quadratic condition.
Q355. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 6$ satisfies the quadratic condition.
Q356. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 7$ satisfies the quadratic condition.
Q357. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 2$ satisfies the quadratic condition.
Q358. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 3$ satisfies the quadratic condition.
Q359. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 4$ satisfies the quadratic condition.
Q360. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 5$ satisfies the quadratic condition.
Q361. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 6$ satisfies the quadratic condition.
Q362. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 7$ satisfies the quadratic condition.
Q363. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 2$ satisfies the quadratic condition.
Q364. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 3$ satisfies the quadratic condition.
Q365. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 4$ satisfies the quadratic condition.
Q366. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 5$ satisfies the quadratic condition.
Q367. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 6$ satisfies the quadratic condition.
Q368. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 7$ satisfies the quadratic condition.
Q369. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 2$ satisfies the quadratic condition.
Q370. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 3$ satisfies the quadratic condition.
Q371. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 4$ satisfies the quadratic condition.
Q372. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 5$ satisfies the quadratic condition.
Q373. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 6$ satisfies the quadratic condition.
Q374. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 7$ satisfies the quadratic condition.
Q375. Product $= \frac{2m^2}{2m^2} = 1$. Yes, $m = 2$ satisfies the quadratic condition.
Q376. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.
Q377. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.
Q378. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.
Q379. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.
Q380. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.
Q381. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.
Q382. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.
Q383. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.
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Q386. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.
Q387. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.
Q388. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.
Q389. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.
Q390. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.
Q391. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$,

no real k.

Q392. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.

Q393. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.

Q394. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.

Q395. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.

Q396. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.

Q397. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.

Q398. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.

Q399. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.

Q400. $\frac{k^2+1}{k-1} = 1 \implies k^2 - k + 2 = 0$, discriminant $D < 0$, no real k.

Pattern 3: Symmetric Transformations of Zeroes — Answer Key

Q401. $\alpha + \beta = 3, \alpha\beta = k \implies \alpha^2 + \beta^2 = 3^2 - 2k = 7 \implies k = 1$.

Q402. $\alpha + \beta = 5, \alpha\beta = k \implies \alpha^2 + \beta^2 = 5^2 - 2k = 21 \implies k = 2$.

Q403. $\alpha + \beta = 7, \alpha\beta = k \implies \alpha^2 + \beta^2 = 7^2 - 2k = 43 \implies k = 3$.

Q404. $\alpha + \beta = 9, \alpha\beta = k \implies \alpha^2 + \beta^2 = 9^2 - 2k = 73 \implies k = 4$.

Q405. $\alpha + \beta = 11, \alpha\beta = k \implies \alpha^2 + \beta^2 = 11^2 - 2k = 111 \implies k = 5$.

Q406. $\alpha + \beta = 3, \alpha\beta = k \implies \alpha^2 + \beta^2 = 3^2 - 2k = -3 \implies k = 6$.

Q407. $\alpha + \beta = 5, \alpha\beta = k \implies \alpha^2 + \beta^2 = 5^2 - 2k = 11 \implies k = 7$.

Q408. $\alpha + \beta = 7, \alpha\beta = k \implies \alpha^2 + \beta^2 = 7^2 - 2k = 33 \implies k = 8$.

Q409. $\alpha + \beta = 9, \alpha\beta = k \implies \alpha^2 + \beta^2 = 9^2 - 2k = 63 \implies k = 9$.

Q410. $\alpha + \beta = 11, \alpha\beta = k \implies \alpha^2 + \beta^2 = 11^2 - 2k = 119 \implies k = 1$.

Q411. $\alpha + \beta = 3, \alpha\beta = k \implies \alpha^2 + \beta^2 = 3^2 - 2k = 5 \implies k = 2$.

Q412. $\alpha + \beta = 5, \alpha\beta = k \implies \alpha^2 + \beta^2 = 5^2 - 2k = 19 \implies k = 3$.

Q413. $\alpha + \beta = 7, \alpha\beta = k \implies \alpha^2 + \beta^2 = 7^2 - 2k = 41 \implies k = 4$.

Q414. $\alpha + \beta = 9, \alpha\beta = k \implies \alpha^2 + \beta^2 = 9^2 - 2k = 71 \implies k = 5$.

Q415. $\alpha + \beta = 11, \alpha\beta = k \implies \alpha^2 + \beta^2 = 11^2 - 2k = 109 \implies k = 6$.

Q416. $\alpha + \beta = 3, \alpha\beta = k \implies \alpha^2 + \beta^2 = 3^2 - 2k = -5 \implies k = 7$.

Q417. $\alpha + \beta = 5, \alpha\beta = k \implies \alpha^2 + \beta^2 = 5^2 - 2k = 9 \implies k = 8$.

Q418. $\alpha + \beta = 7, \alpha\beta = k \implies \alpha^2 + \beta^2 = 7^2 - 2k = 31 \implies k = 9$.

Q419. $\alpha + \beta = 9, \alpha\beta = k \implies \alpha^2 + \beta^2 = 9^2 - 2k = 79 \implies k = 1$.

Q420. $\alpha + \beta = 11, \alpha\beta = k \implies \alpha^2 + \beta^2 = 11^2 - 2k = 117 \implies k = 2$.

Q421. $\alpha + \beta = 3, \alpha\beta = k \implies \alpha^2 + \beta^2 = 3^2 - 2k = 3 \implies k = 3$.

Q422. $\alpha + \beta = 5, \alpha\beta = k \implies \alpha^2 + \beta^2 = 5^2 - 2k = 17 \implies k = 4$.

Q423. $\alpha + \beta = 7, \alpha\beta = k \implies \alpha^2 + \beta^2 = 7^2 - 2k = 39 \implies k = 5$.

Q424. $\alpha + \beta = 9, \alpha\beta = k \implies \alpha^2 + \beta^2 = 9^2 - 2k = 69 \implies k = 6$.

Q425. $\alpha + \beta = 11, \alpha\beta = k \implies \alpha^2 + \beta^2 = 11^2 - 2k = 107 \implies k = 7$.

Q426. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 3^2 - 4k \implies 4k = 8 \implies k = \frac{8}{4}$.

Q427. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 5^2 - 4k \implies 4k = 24 \implies k = \frac{24}{4}$.

Q428. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 7^2 - 4k \implies 4k = 48 \implies k = \frac{48}{4}$.

Q429. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 9^2 - 4k \implies 4k = 80 \implies k = \frac{80}{4}$.

Q430. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 11^2 - 4k \implies 4k = 120 \implies k = \frac{120}{4}$.

Q431. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 3^2 - 4k \implies 4k = 8 \implies k = \frac{8}{4}$.

Q432. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 5^2 - 4k \implies 4k = 24 \implies k = \frac{24}{4}$.

Q433. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 7^2 - 4k \implies 4k = 48 \implies k = \frac{48}{4}$.

Q434. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 9^2 - 4k \implies 4k = 80 \implies k = \frac{80}{4}$.

Q435. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 11^2 - 4k \implies 4k = 120 \implies k = \frac{120}{4}$.

Q436. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 3^2 - 4k \implies 4k = 8 \implies k = \frac{8}{4}$.

Q437. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 5^2 - 4k \implies 4k = 24 \implies k = \frac{24}{4}$.

Q438. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 7^2 - 4k \implies 4k = 48 \implies k = \frac{48}{4}$.

Q439. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 9^2 - 4k \implies 4k = 80 \implies k = \frac{80}{4}$.

Q440. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 11^2 - 4k \implies 4k = 120 \implies k = \frac{120}{4}$.

Q441. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = 3^2 - 4k \implies 4k = 8 \implies k = \frac{8}{4}$.

Q442. $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta \implies 1=5^2-4k \implies 4k=24 \implies k=\frac{24}{4}$.

Q443. $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta \implies 1=7^2-4k \implies 4k=48 \implies k=\frac{48}{4}$.

Q444. $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta \implies 1=9^2-4k \implies 4k=80 \implies k=\frac{80}{4}$.

Q445. $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta \implies 1=11^2-4k \implies 4k=120 \implies k=\frac{120}{4}$.

Q446. $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta \implies 1=3^2-4k \implies 4k=8 \implies k=\frac{8}{4}$.

Q447. $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta \implies 1=5^2-4k \implies 4k=24 \implies k=\frac{24}{4}$.

Q448. $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta \implies 1=7^2-4k \implies 4k=48 \implies k=\frac{48}{4}$.

Q449. $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta \implies 1=9^2-4k \implies 4k=80 \implies k=\frac{80}{4}$.

Q450. $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta \implies 1=11^2-4k \implies 4k=120 \implies k=\frac{120}{4}$.

Q451. $(\alpha+1)(\beta+1)=\alpha\beta+(\alpha+\beta)+1=-(p+c)+p+1=1-c$.

Q452. $(\alpha+1)(\beta+1)=\alpha\beta+(\alpha+\beta)+1=-(p+c)+p+1=1-c$.

Q453. $(\alpha+1)(\beta+1)=\alpha\beta+(\alpha+\beta)+1=-(p+c)+p+1=1-c$.

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Q464. $(\alpha+1)(\beta+1)=\alpha\beta+(\alpha+\beta)+1=-(p+c)+p+1=1-c$.

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Q472. $(\alpha+1)(\beta+1)=\alpha\beta+(\alpha+\beta)+1=-(p+c)+p+1=1-c$.

Q473. $(\alpha+1)(\beta+1)=\alpha\beta+(\alpha+\beta)+1=-(p+c)+p+1=1-c$.

Q474. $(\alpha+1)(\beta+1)=\alpha\beta+(\alpha+\beta)+1=-(p+c)+p+1=1-c$.

Q475. $(\alpha+1)(\beta+1)=\alpha\beta+(\alpha+\beta)+1=-(p+c)+p+1=1-c$.

Q476. Correct Option: (A) $-b/c$, since $\frac{\alpha+\beta}{\alpha\beta}=\frac{-b/a}{c/a}=-b/c$.

Q477. Correct Option: (A) $-b/c$, since $\frac{\alpha+\beta}{\alpha\beta}=\frac{-b/a}{c/a}=-b/c$.

Q478. Correct Option: (A) $-b/c$, since $\frac{\alpha+\beta}{\alpha\beta}=\frac{-b/a}{c/a}=-b/c$.

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Q493. Correct Option: (A) $-b/c$, since $\frac{\alpha+\beta}{\alpha\beta} = \frac{-b/a}{c/a} = -b/c$.

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Q499. Correct Option: (A) $-b/c$, since $\frac{\alpha+\beta}{\alpha\beta} = \frac{-b/a}{c/a} = -b/c$.

Q500. Correct Option: (A) $-b/c$, since $\frac{\alpha+\beta}{\alpha\beta} = \frac{-b/a}{c/a} = -b/c$.

Q501. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q502. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q503. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q504. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

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Q507. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q508. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q509. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q510. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q511. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q512. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

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Q514. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q515. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q516. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

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Q518. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q519. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q520. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q521. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q522. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q523. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q524. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q525. $(\alpha+\beta)^2 - \alpha\beta = 25/4 - k/2 = 21/4 \implies k/2 = 1 \implies k = 2$.

Q526. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{9-2k}{k} = 3 \implies 5k = 9 \implies k = 9/5$.

Q527. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{25-2k}{k} = 3 \implies 5k = 25 \implies k = 25/5$.

Q528. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{49-2k}{k} = 3 \implies 5k = 49 \implies k = 49/5$.

Q529. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{81-2k}{k} = 3 \implies 5k = 81 \implies k = 81/5$.

Q530. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{121-2k}{k} = 3 \implies 5k = 121 \implies k = 121/5$.

Q531. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{9-2k}{k} = 3 \implies 5k = 9 \implies k = 9/5$.

Q532. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{25-2k}{k} = 3 \implies 5k = 25 \implies k = 25/5$.

Q533. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{49-2k}{k} = 3 \implies 5k = 49 \implies k = 49/5$.

Q534. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{81-2k}{k} = 3 \implies 5k = 81 \implies k = 81/5$.

Q535. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{121-2k}{k} = 3 \implies 5k = 121 \implies k = 121/5$.

Q536. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{9-2k}{k} = 3 \implies 5k = 9 \implies k = 9/5$.

Q537. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{25-2k}{k} = 3 \implies 5k = 25 \implies k = 25/5$.

Q538. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{49-2k}{k} = 3 \implies 5k = 49 \implies k = 49/5$.

Q539. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{81-2k}{k} = 3 \implies 5k = 81 \implies k = 81/5$.

Q540. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{121-2k}{k} = 3 \implies 5k = 121 \implies k = 121/5$.

Q541. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{9-2k}{k} = 3 \implies 5k = 9 \implies k = 9/5$.

Q542. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{25-2k}{k} = 3 \implies 5k = 25 \implies k = 25/5$.

Q543. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{49-2k}{k} = 3 \implies 5k = 49 \implies k = 49/5$.

Q544. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{81-2k}{k} = 3 \implies 5k = 81 \implies k = 81/5$.

Q545. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{121-2k}{k} = 3 \implies 5k = 121 \implies k = 121/5$.

Q546. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{9-2k}{k} = 3 \implies 5k = 9 \implies k = 9/5$.

Q547. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{25-2k}{k} = 3 \implies 5k = 25 \implies k = 25/5$.

Q548. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{49-2k}{k} = 3 \implies 5k = 49 \implies k = 49/5$.

Q549. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{81-2k}{k} = 3 \implies 5k = 81 \implies k = 81/5$.

Q550. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{121-2k}{k} = 3 \implies 5k = 121 \implies k = 121/5$.

Q551. $\alpha+\beta = 6 \implies 2\alpha+2\beta = 12$. Subtracting gives $\alpha = 8 \implies \beta = -2 \implies k = -16$.

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Q566. $\alpha+\beta = 6 \implies 2\alpha+2\beta = 12$. Subtracting gives $\alpha = 8 \implies \beta = -2 \implies k = -16$.

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Q568. $\alpha+\beta = 6 \implies 2\alpha+2\beta = 12$. Subtracting gives $\alpha = 8 \implies \beta = -2 \implies k = -16$.

Q569. $\alpha+\beta = 6 \implies 2\alpha+2\beta = 12$. Subtracting gives $\alpha = 8 \implies \beta = -2 \implies k = -16$.

Q570. $\alpha+\beta = 6 \implies 2\alpha+2\beta = 12$. Subtracting gives $\alpha = 8 \implies \beta = -2 \implies k = -16$.

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Q573. $\alpha+\beta = 6 \implies 2\alpha+2\beta = 12$. Subtracting gives $\alpha = 8 \implies \beta = -2 \implies k = -16$.

Q574. $\alpha+\beta = 6 \implies 2\alpha+2\beta = 12$. Subtracting gives $\alpha = 8 \implies \beta = -2 \implies k = -16$.

Q575. $\alpha+\beta = 6 \implies 2\alpha+2\beta = 12$. Subtracting gives $\alpha = 8 \implies \beta = -2 \implies k = -16$.

Q576. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 3(-6) = -18$.

Q577. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 5(7) = 35$.

Q578. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 7(28) = 196$.

Q579. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 9(57) = 513$.

Q580. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 11(94) = 1034$.

Q581. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 3(6) = 18$.

Q582. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 5(19) = 95$.

Q583. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 7(40) = 280$.

Q584. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 9(69) = 621$.

Q585. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 11(106) = 1166$.

Q586. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 3(-9) = -27$.

Q587. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 5(4) = 20$.

Q588. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 7(25) = 175$.

Q589. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 9(54) = 486$.

Q590. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 11(118) = 1298$.

Q591. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 3(3) = 9$.

Q592. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 5(16) = 80$.

Q593. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 7(37) = 259$.

Q594. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 9(66) = 594$.

Q595. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 11(103) = 1133$.

Q596. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 3(-12) = -36$.

Q597. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 5(1) = 5$.

Q598. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 7(22) = 154$.

Q599. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 9(78) = 702$.

Q600. $\alpha^3+\beta^3 = (\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2) = 11(115) = 1265$.

Pattern 4: Bi-Quadratic Polynomials with Radical Zeroes — Answer Key

Q601. Dividing by x^2-2 gives quotient $x^2-1x+2=0$. Other zeroes depend on quadratic formula.

Q602. Dividing by x^2-3 gives quotient $x^2-2x+3=0$. Other zeroes depend on quadratic formula.

Q603. Dividing by x^2-5 gives quotient $x^2-3x+4=0$. Other zeroes depend on quadratic formula.

Q604. Dividing by x^2-2 gives quotient $x^2-4x+5=0$. Other zeroes depend on quadratic formula.

Q605. Dividing by x^2-3 gives quotient $x^2-1x+6=0$. Other zeroes depend on quadratic formula.

Q606. Dividing by x^2-5 gives quotient $x^2-2x+2=0$. Other zeroes depend on quadratic formula.

Q607. Dividing by x^2-2 gives quotient $x^2-3x+3=0$. Other zeroes depend on quadratic formula.

Q608. Dividing by x^2-3 gives quotient $x^2-4x+4=0$. Other zeroes depend on quadratic formula.

Q609. Dividing by x^2-5 gives quotient $x^2-1x+5=0$. Other zeroes depend on quadratic formula.

Q610. Dividing by x^2-2 gives quotient $x^2-2x+6=0$. Other zeroes depend on quadratic formula.

Q611. Dividing by x^2-3 gives quotient $x^2-3x+2=0$. Other zeroes depend on quadratic formula.

Q612. Dividing by x^2-5 gives quotient $x^2-4x+3=0$.

Other zeroes depend on quadratic formula.

Q613. Dividing by $x^2 - 2$ gives quotient $x^2 - 1x + 4 = 0$. Other zeroes depend on quadratic formula.

Q614. Dividing by $x^2 - 3$ gives quotient $x^2 - 2x + 5 = 0$. Other zeroes depend on quadratic formula.

Q615. Dividing by $x^2 - 5$ gives quotient $x^2 - 3x + 6 = 0$. Other zeroes depend on quadratic formula.

Q616. Dividing by $x^2 - 2$ gives quotient $x^2 - 4x + 2 = 0$. Other zeroes depend on quadratic formula.

Q617. Dividing by $x^2 - 3$ gives quotient $x^2 - 1x + 3 = 0$. Other zeroes depend on quadratic formula.

Q618. Dividing by $x^2 - 5$ gives quotient $x^2 - 2x + 4 = 0$. Other zeroes depend on quadratic formula.

Q619. Dividing by $x^2 - 2$ gives quotient $x^2 - 3x + 5 = 0$. Other zeroes depend on quadratic formula.

Q620. Dividing by $x^2 - 3$ gives quotient $x^2 - 4x + 6 = 0$. Other zeroes depend on quadratic formula.

Q621. Dividing by $x^2 - 5$ gives quotient $x^2 - 1x + 2 = 0$. Other zeroes depend on quadratic formula.

Q622. Dividing by $x^2 - 2$ gives quotient $x^2 - 2x + 3 = 0$. Other zeroes depend on quadratic formula.

Q623. Dividing by $x^2 - 3$ gives quotient $x^2 - 3x + 4 = 0$. Other zeroes depend on quadratic formula.

Q624. Dividing by $x^2 - 5$ gives quotient $x^2 - 4x + 5 = 0$. Other zeroes depend on quadratic formula.

Q625. Dividing by $x^2 - 2$ gives quotient $x^2 - 1x + 6 = 0$. Other zeroes depend on quadratic formula.

Q626. $p(x) = (x^2 - 3)(x^2 - 2x + 2)$.

Q627. $p(x) = (x^2 - 5)(x^2 - 3x + 3)$.

Q628. $p(x) = (x^2 - 2)(x^2 - 4x + 4)$.

Q629. $p(x) = (x^2 - 3)(x^2 - 1x + 5)$.

Q630. $p(x) = (x^2 - 5)(x^2 - 2x + 6)$.

Q631. $p(x) = (x^2 - 2)(x^2 - 3x + 2)$.

Q632. $p(x) = (x^2 - 3)(x^2 - 4x + 3)$.

Q633. $p(x) = (x^2 - 5)(x^2 - 1x + 4)$.

Q634. $p(x) = (x^2 - 2)(x^2 - 2x + 5)$.

Q635. $p(x) = (x^2 - 3)(x^2 - 3x + 6)$.

Q636. $p(x) = (x^2 - 5)(x^2 - 4x + 2)$.

Q637. $p(x) = (x^2 - 2)(x^2 - 1x + 3)$.

Q638. $p(x) = (x^2 - 3)(x^2 - 2x + 4)$.

Q639. $p(x) = (x^2 - 5)(x^2 - 3x + 5)$.

Q640. $p(x) = (x^2 - 2)(x^2 - 4x + 6)$.

Q641. $p(x) = (x^2 - 3)(x^2 - 1x + 2)$.

Q642. $p(x) = (x^2 - 5)(x^2 - 2x + 3)$.

Q643. $p(x) = (x^2 - 2)(x^2 - 3x + 4)$.

Q644. $p(x) = (x^2 - 3)(x^2 - 4x + 5)$.

Q645. $p(x) = (x^2 - 5)(x^2 - 1x + 6)$.

Q646. $p(x) = (x^2 - 2)(x^2 - 2x + 2)$.

Q647. $p(x) = (x^2 - 3)(x^2 - 3x + 3)$.

Q648. $p(x) = (x^2 - 5)(x^2 - 4x + 4)$.

Q649. $p(x) = (x^2 - 2)(x^2 - 1x + 5)$.

Q650. $p(x) = (x^2 - 3)(x^2 - 2x + 6)$.

Q651. Factor is $(x-2)^2 - 5 = x^2 - 4x + (4 - 5)$.

Q652. Factor is $(x-2)^2 - 2 = x^2 - 4x + (4 - 2)$.

Q653. Factor is $(x-2)^2 - 3 = x^2 - 4x + (4 - 3)$.

Q654. Factor is $(x-2)^2 - 5 = x^2 - 4x + (4 - 5)$.

Q655. Factor is $(x-2)^2 - 2 = x^2 - 4x + (4 - 2)$.

Q656. Factor is $(x-2)^2 - 3 = x^2 - 4x + (4 - 3)$.

Q657. Factor is $(x-2)^2 - 5 = x^2 - 4x + (4 - 5)$.

Q658. Factor is $(x-2)^2 - 2 = x^2 - 4x + (4 - 2)$.

Q659. Factor is $(x-2)^2 - 3 = x^2 - 4x + (4 - 3)$.

Q660. Factor is $(x-2)^2 - 5 = x^2 - 4x + (4 - 5)$.

Q661. Factor is $(x-2)^2 - 2 = x^2 - 4x + (4 - 2)$.

Q662. Factor is $(x-2)^2 - 3 = x^2 - 4x + (4 - 3)$.

Q663. Factor is $(x-2)^2 - 5 = x^2 - 4x + (4 - 5)$.

Q664. Factor is $(x-2)^2 - 2 = x^2 - 4x + (4 - 2)$.

Q665. Factor is $(x-2)^2 - 3 = x^2 - 4x + (4 - 3)$.

Q666. Factor is $(x-2)^2 - 5 = x^2 - 4x + (4 - 5)$.

Q667. Factor is $(x-2)^2 - 2 = x^2 - 4x + (4 - 2)$.

Q668. Factor is $(x-2)^2 - 3 = x^2 - 4x + (4 - 3)$.

Q669. Factor is $(x-2)^2 - 5 = x^2 - 4x + (4 - 5)$.

Q670. Factor is $(x-2)^2 - 2 = x^2 - 4x + (4 - 2)$.

Q671. Factor is $(x-2)^2 - 3 = x^2 - 4x + (4 - 3)$.

Q672. Factor is $(x-2)^2 - 5 = x^2 - 4x + (4 - 5)$.

Q673. Factor is $(x-2)^2 - 2 = x^2 - 4x + (4 - 2)$.

Q674. Factor is $(x-2)^2 - 3 = x^2 - 4x + (4 - 3)$.

Q675. Factor is $(x-2)^2 - 5 = x^2 - 4x + (4 - 5)$.

Q676. Correct Option: (A) $2, -2$.

Q677. Correct Option: (A) $2, -2$.

Q678. Correct Option: (A) $2, -2$.

Q679. Correct Option: (A) $2, -2$.

Q680. Correct Option: (A) $2, -2$.

Q681. Correct Option: (A) $2, -2$.

Q682. Correct Option: (A) $2, -2$.

Q683. Correct Option: (A) $2, -2$.

Q684. Correct Option: (A) $2, -2$.

Q685. Correct Option: (A) $2, -2$.

Q686. Correct Option: (A) $2, -2$.

Q687. Correct Option: (A) $2, -2$.

Q688. Correct Option: (A) $2, -2$.

Q689. Correct Option: (A) $2, -2$.

Q690. Correct Option: (A) $2, -2$.

Q691. Correct Option: (A) $2, -2$.

Q692. Correct Option: (A) $2, -2$.

Q693. Correct Option: (A) $2, -2$.

Q694. Correct Option: (A) $2, -2$.

Q695. Correct Option: (A) $2, -2$.

Q696. Correct Option: (A) $2, -2$.

Q697. Correct Option: (A) $2, -2$.

Q698. Correct Option: (A) $2, -2$.

Q699. Correct Option: (A) $2, -2$.
Q700. Correct Option: (A) $2, -2$.
Q701. $p(\sqrt{3}) = (3)^2 - 3(3) = 0$. Multiplicity is 1.
Q702. $p(\sqrt{5}) = (5)^2 - 5(5) = 0$. Multiplicity is 1.
Q703. $p(\sqrt{2}) = (2)^2 - 2(2) = 0$. Multiplicity is 1.
Q704. $p(\sqrt{3}) = (3)^2 - 3(3) = 0$. Multiplicity is 1.
Q705. $p(\sqrt{5}) = (5)^2 - 5(5) = 0$. Multiplicity is 1.
Q706. $p(\sqrt{2}) = (2)^2 - 2(2) = 0$. Multiplicity is 1.
Q707. $p(\sqrt{3}) = (3)^2 - 3(3) = 0$. Multiplicity is 1.
Q708. $p(\sqrt{5}) = (5)^2 - 5(5) = 0$. Multiplicity is 1.
Q709. $p(\sqrt{2}) = (2)^2 - 2(2) = 0$. Multiplicity is 1.
Q710. $p(\sqrt{3}) = (3)^2 - 3(3) = 0$. Multiplicity is 1.
Q711. $p(\sqrt{5}) = (5)^2 - 5(5) = 0$. Multiplicity is 1.
Q712. $p(\sqrt{2}) = (2)^2 - 2(2) = 0$. Multiplicity is 1.
Q713. $p(\sqrt{3}) = (3)^2 - 3(3) = 0$. Multiplicity is 1.
Q714. $p(\sqrt{5}) = (5)^2 - 5(5) = 0$. Multiplicity is 1.
Q715. $p(\sqrt{2}) = (2)^2 - 2(2) = 0$. Multiplicity is 1.
Q716. $p(\sqrt{3}) = (3)^2 - 3(3) = 0$. Multiplicity is 1.
Q717. $p(\sqrt{5}) = (5)^2 - 5(5) = 0$. Multiplicity is 1.
Q718. $p(\sqrt{2}) = (2)^2 - 2(2) = 0$. Multiplicity is 1.
Q719. $p(\sqrt{3}) = (3)^2 - 3(3) = 0$. Multiplicity is 1.
Q720. $p(\sqrt{5}) = (5)^2 - 5(5) = 0$. Multiplicity is 1.
Q721. $p(\sqrt{2}) = (2)^2 - 2(2) = 0$. Multiplicity is 1.
Q722. $p(\sqrt{3}) = (3)^2 - 3(3) = 0$. Multiplicity is 1.
Q723. $p(\sqrt{5}) = (5)^2 - 5(5) = 0$. Multiplicity is 1.
Q724. $p(\sqrt{2}) = (2)^2 - 2(2) = 0$. Multiplicity is 1.
Q725. $p(\sqrt{3}) = (3)^2 - 3(3) = 0$. Multiplicity is 1.
Q726. $(x^2-5)(x^2-2) = 0 \implies x = \pm\sqrt{5}, \pm\sqrt{2}$.
Q727. $(x^2-2)(x^2-3) = 0 \implies x = \pm\sqrt{2}, \pm\sqrt{3}$.
Q728. $(x^2-3)(x^2-4) = 0 \implies x = \pm\sqrt{3}, \pm\sqrt{4}$.
Q729. $(x^2-5)(x^2-1) = 0 \implies x = \pm\sqrt{5}, \pm\sqrt{1}$.
Q730. $(x^2-2)(x^2-2) = 0 \implies x = \pm\sqrt{2}, \pm\sqrt{2}$.
Q731. $(x^2-3)(x^2-3) = 0 \implies x = \pm\sqrt{3}, \pm\sqrt{3}$.
Q732. $(x^2-5)(x^2-4) = 0 \implies x = \pm\sqrt{5}, \pm\sqrt{4}$.
Q733. $(x^2-2)(x^2-1) = 0 \implies x = \pm\sqrt{2}, \pm\sqrt{1}$.
Q734. $(x^2-3)(x^2-2) = 0 \implies x = \pm\sqrt{3}, \pm\sqrt{2}$.
Q735. $(x^2-5)(x^2-3) = 0 \implies x = \pm\sqrt{5}, \pm\sqrt{3}$.
Q736. $(x^2-2)(x^2-4) = 0 \implies x = \pm\sqrt{2}, \pm\sqrt{4}$.
Q737. $(x^2-3)(x^2-1) = 0 \implies x = \pm\sqrt{3}, \pm\sqrt{1}$.
Q738. $(x^2-5)(x^2-2) = 0 \implies x = \pm\sqrt{5}, \pm\sqrt{2}$.
Q739. $(x^2-2)(x^2-3) = 0 \implies x = \pm\sqrt{2}, \pm\sqrt{3}$.
Q740. $(x^2-3)(x^2-4) = 0 \implies x = \pm\sqrt{3}, \pm\sqrt{4}$.
Q741. $(x^2-5)(x^2-1) = 0 \implies x = \pm\sqrt{5}, \pm\sqrt{1}$.
Q742. $(x^2-2)(x^2-2) = 0 \implies x = \pm\sqrt{2}, \pm\sqrt{2}$.
Q743. $(x^2-3)(x^2-3) = 0 \implies x = \pm\sqrt{3}, \pm\sqrt{3}$.
Q744. $(x^2-5)(x^2-4) = 0 \implies x = \pm\sqrt{5}, \pm\sqrt{4}$.
Q745. $(x^2-2)(x^2-1) = 0 \implies x = \pm\sqrt{2}, \pm\sqrt{1}$.
Q746. $(x^2-3)(x^2-2) = 0 \implies x = \pm\sqrt{3}, \pm\sqrt{2}$.
Q747. $(x^2-5)(x^2-3) = 0 \implies x = \pm\sqrt{5}, \pm\sqrt{3}$.
Q748. $(x^2-2)(x^2-4) = 0 \implies x = \pm\sqrt{2}, \pm\sqrt{4}$.
Q749. $(x^2-3)(x^2-1) = 0 \implies x = \pm\sqrt{3}, \pm\sqrt{1}$.
Q750. $(x^2-5)(x^2-2) = 0 \implies x = \pm\sqrt{5}, \pm\sqrt{2}$.
Q751. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q752. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q753. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q754. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q755. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q756. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q757. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q758. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q759. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q760. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q761. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q762. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q763. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q764. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q765. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q766. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q767. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q768. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q769. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q770. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q771. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q772. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q773. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q774. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.
Q775. Dividing by x^2-2 gives $2x^2-3x+1=0 \implies x = 1, 1/2$. All zeroes: $\sqrt{2}, -\sqrt{2}, 1, 1/2$.

Q776. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
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Q779. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q780. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q781. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q782. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q783. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q784. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q785. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q786. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q787. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q788. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q789. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q790. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q791. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q792. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q793. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q794. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q795. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q796. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q797. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q798. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q799. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.
Q800. Irrational roots of polynomials with rational coefficients occur in conjugate pairs.

Pattern 5: Geometrical & Graphical Zeroes — Answer Key

Q801. It has 2 coincident (equal) real zeroes: $x = -3, -3$.
Q802. It has 2 coincident (equal) real zeroes: $x = -2, -2$.
Q803. It has 2 coincident (equal) real zeroes: $x = -1, -1$.
Q804. It has 2 coincident (equal) real zeroes: $x = 0, 0$.
Q805. It has 2 coincident (equal) real zeroes: $x = 1, 1$.
Q806. It has 2 coincident (equal) real zeroes: $x = 2, 2$.
Q807. It has 2 coincident (equal) real zeroes: $x = 3, 3$.
Q808. It has 2 coincident (equal) real zeroes: $x = -3, -3$.
Q809. It has 2 coincident (equal) real zeroes: $x = -2, -2$.
Q810. It has 2 coincident (equal) real zeroes: $x = -1, -1$.
Q811. It has 2 coincident (equal) real zeroes: $x = 0, 0$.
Q812. It has 2 coincident (equal) real zeroes: $x = 1, 1$.
Q813. It has 2 coincident (equal) real zeroes: $x = 2, 2$.
Q814. It has 2 coincident (equal) real zeroes: $x = 3, 3$.
Q815. It has 2 coincident (equal) real zeroes: $x = -3, -3$.
Q816. It has 2 coincident (equal) real zeroes: $x = -2, -2$.
Q817. It has 2 coincident (equal) real zeroes: $x = -1, -1$.
Q818. It has 2 coincident (equal) real zeroes: $x = 0, 0$.
Q819. It has 2 coincident (equal) real zeroes: $x = 1, 1$.
Q820. It has 2 coincident (equal) real zeroes: $x = 2, 2$.
Q821. It has 2 coincident (equal) real zeroes: $x = 3, 3$.
Q822. It has 2 coincident (equal) real zeroes: $x = -3, -3$.
Q823. It has 2 coincident (equal) real zeroes: $x = -2, -2$.
Q824. It has 2 coincident (equal) real zeroes: $x = -1, -1$.
Q825. It has 2 coincident (equal) real zeroes: $x = 0, 0$.
Q826. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q827. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q828. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q829. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q830. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q831. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q832. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q833. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q834. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q835. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q836. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q837. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q838. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q839. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q840. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q841. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q842. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q843. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q844. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q845. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q846. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.
Q847. Minimum degree $= 3$. Zeroes: $x = -2, 1, 4$.

Q848. Minimum degree = 3. Zeroes: $x = -2, 1, 4$.
Q849. Minimum degree = 3. Zeroes: $x = -2, 1, 4$.
Q850. Minimum degree = 3. Zeroes: $x = -2, 1, 4$.
Q851. Zero real zeroes; Discriminant $D < 0$.
Q852. Zero real zeroes; Discriminant $D < 0$.
Q853. Zero real zeroes; Discriminant $D < 0$.
Q854. Zero real zeroes; Discriminant $D < 0$.
Q855. Zero real zeroes; Discriminant $D < 0$.
Q856. Zero real zeroes; Discriminant $D < 0$.
Q857. Zero real zeroes; Discriminant $D < 0$.
Q858. Zero real zeroes; Discriminant $D < 0$.
Q859. Zero real zeroes; Discriminant $D < 0$.
Q860. Zero real zeroes; Discriminant $D < 0$.
Q861. Zero real zeroes; Discriminant $D < 0$.
Q862. Zero real zeroes; Discriminant $D < 0$.
Q863. Zero real zeroes; Discriminant $D < 0$.
Q864. Zero real zeroes; Discriminant $D < 0$.
Q865. Zero real zeroes; Discriminant $D < 0$.
Q866. Zero real zeroes; Discriminant $D < 0$.
Q867. Zero real zeroes; Discriminant $D < 0$.
Q868. Zero real zeroes; Discriminant $D < 0$.
Q869. Zero real zeroes; Discriminant $D < 0$.
Q870. Zero real zeroes; Discriminant $D < 0$.
Q871. Zero real zeroes; Discriminant $D < 0$.
Q872. Zero real zeroes; Discriminant $D < 0$.
Q873. Zero real zeroes; Discriminant $D < 0$.
Q874. Zero real zeroes; Discriminant $D < 0$.
Q875. Zero real zeroes; Discriminant $D < 0$.
Q876. Correct Option: (A) 3.
Q877. Correct Option: (A) 3.
Q878. Correct Option: (A) 3.
Q879. Correct Option: (A) 3.
Q880. Correct Option: (A) 3.
Q881. Correct Option: (A) 3.
Q882. Correct Option: (A) 3.
Q883. Correct Option: (A) 3.
Q884. Correct Option: (A) 3.
Q885. Correct Option: (A) 3.
Q886. Correct Option: (A) 3.
Q887. Correct Option: (A) 3.
Q888. Correct Option: (A) 3.
Q889. Correct Option: (A) 3.
Q890. Correct Option: (A) 3.
Q891. Correct Option: (A) 3.
Q892. Correct Option: (A) 3.
Q893. Correct Option: (A) 3.
Q894. Correct Option: (A) 3.
Q895. Correct Option: (A) 3.
Q896. Correct Option: (A) 3.
Q897. Correct Option: (A) 3.
Q898. Correct Option: (A) 3.
Q899. Correct Option: (A) 3.
Q900. Correct Option: (A) 3.
Q901. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q902. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q903. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q904. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q905. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q906. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q907. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q908. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q909. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q910. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q911. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q912. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q913. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q914. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q915. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q916. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q917. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q918. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q919. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q920. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q921. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q922. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q923. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q924. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q925. Distance from -1 to 2 is 3. Other zero is $2 + 3 = 5$.
Q926. $-x^2 + 4 = 0 \implies x = \pm 2$. Width $= 2 - (-2) = 4$ units.
Q927. $-x^2 + 4 = 0 \implies x = \pm 2$. Width $= 2 - (-2) = 4$ units.
Q928. $-x^2 + 4 = 0 \implies x = \pm 2$. Width $= 2 - (-2) = 4$ units.
Q929. $-x^2 + 4 = 0 \implies x = \pm 2$. Width $= 2 - (-2) = 4$ units.
Q930. $-x^2 + 4 = 0 \implies x = \pm 2$. Width $= 2 - (-2) = 4$ units.
Q931. $-x^2 + 4 = 0 \implies x = \pm 2$. Width $= 2 - (-2) = 4$ units.
Q932. $-x^2 + 4 = 0 \implies x = \pm 2$. Width $= 2 - (-2) = 4$ units.
Q933. $-x^2 + 4 = 0 \implies x = \pm 2$. Width $= 2 - (-2) = 4$ units.
Q934. $-x^2 + 4 = 0 \implies x = \pm 2$. Width $= 2 - (-2) = 4$ units.
Q935. $-x^2 + 4 = 0 \implies x = \pm 2$. Width $= 2 - (-2) = 4$ units.
Q936. $-x^2 + 4 = 0 \implies x = \pm 2$. Width $= 2 - (-2) = 4$ units.

Q937. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q938. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q939. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q940. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q941. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q942. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q943. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q944. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q945. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q946. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q947. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q948. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q949. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q950. $-x^2+4=0 \implies x=\pm 2$. Width $=2-(-2)=4$ units.

Q951. Zero real zeroes.

Q952. Zero real zeroes.

Q953. Zero real zeroes.

Q954. Zero real zeroes.

Q955. Zero real zeroes.

Q956. Zero real zeroes.

Q957. Zero real zeroes.

Q958. Zero real zeroes.

Q959. Zero real zeroes.

Q960. Zero real zeroes.

Q961. Zero real zeroes.

Q962. Zero real zeroes.

Q963. Zero real zeroes.

Q964. Zero real zeroes.

Q965. Zero real zeroes.

Q966. Zero real zeroes.

Q967. Zero real zeroes.

Q968. Zero real zeroes.

Q969. Zero real zeroes.

Q970. Zero real zeroes.

Q971. Zero real zeroes.

Q972. Zero real zeroes.

Q973. Zero real zeroes.

Q974. Zero real zeroes.

Q975. Zero real zeroes.

Q976. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q977. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q978. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q979. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q980. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q981. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q982. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q983. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q984. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q985. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q986. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q987. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q988. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q989. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q990. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q991. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q992. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q993. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q994. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q995. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q996. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q997. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q998. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q999. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Q1000. $x(x-2)(x+2)=0 \implies x=-2,0,2$. Total 3 zeroes.

Pattern 6: Interchanged Coefficients / Large System Values — Answer Key

Q1001. Adding: $(300)(x+y) = 600 \implies x+y = 2$. Subtracting gives $x-y=0$. Solution: $x=1, y=1$.

Q1002. Adding: $(302)(x+y) = 1208 \implies x+y = 4$. Subtracting gives $x-y=0$. Solution: $x=2, y=2$.

Q1003. Adding: $(304)(x+y) = 1824 \implies x+y = 6$. Subtracting gives $x-y=0$. Solution: $x=3, y=3$.

Q1004. Adding: $(306)(x+y) = 2448 \implies x+y = 8$. Subtracting gives $x-y=0$. Solution: $x=4, y=4$.

Q1005. Adding: $(308)(x+y) = 1848 \implies x+y = 6$. Subtracting gives $x-y=4$. Solution: $x=5, y=1$.

Q1006. Adding: $(310)(x+y) = 930 \implies x+y = 3$. Subtracting gives $x-y=-1$. Solution: $x=1, y=2$.

Q1007. Adding: $(312)(x+y) = 1560 \implies x+y = 5$. Subtracting gives $x-y=-1$. Solution: $x=2, y=3$.

Q1008. Adding: $(314)(x+y) = 2198 \implies x+y = 7$. Subtracting gives $x-y=-1$. Solution: $x=3, y=4$.

Q1009. Adding: $(316)(x+y) = 1580 \implies x+y = 5$. Subtracting gives $x-y=3$. Solution: $x=4, y=1$.

Q1010. Adding: $(318)(x+y) = 2226 \implies x+y = 7$. Subtracting gives $x-y=3$. Solution: $x=5, y=2$.

Q1011. Adding: $(320)(x+y) = 1280 \implies x+y = 4$. Subtracting gives $x-y=-2$. Solution: $x=1, y=3$.

Q1012. Adding: $(322)(x+y) = 1932 \implies x+y = 6$. Subtracting gives $x-y=-2$. Solution: $x=2, y=4$.

Q1013. Adding: $(324)(x+y) = 1296 \implies x+y = 4$. Subtracting gives $x-y=2$. Solution: $x=3, y=1$.

Q1014. Adding: $(326)(x+y) = 1956 \implies x+y = 6$. Subtracting gives $x-y=2$. Solution: $x=4, y=2$.

Q1015. Adding: $(328)(x+y) = 2624 \implies x+y = 8$. Subtracting gives $x-y=2$. Solution: $x=5, y=3$.

Q1016. Adding: $(330)(x+y) = 1650 \implies x+y = 5$. Subtracting gives $x-y=-3$. Solution: $x=1, y=4$.

Q1017. Adding: $(332)(x+y) = 996 \implies x+y = 3$. Subtracting gives $x-y=1$. Solution: $x=2, y=1$.

Q1018. Adding: $(334)(x+y) = 1670 \implies x+y = 5$. Subtracting gives $x-y=1$. Solution: $x=3, y=2$.

Q1019. Adding: $(336)(x+y) = 2352 \implies x+y = 7$. Subtracting gives $x-y=1$. Solution: $x=4, y=3$.

Q1020. Adding: $(338)(x+y) = 3042 \implies x+y = 9$. Subtracting gives $x-y=1$. Solution: $x=5, y=4$.

Q1021. Adding: $(340)(x+y) = 680 \implies x+y = 2$. Subtracting gives $x-y=0$. Solution: $x=1, y=1$.

Q1022. Adding: $(342)(x+y) = 1368 \implies x+y = 4$. Subtracting gives $x-y=0$. Solution: $x=2, y=2$.

Q1023. Adding: $(344)(x+y) = 2064 \implies x+y = 6$. Subtracting gives $x-y=0$. Solution: $x=3, y=3$.

Q1024. Adding: $(346)(x+y) = 2768 \implies x+y = 8$. Subtracting gives $x-y=0$. Solution: $x=4, y=4$.

Q1025. Adding: $(348)(x+y) = 2088 \implies x+y = 6$. Subtracting gives $x-y=4$. Solution: $x=5, y=1$.

Q1026. Add both equations: $(350)x + (350)y = 1050 \implies x+y = \frac{1050}{350} = 3$.

Q1027. Add both equations: $(352)x + (352)y = 1760 \implies x+y = \frac{1760}{352} = 5$.

Q1028. Add both equations: $(354)x + (354)y = 2478 \implies x+y = \frac{2478}{354} = 7$.

Q1029. Add both equations: $(356)x + (356)y = 1780 \implies x+y = \frac{1780}{356} = 5$.

Q1030. Add both equations: $(358)x + (358)y = 2506 \implies x+y = \frac{2506}{358} = 7$.

Q1031. Add both equations: $(360)x + (360)y = 1440 \implies x+y = \frac{1440}{360} = 4$.

Q1032. Add both equations: $(362)x + (362)y = 2172 \implies x+y = \frac{2172}{362} = 6$.

Q1033. Add both equations: $(364)x + (364)y = 1456 \implies x+y = \frac{1456}{364} = 4$.

Q1034. Add both equations: $(366)x + (366)y = 2196 \implies x+y = \frac{2196}{366} = 6$.

Q1035. Add both equations: $(368)x + (368)y = 2944 \implies x+y = \frac{2944}{368} = 8$.

Q1036. Add both equations: $(370)x + (370)y = 1850 \implies x+y = \frac{1850}{370} = 5$.

Q1037. Add both equations: $(372)x + (372)y = 1116 \implies x+y = \frac{1116}{372} = 3$.

Q1038. Add both equations: $(374)x + (374)y = 1870 \implies x+y = \frac{1870}{374} = 5$.

Q1039. Add both equations: $(376)x + (376)y = 2632 \implies x+y = \frac{2632}{376} = 7$.

Q1040. Add both equations: $(378)x + (378)y = 3402 \implies x+y = \frac{3402}{378} = 9$.

Q1041. Add both equations: $(380)x + (380)y = 760 \implies x+y = \frac{760}{380} = 2$.

Q1042. Add both equations: $(382)x + (382)y = 1528 \implies x+y = \frac{1528}{382} = 4$.

Q1043. Add both equations: $(384)x + (384)y = 2304 \implies x+y = \frac{2304}{384} = 6$.

Q1044. Add both equations: $(386)x + (386)y = 3088 \implies x+y = \frac{3088}{386} = 8$.

Q1045. Add both equations: $(388)x + (388)y = 2328 \implies x+y = \frac{2328}{388} = 6$.

Q1046. Add both equations: $(390)x + (390)y = 1170 \implies x+y = \frac{1170}{390} = 3$.

Q1047. Add both equations: $(392)x + (392)y = 1960 \implies x+y = \frac{1960}{392} = 5$.

Q1048. Add both equations: $(394)x + (394)y = 2758 \implies x+y = \frac{2758}{394} = 7$.

Q1049. Add both equations: $(396)x + (396)y = 1980 \implies x+y = \frac{1980}{396} = 5$.

Q1050. Add both equations: $(398)x + (398)y = 2786 \implies x+y = \frac{2786}{398} = 7$.

Q1051. Subtract equations: $(-150)x - (-150)y = 300 \implies x-y = \frac{300}{-150} = -2$.

Q1052. Subtract equations: $(-150)x - (-150)y = 300 \implies x-y = \frac{300}{-150} = -2$.

Q1053. Subtract equations: $(-150)x - (-150)y = -300 \implies x - y = \frac{-300}{-150} = 2$.
Q1054. Subtract equations: $(-150)x - (-150)y = -300 \implies x - y = \frac{-300}{-150} = 2$.
Q1055. Subtract equations: $(-150)x - (-150)y = -300 \implies x - y = \frac{-300}{-150} = 2$.
Q1056. Subtract equations: $(-150)x-(-150)y = 450 \implies x - y = \frac{450}{-150} = -3$.
Q1057. Subtract equations: $(-150)x - (-150)y = -150 \implies x - y = \frac{-150}{-150} = 1$.
Q1058. Subtract equations: $(-150)x - (-150)y = -150 \implies x - y = \frac{-150}{-150} = 1$.
Q1059. Subtract equations: $(-150)x - (-150)y = -150 \implies x - y = \frac{-150}{-150} = 1$.
Q1060. Subtract equations: $(-150)x - (-150)y = -150 \implies x - y = \frac{-150}{-150} = 1$.
Q1061. Subtract equations: $(-90)x - (-90)y = 0 \implies x - y = \frac{0}{-90} = 0$.
Q1062. Subtract equations: $(-90)x - (-90)y = 0 \implies x - y = \frac{0}{-90} = 0$.
Q1063. Subtract equations: $(-90)x - (-90)y = 0 \implies x - y = \frac{0}{-90} = 0$.
Q1064. Subtract equations: $(-90)x - (-90)y = 0 \implies x - y = \frac{0}{-90} = 0$.
Q1065. Subtract equations: $(-90)x-(-90)y = -360 \implies x - y = \frac{-360}{-90} = 4$.
Q1066. Subtract equations: $(-90)x - (-90)y = 90 \implies x - y = \frac{90}{-90} = -1$.
Q1067. Subtract equations: $(-90)x - (-90)y = 90 \implies x - y = \frac{90}{-90} = -1$.
Q1068. Subtract equations: $(-90)x - (-90)y = 90 \implies x - y = \frac{90}{-90} = -1$.
Q1069. Subtract equations: $(-90)x-(-90)y = -270 \implies x - y = \frac{-270}{-90} = 3$.
Q1070. Subtract equations: $(-90)x-(-90)y = -270 \implies x - y = \frac{-270}{-90} = 3$.
Q1071. Subtract equations: $(-90)x - (-90)y = 180 \implies x - y = \frac{180}{-90} = -2$.
Q1072. Subtract equations: $(-90)x - (-90)y = 180 \implies x - y = \frac{180}{-90} = -2$.
Q1073. Subtract equations: $(-90)x-(-90)y = -180 \implies x - y = \frac{-180}{-90} = 2$.
Q1074. Subtract equations: $(-90)x-(-90)y = -180 \implies x - y = \frac{-180}{-90} = 2$.
Q1075. Subtract equations: $(-90)x-(-90)y = -180 \implies x - y = \frac{-180}{-90} = 2$.
Q1076. Correct Option: (A) $x = 2$.
Q1077. Correct Option: (A) $x = 2$.
Q1078. Correct Option: (A) $x = 2$.
Q1079. Correct Option: (A) $x = 2$.
Q1080. Correct Option: (A) $x = 2$.
Q1081. Correct Option: (A) $x = 2$.
Q1082. Correct Option: (A) $x = 2$.
Q1083. Correct Option: (A) $x = 2$.
Q1084. Correct Option: (A) $x = 2$.
Q1085. Correct Option: (A) $x = 2$.
Q1086. Correct Option: (A) $x = 2$.
Q1087. Correct Option: (A) $x = 2$.
Q1088. Correct Option: (A) $x = 2$.
Q1089. Correct Option: (A) $x = 2$.
Q1090. Correct Option: (A) $x = 2$.
Q1091. Correct Option: (A) $x = 2$.
Q1092. Correct Option: (A) $x = 2$.
Q1093. Correct Option: (A) $x = 2$.
Q1094. Correct Option: (A) $x = 2$.
Q1095. Correct Option: (A) $x = 2$.
Q1096. Correct Option: (A) $x = 2$.
Q1097. Correct Option: (A) $x = 2$.
Q1098. Correct Option: (A) $x = 2$.
Q1099. Correct Option: (A) $x = 2$.
Q1100. Correct Option: (A) $x = 2$.
Q1101. $100(1) + 240(1) = 340$ and $240(1) + 100(1) = 340$. Verified.
Q1102. $101(2) + 241(2) = 684$ and $241(2) + 101(2) = 684$. Verified.
Q1103. $102(3)+242(3) = 1032$ and $242(3)+102(3) = 1032$. Verified.
Q1104. $103(4)+243(4) = 1384$ and $243(4)+103(4) = 1384$. Verified.
Q1105. $104(5)+244(1) = 764$ and $244(5)+104(1) = 1324$. Verified.
Q1106. $105(1) + 245(2) = 595$ and $245(1) + 105(2) = 455$. Verified.
Q1107. $106(2) + 246(3) = 950$ and $246(2) + 106(3) = 810$. Verified.
Q1108. $107(3)+247(4) = 1309$ and $247(3)+107(4) = 1169$. Verified.
Q1109. $108(4)+248(1) = 680$ and $248(4)+108(1) = 1100$. Verified.
Q1110. $109(5)+249(2) = 1043$ and $249(5)+109(2) = 1463$. Verified.
Q1111. $110(1) + 250(3) = 860$ and $250(1) + 110(3) = 580$. Verified.
Q1112. $111(2)+251(4) = 1226$ and $251(2)+111(4) = 946$. Verified.
Q1113. $112(3) + 252(1) = 588$ and $252(3) + 112(1) = 868$. Verified.
Q1114. $113(4)+253(2) = 958$ and $253(4)+113(2) = 1238$. Verified.
Q1115. $114(5)+254(3) = 1332$ and $254(5)+114(3) = 1612$. Verified.
Q1116. $115(1)+255(4) = 1135$ and $255(1)+115(4) = 715$. Verified.
Q1117. $116(2) + 256(1) = 488$ and $256(2) + 116(1) = 628$. Verified.

Q1118. $117(3)+257(2)=865$ and $257(3)+117(2)=1005$. Verified.

Q1119. $118(4)+258(3)=1246$ and $258(4)+118(3)=1386$. Verified.

Q1120. $119(5)+259(4)=1631$ and $259(5)+119(4)=1771$. Verified.

Q1121. $120(1)+200(1)=320$ and $200(1)+120(1)=320$. Verified.

Q1122. $121(2)+201(2)=644$ and $201(2)+121(2)=644$. Verified.

Q1123. $122(3)+202(3)=972$ and $202(3)+122(3)=972$. Verified.

Q1124. $123(4)+203(4)=1304$ and $203(4)+123(4)=1304$. Verified.

Q1125. $124(5)+204(1)=824$ and $204(5)+124(1)=1144$. Verified.

Q1126. $x=1, y=2$.

Q1127. $x=2, y=3$.

Q1128. $x=3, y=4$.

Q1129. $x=4, y=1$.

Q1130. $x=5, y=2$.

Q1131. $x=1, y=3$.

Q1132. $x=2, y=4$.

Q1133. $x=3, y=1$.

Q1134. $x=4, y=2$.

Q1135. $x=5, y=3$.

Q1136. $x=1, y=4$.

Q1137. $x=2, y=1$.

Q1138. $x=3, y=2$.

Q1139. $x=4, y=3$.

Q1140. $x=5, y=4$.

Q1141. $x=1, y=1$.

Q1142. $x=2, y=2$.

Q1143. $x=3, y=3$.

Q1144. $x=4, y=4$.

Q1145. $x=5, y=1$.

Q1146. $x=1, y=2$.

Q1147. $x=2, y=3$.

Q1148. $x=3, y=4$.

Q1149. $x=4, y=1$.

Q1150. $x=5, y=2$.

Q1151. $y=\frac{790-100x}{230}$.

Q1152. $y=\frac{1126-101x}{231}$.

Q1153. $y=\frac{538-102x}{232}$.

Q1154. $y=\frac{878-103x}{233}$.

Q1155. $y=\frac{1222-104x}{234}$.

Q1156. $y=\frac{1045-105x}{235}$.

Q1157. $y=\frac{448-106x}{236}$.

Q1158. $y=\frac{795-107x}{237}$.

Q1159. $y=\frac{1146-108x}{238}$.

Q1160. $y=\frac{1501-109x}{239}$.

Q1161. $y=\frac{350-110x}{240}$.

Q1162. $y=\frac{704-111x}{241}$.

Q1163. $y=\frac{1062-112x}{242}$.

Q1164. $y=\frac{1424-113x}{243}$.

Q1165. $y=\frac{814-114x}{244}$.

Q1166. $y=\frac{605-115x}{245}$.

Q1167. $y=\frac{970-116x}{246}$.

Q1168. $y=\frac{1339-117x}{247}$.

Q1169. $y=\frac{720-118x}{248}$.

Q1170. $y=\frac{1093-119x}{249}$.

Q1171. $y=\frac{870-120x}{250}$.

Q1172. $y=\frac{1246-121x}{251}$.

Q1173. $y=\frac{618-122x}{252}$.

Q1174. $y=\frac{998-123x}{253}$.

Q1175. $y=\frac{1382-124x}{254}$.

Q1176. $x^2-y^2=(x+y)(x-y)=(5)(-3)=-15$.

Q1177. $x^2-y^2=(x+y)(x-y)=(3)(1)=3$.

Q1178. $x^2-y^2=(x+y)(x-y)=(5)(1)=5$.

Q1179. $x^2-y^2=(x+y)(x-y)=(7)(1)=7$.

Q1180. $x^2-y^2=(x+y)(x-y)=(9)(1)=9$.

Q1181. $x^2-y^2=(x+y)(x-y)=(2)(0)=0$.

Q1182. $x^2-y^2=(x+y)(x-y)=(4)(0)=0$.

Q1183. $x^2-y^2=(x+y)(x-y)=(6)(0)=0$.

Q1184. $x^2-y^2=(x+y)(x-y)=(8)(0)=0$.

Q1185. $x^2-y^2=(x+y)(x-y)=(6)(4)=24$.

Q1186. $x^2-y^2=(x+y)(x-y)=(3)(-1)=-3$.

Q1187. $x^2-y^2=(x+y)(x-y)=(5)(-1)=-5$.

Q1188. $x^2-y^2=(x+y)(x-y)=(7)(-1)=-7$.

Q1189. $x^2-y^2=(x+y)(x-y)=(5)(3)=15$.

Q1190. $x^2-y^2=(x+y)(x-y)=(7)(3)=21$.

Q1191. $x^2-y^2=(x+y)(x-y)=(4)(-2)=-8$.

Q1192. $x^2-y^2=(x+y)(x-y)=(6)(-2)=-12$.

Q1193. $x^2-y^2=(x+y)(x-y)=(4)(2)=8$.

Q1194. $x^2-y^2=(x+y)(x-y)=(6)(2)=12$.

Q1195. $x^2-y^2=(x+y)(x-y)=(8)(2)=16$.

Q1196. $x^2-y^2=(x+y)(x-y)=(5)(-3)=-15$.

Q1197. $x^2-y^2=(x+y)(x-y)=(3)(1)=3$.

Q1198. $x^2-y^2=(x+y)(x-y)=(5)(1)=5$.

Q1199. $x^2-y^2=(x+y)(x-y)=(7)(1)=7$.

Q1200. $x^2-y^2=(x+y)(x-y)=(9)(1)=9$.

Pattern 7: Fixed Charges & Variable Rate Constraints — Answer Key

Q1201. Equations: $x+3y=26$ and $x+7y=34$. Subtracting gives $4y=8 \implies y=2$. Fixed charge $x=20$.

Q1202. Equations: $x+4y=37$ and $x+8y=49$. Subtracting gives $4y=12 \implies y=3$. Fixed charge

$x = 25$.

Q1203. Equations: $x + 5y = 50$ and $x + 9y = 66$. Subtracting gives $4y = 16 \implies y = 4$. Fixed charge $x = 30$.

Q1204. Equations: $x + 6y = 65$ and $x + 10y = 85$. Subtracting gives $4y = 20 \implies y = 5$. Fixed charge $x = 35$.

Q1205. Equations: $x + 3y = 58$ and $x + 11y = 106$. Subtracting gives $8y = 48 \implies y = 6$. Fixed charge $x = 40$.

Q1206. Equations: $x + 4y = 53$ and $x + 7y = 59$. Subtracting gives $3y = 6 \implies y = 2$. Fixed charge $x = 45$.

Q1207. Equations: $x + 5y = 65$ and $x + 8y = 74$. Subtracting gives $3y = 9 \implies y = 3$. Fixed charge $x = 50$.

Q1208. Equations: $x + 6y = 79$ and $x + 9y = 91$. Subtracting gives $3y = 12 \implies y = 4$. Fixed charge $x = 55$.

Q1209. Equations: $x + 3y = 75$ and $x + 10y = 110$. Subtracting gives $7y = 35 \implies y = 5$. Fixed charge $x = 60$.

Q1210. Equations: $x + 4y = 89$ and $x + 11y = 131$. Subtracting gives $7y = 42 \implies y = 6$. Fixed charge $x = 65$.

Q1211. Equations: $x + 5y = 30$ and $x + 7y = 34$. Subtracting gives $2y = 4 \implies y = 2$. Fixed charge $x = 20$.

Q1212. Equations: $x + 6y = 43$ and $x + 8y = 49$. Subtracting gives $2y = 6 \implies y = 3$. Fixed charge $x = 25$.

Q1213. Equations: $x + 3y = 42$ and $x + 9y = 66$. Subtracting gives $6y = 24 \implies y = 4$. Fixed charge $x = 30$.

Q1214. Equations: $x + 4y = 55$ and $x + 10y = 85$. Subtracting gives $6y = 30 \implies y = 5$. Fixed charge $x = 35$.

Q1215. Equations: $x + 5y = 70$ and $x + 11y = 106$. Subtracting gives $6y = 36 \implies y = 6$. Fixed charge $x = 40$.

Q1216. Equations: $x + 6y = 57$ and $x + 7y = 59$. Subtracting gives $1y = 2 \implies y = 2$. Fixed charge $x = 45$.

Q1217. Equations: $x + 3y = 59$ and $x + 8y = 74$. Subtracting gives $5y = 15 \implies y = 3$. Fixed charge $x = 50$.

Q1218. Equations: $x + 4y = 71$ and $x + 9y = 91$. Subtracting gives $5y = 20 \implies y = 4$. Fixed charge $x = 55$.

Q1219. Equations: $x + 5y = 85$ and $x + 10y = 110$. Subtracting gives $5y = 25 \implies y = 5$. Fixed charge $x = 60$.

Q1220. Equations: $x + 6y = 101$ and $x + 11y = 131$. Subtracting gives $5y = 30 \implies y = 6$. Fixed charge $x = 65$.

Q1221. Equations: $x + 3y = 26$ and $x + 7y = 34$. Subtracting gives $4y = 8 \implies y = 2$. Fixed charge $x = 20$.

Q1222. Equations: $x + 4y = 37$ and $x + 8y = 49$. Subtracting gives $4y = 12 \implies y = 3$. Fixed charge $x = 25$.

Q1223. Equations: $x + 5y = 50$ and $x + 9y = 66$. Subtracting gives $4y = 16 \implies y = 4$. Fixed charge $x = 30$.

Q1224. Equations: $x + 6y = 65$ and $x + 10y = 85$. Subtracting gives $4y = 20 \implies y = 5$. Fixed charge $x = 35$.

Q1225. Equations: $x + 3y = 58$ and $x + 11y = 106$. Subtracting gives $8y = 48 \implies y = 6$. Fixed charge $x = 40$.

Q1226. $x + 4y = 53$ and $x + 7y = 59$.

Q1227. $x + 5y = 65$ and $x + 8y = 74$.

Q1228. $x + 6y = 79$ and $x + 9y = 91$.

Q1229. $x + 3y = 75$ and $x + 10y = 110$.

Q1230. $x + 4y = 89$ and $x + 11y = 131$.

Q1231. $x + 5y = 30$ and $x + 7y = 34$.

Q1232. $x + 6y = 43$ and $x + 8y = 49$.

Q1233. $x + 3y = 42$ and $x + 9y = 66$.

Q1234. $x + 4y = 55$ and $x + 10y = 85$.

Q1235. $x + 5y = 70$ and $x + 11y = 106$.

Q1236. $x + 6y = 57$ and $x + 7y = 59$.

Q1237. $x + 3y = 59$ and $x + 8y = 74$.

Q1238. $x + 4y = 71$ and $x + 9y = 91$.

Q1239. $x + 5y = 85$ and $x + 10y = 110$.

Q1240. $x + 6y = 101$ and $x + 11y = 131$.

Q1241. $x + 3y = 26$ and $x + 7y = 34$.

Q1242. $x + 4y = 37$ and $x + 8y = 49$.

Q1243. $x + 5y = 50$ and $x + 9y = 66$.

Q1244. $x + 6y = 65$ and $x + 10y = 85$.

Q1245. $x + 3y = 58$ and $x + 11y = 106$.

Q1246. $x + 4y = 53$ and $x + 7y = 59$.

Q1247. $x + 5y = 65$ and $x + 8y = 74$.

Q1248. $x + 6y = 79$ and $x + 9y = 91$.

Q1249. $x + 3y = 75$ and $x + 10y = 110$.

Q1250. $x + 4y = 89$ and $x + 11y = 131$.

Q1251. $x + 5y = 30$.

Q1252. $x + 6y = 43$.

Q1253. $x + 3y = 42$.

Q1254. $x + 4y = 55$.

Q1255. $x + 5y = 70$.

Q1256. $x + 6y = 57$.

Q1257. $x + 3y = 59$.

Q1258. $x + 4y = 71$.

Q1259. $x + 5y = 85$.

Q1260. $x + 6y = 101$.

Q1261. $x + 3y = 26$.

Q1262. $x + 4y = 37$.

Q1263. $x + 5y = 50$.

Q1264. $x + 6y = 65$.

Q1265. $x + 3y = 58$.

Q1266. $x + 4y = 53$.

Q1267. $x + 5y = 65$.

Q1268. $x + 6y = 79$.

Q1269. $x + 3y = 75$.

Q1270. $x + 4y = 89$.

Q1271. $x + 5y = 30$.

Q1272. $x + 6y = 43$.

Q1273. $x + 3y = 42$.

Q1274. $x + 4y = 55$.

Q1275. $x + 5y = 70$.

Q1276. Correct Option: (A) Rs. 75.

Q1277. Correct Option: (A) Rs. 95.

Q1278. Correct Option: (A) Rs. 115.

Q1279. Correct Option: (A) Rs. 135.

Q1280. Correct Option: (A) Rs. 155.

Q1281. Correct Option: (A) Rs. 50.

Q1282. Correct Option: (A) Rs. 70.

Q1283. Correct Option: (A) Rs. 90.

Q1284. Correct Option: (A) Rs. 110.

Q1285. Correct Option: (A) Rs. 130.

Q1286. Correct Option: (A) Rs. 75.

Q1287. Correct Option: (A) Rs. 95.

Q1288. Correct Option: (A) Rs. 115.

Q1289. Correct Option: (A) Rs. 135.

Q1290. Correct Option: (A) Rs. 155.

Q1291. Correct Option: (A) Rs. 50.

Q1292. Correct Option: (A) Rs. 70.

Q1293. Correct Option: (A) Rs. 90.

Q1294. Correct Option: (A) Rs. 110.

Q1295. Correct Option: (A) Rs. 130.

Q1296. Correct Option: (A) Rs. 75.

Q1297. Correct Option: (A) Rs. 95.

Q1298. Correct Option: (A) Rs. 115.

Q1299. Correct Option: (A) Rs. 135.

Q1300. Correct Option: (A) Rs. 155.

Q1301. New rate $= 2 \times 2 = 4$. Total cost $= 20 + 3 \times 4 = 32$.

Q1302. New rate $= 2 \times 3 = 6$. Total cost $= 25 + 4 \times 6 = 49$.

Q1303. New rate $= 2 \times 4 = 8$. Total cost $= 30 + 5 \times 8 = 70$.

Q1304. New rate $= 2 \times 5 = 10$. Total cost $= 35 + 6 \times 10 = 95$.

Q1305. New rate $= 2 \times 6 = 12$. Total cost $= 40 + 3 \times 12 = 76$.

Q1306. New rate $= 2 \times 2 = 4$. Total cost $= 45 + 4 \times 4 = 61$.

Q1307. New rate $= 2 \times 3 = 6$. Total cost $= 50 + 5 \times 6 = 80$.

Q1308. New rate $= 2 \times 4 = 8$. Total cost $= 55 + 6 \times 8 = 103$.

Q1309. New rate $= 2 \times 5 = 10$. Total cost $= 60 + 3 \times 10 = 90$.

Q1310. New rate $= 2 \times 6 = 12$. Total cost $= 65 + 4 \times 12 = 113$.

Q1311. New rate $= 2 \times 2 = 4$. Total cost $= 20 + 5 \times 4 = 40$.

Q1312. New rate $= 2 \times 3 = 6$. Total cost $= 25 + 6 \times 6 = 61$.

Q1313. New rate $= 2 \times 4 = 8$. Total cost $= 30 + 3 \times 8 = 54$.

Q1314. New rate $= 2 \times 5 = 10$. Total cost $= 35 + 4 \times 10 = 75$.

Q1315. New rate $= 2 \times 6 = 12$. Total cost $= 40 + 5 \times 12 = 100$.

Q1316. New rate $= 2 \times 2 = 4$. Total cost $= 45 + 6 \times 4 = 69$.

Q1317. New rate $= 2 \times 3 = 6$. Total cost $= 50 + 3 \times 6 = 68$.

Q1318. New rate $= 2 \times 4 = 8$. Total cost $= 55 + 4 \times 8 = 87$.

Q1319. New rate $= 2 \times 5 = 10$. Total cost $= 60 + 5 \times 10 = 110$.

Q1320. New rate $= 2 \times 6 = 12$. Total cost $= 65 + 6 \times 12 = 137$.

Q1321. New rate $= 2 \times 2 = 4$. Total cost $= 20 + 3 \times 4 = 32$.

Q1322. New rate $= 2 \times 3 = 6$. Total cost $= 25 + 4 \times 6 = 49$.

Q1323. New rate $= 2 \times 4 = 8$. Total cost $= 30 + 5 \times 8 = 70$.

Q1324. New rate $= 2 \times 5 = 10$. Total cost $= 35 + 6 \times 10 = 95$.

Q1325. New rate $= 2 \times 6 = 12$. Total cost $= 40 + 3 \times 12 = 76$.

Q1326. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1327. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1328. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1329. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1330. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1331. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1332. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1333. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1334. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1335. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1336. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1337. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1338. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1339. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1340. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1341. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1342. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1343. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1344. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1345. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1346. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1347. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1348. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1349. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1350. Let x be fixed room fee, y be dining fee per person: $x + n_1 y = C_1, x + n_2 y = C_2$.

Q1351. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1352. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1353. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1354. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1355. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1356. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1357. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1358. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1359. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1360. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1361. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1362. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1363. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1364. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1365. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1366. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1367. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1368. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1369. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1370. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1371. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1372. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1373. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1374. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1375. A negative variable rate would mean the service provider pays the customer for additional distance/days.

Q1376. Total Charge $= 45 + 2D$.

Q1377. Total Charge $= 50 + 3D$.

Q1378. Total Charge $= 55 + 4D$.

Q1379. Total Charge $= 60 + 5D$.

Q1380. Total Charge $= 65 + 6D$.

Q1381. Total Charge $= 20 + 2D$.

Q1382. Total Charge $= 25 + 3D$.

Q1383. Total Charge $= 30 + 4D$.

Q1384. Total Charge $= 35 + 5D$.

Q1385. Total Charge $= 40 + 6D$.

Q1386. Total Charge $= 45 + 2D$.

Q1387. Total Charge $= 50 + 3D$.

Q1388. Total Charge $= 55 + 4D$.

Q1389. Total Charge $= 60 + 5D$.

Q1390. Total Charge $= 65 + 6D$.

Q1391. Total Charge $= 20 + 2D$.

Q1392. Total Charge $= 25 + 3D$.

Q1393. Total Charge $= 30 + 4D$.

Q1394. Total Charge $= 35 + 5D$.

Q1395. Total Charge $= 40 + 6D$.

Q1396. Total Charge $= 45 + 2D$.

Q1397. Total Charge $= 50 + 3D$.

Q1398. Total Charge $= 55 + 4D$.

Q1399. Total Charge $= 60 + 5D$.

Q1400. Total Charge $= 65 + 6D$.

Pattern 8: Time & Work Reciprocal Rate Modeling — Answer Key

Q1401. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $2u + 5v = 1/4$ and $(3)u + (4)v = 1/3$.

Q1402. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $3u + 6v = 1/5$ and $(4)u + (5)v = 1/4$.

Q1403. Let 1 man take x days ($u = 1/x$) and 1 woman

take y days ($v = 1/y$). Solve system $4u + 7v = 1/6$ and $(5)u + (6)v = 1/3$.

Q1404. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $2u + 8v = 1/4$ and $(3)u + (7)v = 1/4$.

Q1405. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $3u + 5v = 1/5$ and $(4)u + (4)v = 1/3$.

Q1406. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $4u + 6v = 1/6$ and $(5)u + (5)v = 1/4$.

Q1407. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $2u + 7v = 1/4$ and $(3)u + (6)v = 1/3$.

Q1408. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $3u + 8v = 1/5$ and $(4)u + (7)v = 1/4$.

Q1409. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $4u + 5v = 1/6$ and $(5)u + (4)v = 1/3$.

Q1410. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $2u + 6v = 1/4$ and $(3)u + (5)v = 1/4$.

Q1411. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $3u + 7v = 1/5$ and $(4)u + (6)v = 1/3$.

Q1412. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $4u + 8v = 1/6$ and $(5)u + (7)v = 1/4$.

Q1413. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $2u + 5v = 1/4$ and $(3)u + (4)v = 1/3$.

Q1414. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $3u + 6v = 1/5$ and $(4)u + (5)v = 1/4$.

Q1415. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $4u + 7v = 1/6$ and $(5)u + (6)v = 1/3$.

Q1416. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $2u + 8v = 1/4$ and $(3)u + (7)v = 1/4$.

Q1417. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $3u + 5v = 1/5$ and $(4)u + (4)v = 1/3$.

Q1418. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $4u + 6v = 1/6$ and $(5)u + (5)v = 1/4$.

Q1419. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $2u + 7v = 1/4$ and $(3)u + (6)v = 1/3$.

Q1420. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $3u + 8v = 1/5$ and $(4)u + (7)v = 1/4$.

Q1421. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $4u + 5v = 1/6$ and $(5)u + (4)v = 1/3$.

Q1422. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $2u + 6v = 1/4$ and $(3)u + (5)v = 1/4$.

Q1423. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $3u + 7v = 1/5$ and $(4)u + (6)v = 1/3$.

Q1424. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $4u + 8v = 1/6$ and $(5)u + (7)v = 1/4$.

Q1425. Let 1 man take x days ($u = 1/x$) and 1 woman take y days ($v = 1/y$). Solve system $2u + 5v = 1/4$ and $(3)u + (4)v = 1/3$.

Q1426. $3u + 6v = \frac{1}{5}$.

Q1427. $4u + 7v = \frac{1}{6}$.

Q1428. $2u + 8v = \frac{1}{4}$.

Q1429. $3u + 5v = \frac{1}{5}$.

Q1430. $4u + 6v = \frac{1}{6}$.

Q1431. $2u + 7v = \frac{1}{4}$.

Q1432. $3u + 8v = \frac{1}{5}$.

Q1433. $4u + 5v = \frac{1}{6}$.

Q1434. $2u + 6v = \frac{1}{4}$.

Q1435. $3u + 7v = \frac{1}{5}$.

Q1436. $4u + 8v = \frac{1}{6}$.

Q1437. $2u + 5v = \frac{1}{4}$.

Q1438. $3u + 6v = \frac{1}{5}$.

Q1439. $4u + 7v = \frac{1}{6}$.

Q1440. $2u + 8v = \frac{1}{4}$.

Q1441. $3u + 5v = \frac{1}{5}$.

Q1442. $4u + 6v = \frac{1}{6}$.

Q1443. $2u + 7v = \frac{1}{4}$.

Q1444. $3u + 8v = \frac{1}{5}$.

Q1445. $4u + 5v = \frac{1}{6}$.

Q1446. $2u + 6v = \frac{1}{4}$.

Q1447. $3u + 7v = \frac{1}{5}$.

Q1448. $4u + 8v = \frac{1}{6}$.

Q1449. $2u + 5v = \frac{1}{4}$.

Q1450. $3u + 6v = \frac{1}{5}$.

Q1451. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.

Q1452. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.

Q1453. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.

Q1454. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.

Q1455. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.

Q1456. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.

Q1457. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.

Q1458. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.

Q1459. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.

Q1460. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.

Q1461. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.

Q1462. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.

Q1463. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.
Q1464. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.
Q1465. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.
Q1466. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.
Q1467. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.
Q1468. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.
Q1469. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.
Q1470. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.
Q1471. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.
Q1472. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.
Q1473. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.
Q1474. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.
Q1475. Work in 1 day $= \frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$.
Q1476. Correct Option: (A) 1/20.
Q1477. Correct Option: (A) 1/20.
Q1478. Correct Option: (A) 1/20.
Q1479. Correct Option: (A) 1/20.
Q1480. Correct Option: (A) 1/20.
Q1481. Correct Option: (A) 1/20.
Q1482. Correct Option: (A) 1/20.
Q1483. Correct Option: (A) 1/20.
Q1484. Correct Option: (A) 1/20.
Q1485. Correct Option: (A) 1/20.
Q1486. Correct Option: (A) 1/20.
Q1487. Correct Option: (A) 1/20.
Q1488. Correct Option: (A) 1/20.
Q1489. Correct Option: (A) 1/20.
Q1490. Correct Option: (A) 1/20.
Q1491. Correct Option: (A) 1/20.
Q1492. Correct Option: (A) 1/20.
Q1493. Correct Option: (A) 1/20.
Q1494. Correct Option: (A) 1/20.
Q1495. Correct Option: (A) 1/20.
Q1496. Correct Option: (A) 1/20.
Q1497. Correct Option: (A) 1/20.
Q1498. Correct Option: (A) 1/20.
Q1499. Correct Option: (A) 1/20.
Q1500. Correct Option: (A) 1/20.
Q1501. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1502. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1503. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1504. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1505. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1506. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1507. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1508. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1509. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1510. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1511. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1512. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1513. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1514. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1515. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1516. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1517. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1518. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1519. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1520. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1521. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1522. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1523. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1524. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1525. $2(1/18) + 5(1/36) = 1/9 + 5/36 = 9/36 = 1/4$. Verified.
Q1526. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.
Q1527. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.
Q1528. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.
Q1529. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.
Q1530. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.
Q1531. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.
Q1532. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.
Q1533. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.
Q1534. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.
Q1535. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.
Q1536. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.
Q1537. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1538. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1539. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1540. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1541. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1542. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1543. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1544. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1545. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1546. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1547. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1548. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1549. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1550. $\frac{1}{T_{auto}} + \frac{1}{T_{manual}} = \frac{1}{T_{total}}$.

Q1551. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1552. Time required does not add linearly, but work completed per unit time (work rate) is additive.

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Q1558. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1559. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1560. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1561. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1562. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1563. Time required does not add linearly, but work completed per unit time (work rate) is additive.

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Q1565. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1566. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1567. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1568. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1569. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1570. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1571. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1572. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1573. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1574. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1575. Time required does not add linearly, but work completed per unit time (work rate) is additive.

Q1576. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1577. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1578. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1579. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1580. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1581. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1582. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1583. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1584. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1585. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1586. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1587. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1588. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1589. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1590. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1591. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1592. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1593. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1594. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1595. Combined daily rate $= 3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1596. Combined daily rate = $3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1597. Combined daily rate = $3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1598. Combined daily rate = $3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1599. Combined daily rate = $3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Q1600. Combined daily rate = $3(1/18) + 6(1/36) = 1/6 + 1/6 = 1/3 \implies 3$ days.

Pattern 9: Two-Digit Numbers & Digit Reversal Patterns — Answer Key

Q1601. Let digits be x and y. Equations: $x + y = 3$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 2, y = 1$. Number is 21.

Q1602. Let digits be x and y. Equations: $x + y = 5$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 3, y = 2$. Number is 32.

Q1603. Let digits be x and y. Equations: $x + y = 7$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 4, y = 3$. Number is 43.

Q1604. Let digits be x and y. Equations: $x + y = 9$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 5, y = 4$. Number is 54.

Q1605. Let digits be x and y. Equations: $x + y = 11$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 6, y = 5$. Number is 65.

Q1606. Let digits be x and y. Equations: $x + y = 13$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 7, y = 6$. Number is 76.

Q1607. Let digits be x and y. Equations: $x + y = 9$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 8, y = 1$. Number is 81.

Q1608. Let digits be x and y. Equations: $x + y = 5$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 3, y = 2$. Number is 32.

Q1609. Let digits be x and y. Equations: $x + y = 7$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 4, y = 3$. Number is 43.

Q1610. Let digits be x and y. Equations: $x + y = 9$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 5, y = 4$. Number is 54.

Q1611. Let digits be x and y. Equations: $x + y = 11$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 6, y = 5$. Number is 65.

Q1612. Let digits be x and y. Equations: $x + y = 13$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 7, y = 6$. Number is 76.

Q1613. Let digits be x and y. Equations: $x + y = 8$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 7, y = 1$. Number is 71.

Q1614. Let digits be x and y. Equations: $x + y = 10$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 8, y = 2$. Number is 82.

Q1615. Let digits be x and y. Equations: $x + y = 5$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 2, y = 3$. Number is 23.

Q1616. Let digits be x and y. Equations: $x + y = 7$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 3, y = 4$. Number is 34.

Q1617. Let digits be x and y. Equations: $x + y = 9$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 4, y = 5$. Number is 45.

Q1618. Let digits be x and y. Equations: $x + y = 11$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 5, y = 6$. Number is 56.

Q1619. Let digits be x and y. Equations: $x + y = 7$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 6, y = 1$. Number is 61.

Q1620. Let digits be x and y. Equations: $x + y = 9$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 7, y = 2$. Number is 72.

Q1621. Let digits be x and y. Equations: $x + y = 11$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 8, y = 3$. Number is 83.

Q1622. Let digits be x and y. Equations: $x + y = 6$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 2, y = 4$. Number is 24.

Q1623. Let digits be x and y. Equations: $x + y = 8$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 3, y = 5$. Number is 35.

Q1624. Let digits be x and y. Equations: $x + y = 10$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 4, y = 6$. Number is 46.

Q1625. Let digits be x and y. Equations: $x + y = 6$ and $9(10x + y) = 2(10y + x) \implies 88x = 11y \implies y = 8x$. Solving gives $x = 5, y = 1$. Number is 51.

Q1626. $(10x + y) + (10y + x) = 11(x + y) = 88 \implies x + y = 8$. Given $|x - y| = 4$. Number is 62 or 26.

Q1627. $(10x + y) + (10y + x) = 11(x + y) = 110 \implies x + y = 10$. Given $|x - y| = 4$. Number is 73 or 37.

Q1628. $(10x + y) + (10y + x) = 11(x + y) = 132 \implies x + y = 12$. Given $|x - y| = 4$. Number is 84 or 48.

Q1629. $(10x + y) + (10y + x) = 11(x + y) = 77 \implies x + y = 7$. Given $|x - y| = 3$. Number is 25 or 52.

Q1630. $(10x + y) + (10y + x) = 11(x + y) = 99 \implies x + y = 9$. Given $|x - y| = 3$. Number is 36 or 63.

Q1631. $(10x + y) + (10y + x) = 11(x + y) = 55 \implies x + y = 5$. Given $|x - y| = 3$. Number is 41 or 14.

Q1632. $(10x + y) + (10y + x) = 11(x + y) = 77 \implies x + y = 7$. Given $|x - y| = 3$. Number is 52 or 25.

Q1633. $(10x + y) + (10y + x) = 11(x + y) = 99 \implies x + y = 9$. Given $|x - y| = 3$. Number is 63 or 36.

Q1634. $(10x + y) + (10y + x) = 11(x + y) = 121 \implies x + y = 11$. Given $|x - y| = 3$. Number is 74 or 47.

Q1635. $(10x+y)+(10y+x)=11(x+y)=143 \implies x+y=13$. Given $|x-y|=3$. Number is 85 or 58.

Q1636. $(10x+y)+(10y+x)=11(x+y)=88 \implies x+y=8$. Given $|x-y|=4$. Number is 26 or 62.

Q1637. $(10x+y)+(10y+x)=11(x+y)=44 \implies x+y=4$. Given $|x-y|=2$. Number is 31 or 13.

Q1638. $(10x+y)+(10y+x)=11(x+y)=66 \implies x+y=6$. Given $|x-y|=2$. Number is 42 or 24.

Q1639. $(10x+y)+(10y+x)=11(x+y)=88 \implies x+y=8$. Given $|x-y|=2$. Number is 53 or 35.

Q1640. $(10x+y)+(10y+x)=11(x+y)=110 \implies x+y=10$. Given $|x-y|=2$. Number is 64 or 46.

Q1641. $(10x+y)+(10y+x)=11(x+y)=132 \implies x+y=12$. Given $|x-y|=2$. Number is 75 or 57.

Q1642. $(10x+y)+(10y+x)=11(x+y)=154 \implies x+y=14$. Given $|x-y|=2$. Number is 86 or 68.

Q1643. $(10x+y)+(10y+x)=11(x+y)=33 \implies x+y=3$. Given $|x-y|=1$. Number is 21 or 12.

Q1644. $(10x+y)+(10y+x)=11(x+y)=55 \implies x+y=5$. Given $|x-y|=1$. Number is 32 or 23.

Q1645. $(10x+y)+(10y+x)=11(x+y)=77 \implies x+y=7$. Given $|x-y|=1$. Number is 43 or 34.

Q1646. $(10x+y)+(10y+x)=11(x+y)=99 \implies x+y=9$. Given $|x-y|=1$. Number is 54 or 45.

Q1647. $(10x+y)+(10y+x)=11(x+y)=121 \implies x+y=11$. Given $|x-y|=1$. Number is 65 or 56.

Q1648. $(10x+y)+(10y+x)=11(x+y)=143 \implies x+y=13$. Given $|x-y|=1$. Number is 76 or 67.

Q1649. $(10x+y)+(10y+x)=11(x+y)=99 \implies x+y=9$. Given $|x-y|=7$. Number is 81 or 18.

Q1650. $(10x+y)+(10y+x)=11(x+y)=55 \implies x+y=5$. Given $|x-y|=1$. Number is 32 or 23.

Q1651. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1652. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1653. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1654. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1655. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1656. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1657. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1658. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1659. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1660. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1661. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1662. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1663. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1664. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1665. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1666. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1667. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1668. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1669. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1670. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1671. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1672. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1673. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1674. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1675. $10x+y=4(x+y) \implies 6x=3y \implies y=2x$. $10x+y=2xy \implies 12x=4x^2 \implies x=3, y=6$. Number is 36.

Q1676. Correct Option: (A) 63 $(x=6, y=3)$.

Q1677. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1678. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1679. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1680. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1681. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1682. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1683. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1684. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1685. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1686. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1687. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1688. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1689. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1690. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1691. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1692. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1693. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1694. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1695. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1696. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1697. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1698. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1699. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1700. Correct Option: (A) 63 ($x = 6, y = 3$).
Q1701. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1702. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1703. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1704. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1705. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1706. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1707. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1708. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1709. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1710. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1711. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1712. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1713. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1714. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1715. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1716. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1717. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1718. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1719. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1720. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1721. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1722. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1723. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1724. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1725. $(10x + y) + (10y + x) = 11x + 11y = 11(x + y)$, which is a multiple of 11.
Q1726. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1727. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1728. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1729. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1730. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1731. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1732. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1733. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1734. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1735. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1736. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1737. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1738. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1739. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.
Q1740. $(10x + y) + 45 = 10y + x \implies 9y - 9x = 45 \implies y - x = 5$. Since $xy = 14$, $x = 2, y = 7$. Passcode is 27.

Q1741. $(10x+y)+45=10y+x \implies 9y-9x=45 \implies y-x=5$. Since $xy=14$, $x=2, y=7$. Passcode is 27.

Q1742. $(10x+y)+45=10y+x \implies 9y-9x=45 \implies y-x=5$. Since $xy=14$, $x=2, y=7$. Passcode is 27.

Q1743. $(10x+y)+45=10y+x \implies 9y-9x=45 \implies y-x=5$. Since $xy=14$, $x=2, y=7$. Passcode is 27.

Q1744. $(10x+y)+45=10y+x \implies 9y-9x=45 \implies y-x=5$. Since $xy=14$, $x=2, y=7$. Passcode is 27.

Q1745. $(10x+y)+45=10y+x \implies 9y-9x=45 \implies y-x=5$. Since $xy=14$, $x=2, y=7$. Passcode is 27.

Q1746. $(10x+y)+45=10y+x \implies 9y-9x=45 \implies y-x=5$. Since $xy=14$, $x=2, y=7$. Passcode is 27.

Q1747. $(10x+y)+45=10y+x \implies 9y-9x=45 \implies y-x=5$. Since $xy=14$, $x=2, y=7$. Passcode is 27.

Q1748. $(10x+y)+45=10y+x \implies 9y-9x=45 \implies y-x=5$. Since $xy=14$, $x=2, y=7$. Passcode is 27.

Q1749. $(10x+y)+45=10y+x \implies 9y-9x=45 \implies y-x=5$. Since $xy=14$, $x=2, y=7$. Passcode is 27.

Q1750. $(10x+y)+45=10y+x \implies 9y-9x=45 \implies y-x=5$. Since $xy=14$, $x=2, y=7$. Passcode is 27.

Q1751. $(10x+y)-(10y+x)=9(x-y)$.

Q1752. $(10x+y)-(10y+x)=9(x-y)$.

Q1753. $(10x+y)-(10y+x)=9(x-y)$.

Q1754. $(10x+y)-(10y+x)=9(x-y)$.

Q1755. $(10x+y)-(10y+x)=9(x-y)$.

Q1756. $(10x+y)-(10y+x)=9(x-y)$.

Q1757. $(10x+y)-(10y+x)=9(x-y)$.

Q1758. $(10x+y)-(10y+x)=9(x-y)$.

Q1759. $(10x+y)-(10y+x)=9(x-y)$.

Q1760. $(10x+y)-(10y+x)=9(x-y)$.

Q1761. $(10x+y)-(10y+x)=9(x-y)$.

Q1762. $(10x+y)-(10y+x)=9(x-y)$.

Q1763. $(10x+y)-(10y+x)=9(x-y)$.

Q1764. $(10x+y)-(10y+x)=9(x-y)$.

Q1765. $(10x+y)-(10y+x)=9(x-y)$.

Q1766. $(10x+y)-(10y+x)=9(x-y)$.

Q1767. $(10x+y)-(10y+x)=9(x-y)$.

Q1768. $(10x+y)-(10y+x)=9(x-y)$.

Q1769. $(10x+y)-(10y+x)=9(x-y)$.

Q1770. $(10x+y)-(10y+x)=9(x-y)$.

Q1771. $(10x+y)-(10y+x)=9(x-y)$.

Q1772. $(10x+y)-(10y+x)=9(x-y)$.

Q1773. $(10x+y)-(10y+x)=9(x-y)$.

Q1774. $(10x+y)-(10y+x)=9(x-y)$.

Q1775. $(10x+y)-(10y+x)=9(x-y)$.

Q1776. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=1 \implies y=1, x=2$. Number is 21.

Q1777. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=1 \implies y=1, x=2$. Number is 21.

Q1778. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=1 \implies y=1, x=2$. Number is 21.

Q1779. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=1 \implies y=1, x=2$. Number is 21.

Q1780. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=1 \implies y=1, x=2$. Number is 21.

Q1781. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=6 \implies y=6, x=12$. Number is 126.

Q1782. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=6 \implies y=6, x=12$. Number is 126.

Q1783. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=1 \implies y=1, x=2$. Number is 21.

Q1784. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=1 \implies y=1, x=2$. Number is 21.

Q1785. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=1 \implies y=1, x=2$. Number is 21.

Q1786. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=1 \implies y=1, x=2$. Number is 21.

Q1787. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=5 \implies y=5, x=10$. Number is 105.

Q1788. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=5 \implies y=5, x=10$. Number is 105.

Q1789. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=5 \implies y=5, x=10$. Number is 105.

Q1790. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=2 \implies y=2, x=4$. Number is 42.

Q1791. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=2 \implies y=2, x=4$. Number is 42.

Q1792. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=2 \implies y=2, x=4$. Number is 42.

Q1793. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=4 \implies y=4, x=8$. Number is 84.

Q1794. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=4 \implies y=4, x=8$. Number is 84.

Q1795. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=4 \implies y=4, x=8$. Number is 84.

Q1796. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=4 \implies y=4, x=8$. Number is 84.

Q1797. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=3 \implies y=3, x=6$. Number is 63.

Q1798. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=3 \implies y=3, x=6$. Number is 63.

Q1799. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=3 \implies y=3, x=6$. Number is 63.

Q1800. $10x+y=7(x+y) \implies 3x=6y \implies x=2y$. $x-y=3 \implies y=3, x=6$. Number is 63.

Pattern 10: Fraction Modification & Rational Systems — Answer Key

Q1801. Let fraction be x/y. $\frac{x-1}{y}=\frac{1}{3} \implies 3x-y=3$. $\frac{x}{y+8}=\frac{1}{4} \implies 4x-y=8$. Subtracting gives $x=5, y=12$. Fraction is $5/12$.

Q1802. Let fraction be x/y. $\frac{x-1}{y}=\frac{1}{3} \implies 3x-y=3$.

$\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1803. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1804. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1805. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1806. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1807. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1808. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1809. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1810. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1811. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1812. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1813. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1814. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1815. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1816. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1817. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1818. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1819. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1820. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1821. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1822. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1823. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1824. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1825. Let fraction be x/y. $\frac{x-1}{y} = \frac{1}{3} \implies 3x - y = 3$. $\frac{x}{y+8} = \frac{1}{4} \implies 4x - y = 8$. Subtracting gives $x = 5, y = 12$. Fraction is 5/12.

Q1826. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4$. $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3$. Solution: $x = 7, y = 9$. Fraction is 7/9.

Q1827. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4$. $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3$. Solution: $x = 7, y = 9$. Fraction is 7/9.

Q1828. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4$. $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3$. Solution: $x = 7, y = 9$. Fraction is 7/9.

Q1829. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4$. $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3$. Solution: $x = 7, y = 9$. Fraction is 7/9.

Q1830. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4$. $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3$. Solution: $x = 7, y = 9$. Fraction is 7/9.

Q1831. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4$. $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3$. Solution: $x = 7, y = 9$. Fraction is 7/9.

Q1832. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4$. $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3$. Solution: $x = 7, y = 9$. Fraction is 7/9.

Q1833. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4$. $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3$. Solution: $x = 7, y = 9$. Fraction is 7/9.

Q1834. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4$. $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3$. Solution: $x = 7, y = 9$. Fraction is 7/9.

Q1835. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4$. $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3$. Solution: $x = 7, y = 9$.

Fraction is 7/9.

Q1836. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1837. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1838. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1839. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1840. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1841. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1842. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1843. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1844. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1845. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1846. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1847. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1848. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1849. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1850. $\frac{x+2}{y+2} = \frac{9}{11} \implies 11x - 9y = -4.$ $\frac{x+3}{y+3} = \frac{5}{6} \implies 6x - 5y = -3.$ Solution: $x = 7, y = 9.$ Fraction is 7/9.

Q1851. $y = x+4.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+4 \implies x = 2.$ Fraction is $\frac{2}{6}$.

Q1852. $y = x+4.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+4 \implies x = 2.$ Fraction is $\frac{2}{6}$.

Q1853. $y = x+4.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+4 \implies x = 2.$ Fraction is $\frac{2}{6}$.

Q1854. $y = x+4.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+4 \implies x = 2.$ Fraction is $\frac{2}{6}$.

Q1855. $y = x+2.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+2 \implies x = 0.$ Fraction is $\frac{0}{2}$.

Q1856. $y = x+3.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+3 \implies x = 1.$ Fraction is $\frac{1}{4}$.

Q1857. $y = x+3.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+3 \implies x = 1.$ Fraction is $\frac{1}{4}$.

Q1858. $y = x+3.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+3 \implies x = 1.$ Fraction is $\frac{1}{4}$.

Q1859. $y = x+3.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+3 \implies x = 1.$ Fraction is $\frac{1}{4}$.

Q1860. $y = x+3.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+3 \implies x = 1.$ Fraction is $\frac{1}{4}$.

Q1861. $y = x+2.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+2 \implies x = 0.$ Fraction is $\frac{0}{2}$.

Q1862. $y = x+2.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+2 \implies x = 0.$ Fraction is $\frac{0}{2}$.

Q1863. $y = x+2.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+2 \implies x = 0.$ Fraction is $\frac{0}{2}$.

Q1864. $y = x+2.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+2 \implies x = 0.$ Fraction is $\frac{0}{2}$.

Q1865. $y = x+2.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+2 \implies x = 0.$ Fraction is $\frac{0}{2}$.

Q1866. $y = x+7.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+7 \implies x = 5.$ Fraction is $\frac{5}{12}$.

Q1867. $y = x+1.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+1 \implies x = -1.$ Fraction is $\frac{-1}{0}$.

Q1868. $y = x+1.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+1 \implies x = -1.$ Fraction is $\frac{-1}{0}$.

Q1869. $y = x+1.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+1 \implies x = -1.$ Fraction is $\frac{-1}{0}$.

Q1870. $y = x+1.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+1 \implies x = -1.$ Fraction is $\frac{-1}{0}$.

Q1871. $y = x+6.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+6 \implies x = 4.$ Fraction is $\frac{4}{10}$.

Q1872. $y = x+6.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+6 \implies x = 4.$ Fraction is $\frac{4}{10}$.

Q1873. $y = x+2.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+2 \implies x = 0.$ Fraction is $\frac{0}{2}$.

Q1874. $y = x+2.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+2 \implies x = 0.$ Fraction is $\frac{0}{2}$.

Q1875. $y = x+2.$ $\frac{x+1}{y} = \frac{1}{2} \implies 2x+2 = x+2 \implies x = 0.$ Fraction is $\frac{0}{2}$.

Q1876. Correct Option depends on x, y values.

Q1877. Correct Option depends on x, y values.

Q1878. Correct Option depends on x, y values.

Q1879. Correct Option depends on x, y values.
Q1880. Correct Option depends on x, y values.
Q1881. Correct Option depends on x, y values.
Q1882. Correct Option depends on x, y values.
Q1883. Correct Option depends on x, y values.
Q1884. Correct Option depends on x, y values.
Q1885. Correct Option depends on x, y values.
Q1886. Correct Option depends on x, y values.
Q1887. Correct Option depends on x, y values.
Q1888. Correct Option depends on x, y values.
Q1889. Correct Option depends on x, y values.
Q1890. Correct Option depends on x, y values.
Q1891. Correct Option depends on x, y values.
Q1892. Correct Option depends on x, y values.
Q1893. Correct Option depends on x, y values.
Q1894. Correct Option depends on x, y values.
Q1895. Correct Option depends on x, y values.
Q1896. Correct Option depends on x, y values.
Q1897. Correct Option depends on x, y values.
Q1898. Correct Option depends on x, y values.
Q1899. Correct Option depends on x, y values.
Q1900. Correct Option depends on x, y values.
Q1901. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1902. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1903. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1904. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1905. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1906. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1907. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1908. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1909. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1910. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1911. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1912. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1913. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1914. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1915. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1916. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1917. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1918. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1919. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1920. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1921. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1922. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1923. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1924. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1925. $\frac{x+c}{y+c} - \frac{x}{y} = \frac{c(y-x)}{y(y+c)} > 0$ since $y > x$.
Q1926. $\frac{x+1}{y}$ = new ratio.
Q1927. $\frac{x+2}{y}$ = new ratio.
Q1928. $\frac{x+3}{y}$ = new ratio.
Q1929. $\frac{x+4}{y}$ = new ratio.
Q1930. $\frac{x+5}{y}$ = new ratio.
Q1931. $\frac{x+1}{y}$ = new ratio.
Q1932. $\frac{x+2}{y}$ = new ratio.
Q1933. $\frac{x+3}{y}$ = new ratio.
Q1934. $\frac{x+4}{y}$ = new ratio.
Q1935. $\frac{x+5}{y}$ = new ratio.
Q1936. $\frac{x+1}{y}$ = new ratio.
Q1937. $\frac{x+2}{y}$ = new ratio.
Q1938. $\frac{x+3}{y}$ = new ratio.
Q1939. $\frac{x+4}{y}$ = new ratio.
Q1940. $\frac{x+5}{y}$ = new ratio.
Q1941. $\frac{x+1}{y}$ = new ratio.
Q1942. $\frac{x+2}{y}$ = new ratio.
Q1943. $\frac{x+3}{y}$ = new ratio.
Q1944. $\frac{x+4}{y}$ = new ratio.
Q1945. $\frac{x+5}{y}$ = new ratio.
Q1946. $\frac{x+1}{y}$ = new ratio.
Q1947. $\frac{x+2}{y}$ = new ratio.
Q1948. $\frac{x+3}{y}$ = new ratio.
Q1949. $\frac{x+4}{y}$ = new ratio.
Q1950. $\frac{x+5}{y}$ = new ratio.
Q1951. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5$. Fraction is $3/5$.
Q1952. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5$. Fraction is $3/5$.
Q1953. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5$. Fraction is $3/5$.
Q1954. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5$. Fraction is $3/5$.
Q1955. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5$. Fraction is $3/5$.
Q1956. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5$. Fraction is $3/5$.
Q1957. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5$. Fraction is $3/5$.
Q1958. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5$. Fraction is $3/5$.

Q1959. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1960. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1961. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1962. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1963. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1964. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1965. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1966. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1967. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1968. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1969. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1970. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1971. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1972. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1973. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1974. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1975. $x - y = -2$ and $2x - y = 1 \implies x = 3, y = 5.$ Fraction is 3/5.

Q1976. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1977. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1978. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1979. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1980. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1981. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1982. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1983. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1984. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1985. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1986. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1987. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1988. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1989. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1990. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1991. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1992. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1993. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1994. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1995. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1996. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1997. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1998. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q1999. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Q2000. $x + 3 = y - 2 \implies y - x = 5$. Since $x + y = 12$, $x = 3.5, y = 8.5$. Sum $= 12$.

Pattern 11: Ratio Conditions for System Consistency — Answer Key

Q2001. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10.$

Q2002. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10.$

Q2003. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10.$

Q2004. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10.$

Q2005. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10.$

Q2006. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10.$

Q2007. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10.$

Q2008. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} =$

$\frac{2}{k} \implies k = 10$.

Q2009. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2010. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2011. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2012. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2013. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2014. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2015. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2016. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2017. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2018. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2019. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2020. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2021. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2022. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2023. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2024. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2025. Inconsistent condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{1}{5} = \frac{2}{k} \implies k = 10$.

Q2026. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2027. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2028. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2029. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2030. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2031. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2032. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2033. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2034. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2035. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2036. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2037. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2038. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2039. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2040. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2041. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2042. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2043. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2044. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2045. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2046. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2047. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2048. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2049. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2050. Coincident condition: $\frac{2}{4} = \frac{3}{k} = \frac{5}{10} \implies k = 6$.

Q2051. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2052. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2053. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2054. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2055. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2056. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2057. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2058. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2059. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2060. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2061. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2062. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2063. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2064. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2065. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2066. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2067. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2068. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2069. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2070. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2071. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2072. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2073. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2074. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2075. $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b} \implies a = -5, b = -1$.

Q2076. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.

Q2077. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.

Q2078. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.

Q2079. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.

Q2080. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.

Q2081. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2082. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2083. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2084. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2085. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2086. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2087. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2088. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2089. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2090. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2091. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2092. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2093. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2094. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2095. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2096. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2097. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2098. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2099. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2100. Correct Option: (A) $p \neq 4$, since $\frac{4}{2} \neq \frac{p}{2} \implies p \neq 4$.
Q2101. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2102. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2103. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2104. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2105. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2106. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2107. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2108. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2109. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2110. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2111. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2112. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2113. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2114. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2115. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2116. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2117. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2118. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2119. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2120. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2121. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2122. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2123. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2124. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2125. For $k = 3$, $\frac{3}{5} \neq \frac{1}{2}$. Yes, unique solution exists.
Q2126. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2127. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2128. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2129. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2130. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2131. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2132. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2133. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2134. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2135. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2136. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2137. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2138. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2139. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2140. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2141. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2142. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2143. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2144. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2145. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2146. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2147. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2148. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2149. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2150. $\frac{1}{3} = \frac{2}{6} = \frac{5}{c} \implies c = 15$.
Q2151. The lines are parallel and never intersect (no solution).
Q2152. The lines are parallel and never intersect (no solution).
Q2153. The lines are parallel and never intersect (no solution).
Q2154. The lines are parallel and never intersect (no solution).
Q2155. The lines are parallel and never intersect (no solution).
Q2156. The lines are parallel and never intersect (no solution).

Q2157. The lines are parallel and never intersect (no solution).
Q2158. The lines are parallel and never intersect (no solution).
Q2159. The lines are parallel and never intersect (no solution).
Q2160. The lines are parallel and never intersect (no solution).
Q2161. The lines are parallel and never intersect (no solution).
Q2162. The lines are parallel and never intersect (no solution).
Q2163. The lines are parallel and never intersect (no solution).
Q2164. The lines are parallel and never intersect (no solution).
Q2165. The lines are parallel and never intersect (no solution).
Q2166. The lines are parallel and never intersect (no solution).
Q2167. The lines are parallel and never intersect (no solution).
Q2168. The lines are parallel and never intersect (no solution).
Q2169. The lines are parallel and never intersect (no solution).
Q2170. The lines are parallel and never intersect (no solution).
Q2171. The lines are parallel and never intersect (no solution).
Q2172. The lines are parallel and never intersect (no solution).
Q2173. The lines are parallel and never intersect (no solution).
Q2174. The lines are parallel and never intersect (no solution).
Q2175. The lines are parallel and never intersect (no solution).
Q2176. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2177. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2178. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2179. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2180. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2181. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2182. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2183. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2184. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2185. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2186. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2187. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2188. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2189. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2190. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2191. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2192. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2193. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2194. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2195. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2196. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2197. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2198. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2199. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$
Q2200. $\frac{2m-1}{3} = \frac{m-1}{1} \implies 2m-1 = 3m-3 \implies m = 2.$

Pattern 12: Upstream / Downstream Rate Modeling — Answer Key

Q2201. $\frac{24}{15-y} - \frac{24}{15+y} = 1 \implies 24(2y) = 15^2 - y^2 \implies y^2 + 48y - 225 = 0 \implies y = 2$ km/h.
Q2202. $\frac{30}{16-y} - \frac{30}{16+y} = 1 \implies 30(2y) = 16^2 - y^2 \implies y^2 + 60y - 256 = 0 \implies y = 3$ km/h.
Q2203. $\frac{36}{17-y} - \frac{36}{17+y} = 1 \implies 36(2y) = 17^2 - y^2 \implies y^2 + 72y - 289 = 0 \implies y = 4$ km/h.
Q2204. $\frac{42}{18-y} - \frac{42}{18+y} = 1 \implies 42(2y) = 18^2 - y^2 \implies y^2 + 84y - 324 = 0 \implies y = 5$ km/h.
Q2205. $\frac{48}{19-y} - \frac{48}{19+y} = 1 \implies 48(2y) = 19^2 - y^2 \implies y^2 + 96y - 361 = 0 \implies y = 6$ km/h.
Q2206. $\frac{54}{20-y} - \frac{54}{20+y} = 1 \implies 54(2y) = 20^2 - y^2 \implies y^2 + 108y - 400 = 0 \implies y = 2$ km/h.
Q2207. $\frac{24}{21-y} - \frac{24}{21+y} = 1 \implies 24(2y) = 21^2 - y^2 \implies y^2 + 48y - 441 = 0 \implies y = 3$ km/h.
Q2208. $\frac{30}{22-y} - \frac{30}{22+y} = 1 \implies 30(2y) = 22^2 - y^2 \implies y^2 + 60y - 484 = 0 \implies y = 4$ km/h.
Q2209. $\frac{36}{23-y} - \frac{36}{23+y} = 1 \implies 36(2y) = 23^2 - y^2 \implies y^2 + 72y - 529 = 0 \implies y = 5$ km/h.
Q2210. $\frac{42}{24-y} - \frac{42}{24+y} = 1 \implies 42(2y) = 24^2 - y^2 \implies y^2 + 84y - 576 = 0 \implies y = 6$ km/h.
Q2211. $\frac{48}{15-y} - \frac{48}{15+y} = 1 \implies 48(2y) = 15^2 - y^2 \implies y^2 + 96y - 225 = 0 \implies y = 2$ km/h.
Q2212. $\frac{54}{16-y} - \frac{54}{16+y} = 1 \implies 54(2y) = 16^2 - y^2 \implies y^2 + 108y - 256 = 0 \implies y = 3$ km/h.
Q2213. $\frac{24}{17-y} - \frac{24}{17+y} = 1 \implies 24(2y) = 17^2 - y^2 \implies y^2 + 48y - 289 = 0 \implies y = 4$ km/h.
Q2214. $\frac{30}{18-y} - \frac{30}{18+y} = 1 \implies 30(2y) = 18^2 - y^2 \implies y^2 + 60y - 324 = 0 \implies y = 5$ km/h.
Q2215. $\frac{36}{19-y} - \frac{36}{19+y} = 1 \implies 36(2y) = 19^2 - y^2 \implies y^2 + 72y - 361 = 0 \implies y = 6$ km/h.
Q2216. $\frac{42}{20-y} - \frac{42}{20+y} = 1 \implies 42(2y) = 20^2 - y^2 \implies y^2 + 84y - 400 = 0 \implies y = 2$ km/h.
Q2217. $\frac{48}{21-y} - \frac{48}{21+y} = 1 \implies 48(2y) = 21^2 - y^2 \implies y^2 + 96y - 441 = 0 \implies y = 3$ km/h.

Q2218. $\frac{54}{22-y} - \frac{54}{22+y} = 1 \implies 54(2y) = 22^2 - y^2 \implies y^2 + 108y - 484 = 0 \implies y = 4$ km/h.

Q2219. $\frac{24}{23-y} - \frac{24}{23+y} = 1 \implies 24(2y) = 23^2 - y^2 \implies y^2 + 48y - 529 = 0 \implies y = 5$ km/h.

Q2220. $\frac{30}{24-y} - \frac{30}{24+y} = 1 \implies 30(2y) = 24^2 - y^2 \implies y^2 + 60y - 576 = 0 \implies y = 6$ km/h.

Q2221. $\frac{36}{15-y} - \frac{36}{15+y} = 1 \implies 36(2y) = 15^2 - y^2 \implies y^2 + 72y - 225 = 0 \implies y = 2$ km/h.

Q2222. $\frac{42}{16-y} - \frac{42}{16+y} = 1 \implies 42(2y) = 16^2 - y^2 \implies y^2 + 84y - 256 = 0 \implies y = 3$ km/h.

Q2223. $\frac{48}{17-y} - \frac{48}{17+y} = 1 \implies 48(2y) = 17^2 - y^2 \implies y^2 + 96y - 289 = 0 \implies y = 4$ km/h.

Q2224. $\frac{54}{18-y} - \frac{54}{18+y} = 1 \implies 54(2y) = 18^2 - y^2 \implies y^2 + 108y - 324 = 0 \implies y = 5$ km/h.

Q2225. $\frac{24}{19-y} - \frac{24}{19+y} = 1 \implies 24(2y) = 19^2 - y^2 \implies y^2 + 48y - 361 = 0 \implies y = 6$ km/h.

Q2226. $\frac{30}{x+y} + \frac{15.0}{x-y} = T$.

Q2227. $\frac{36}{x+y} + \frac{18.0}{x-y} = T$.

Q2228. $\frac{42}{x+y} + \frac{21.0}{x-y} = T$.

Q2229. $\frac{48}{x+y} + \frac{24.0}{x-y} = T$.

Q2230. $\frac{54}{x+y} + \frac{27.0}{x-y} = T$.

Q2231. $\frac{24}{x+y} + \frac{12.0}{x-y} = T$.

Q2232. $\frac{30}{x+y} + \frac{15.0}{x-y} = T$.

Q2233. $\frac{36}{x+y} + \frac{18.0}{x-y} = T$.

Q2234. $\frac{42}{x+y} + \frac{21.0}{x-y} = T$.

Q2235. $\frac{48}{x+y} + \frac{24.0}{x-y} = T$.

Q2236. $\frac{54}{x+y} + \frac{27.0}{x-y} = T$.

Q2237. $\frac{24}{x+y} + \frac{12.0}{x-y} = T$.

Q2238. $\frac{30}{x+y} + \frac{15.0}{x-y} = T$.

Q2239. $\frac{36}{x+y} + \frac{18.0}{x-y} = T$.

Q2240. $\frac{42}{x+y} + \frac{21.0}{x-y} = T$.

Q2241. $\frac{48}{x+y} + \frac{24.0}{x-y} = T$.

Q2242. $\frac{54}{x+y} + \frac{27.0}{x-y} = T$.

Q2243. $\frac{24}{x+y} + \frac{12.0}{x-y} = T$.

Q2244. $\frac{30}{x+y} + \frac{15.0}{x-y} = T$.

Q2245. $\frac{36}{x+y} + \frac{18.0}{x-y} = T$.

Q2246. $\frac{42}{x+y} + \frac{21.0}{x-y} = T$.

Q2247. $\frac{48}{x+y} + \frac{24.0}{x-y} = T$.

Q2248. $\frac{54}{x+y} + \frac{27.0}{x-y} = T$.

Q2249. $\frac{24}{x+y} + \frac{12.0}{x-y} = T$.

Q2250. $\frac{30}{x+y} + \frac{15.0}{x-y} = T$.

Q2251. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2252. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2253. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2254. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2255. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2256. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2257. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2258. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2259. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2260. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2261. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2262. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2263. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2264. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2265. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2266. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2267. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2268. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2269. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2270. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2271. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2272. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2273. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2274. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2275. Upstream Speed $= x - y$ km/h; Downstream Speed $= x + y$ km/h.

Q2276. Correct Option: (A) 16 km/h (since $(20 + 12)/2 = 16$).

Q2277. Correct Option: (A) 16 km/h (since $(20 + 12)/2 = 16$).

Q2278. Correct Option: (A) 16 km/h (since $(20 + 12)/2 = 16$).

Q2279. Correct Option: (A) 16 km/h (since $(20 + 12)/2 = 16$).

Q2280. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2281. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2282. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2283. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2284. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2285. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2286. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2287. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2288. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2289. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2290. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2291. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2292. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2293. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2294. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2295. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2296. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2297. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2298. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2299. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2300. Correct Option: (A) 16 km/h (since $(20+12)/2 = 16$).

Q2301. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2302. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2303. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2304. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2305. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2306. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2307. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2308. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2309. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2310. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2311. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2312. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2313. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2314. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2315. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2316. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2317. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2318. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2319. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2320. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2321. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2322. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2323. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2324. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2325. If $x \le y$, upstream speed $x - y \le 0$, meaning the boat cannot move upstream.

Q2326. Time $= \frac{54}{20-2} + \frac{54}{20+2} = \frac{54}{18} + \frac{54}{22}$ hours.

Q2327. Time $= \frac{24}{21-3} + \frac{24}{21+3} = \frac{24}{18} + \frac{24}{24}$ hours.

Q2328. Time $= \frac{30}{22-4} + \frac{30}{22+4} = \frac{30}{18} + \frac{30}{26}$ hours.

Q2329. Time $= \frac{36}{23-5} + \frac{36}{23+5} = \frac{36}{18} + \frac{36}{28}$ hours.

Q2330. Time $= \frac{42}{24-6} + \frac{42}{24+6} = \frac{42}{18} + \frac{42}{30}$ hours.

Q2331. Time $= \frac{48}{15-2} + \frac{48}{15+2} = \frac{48}{13} + \frac{48}{17}$ hours.

Q2332. Time $= \frac{54}{16-3} + \frac{54}{16+3} = \frac{54}{13} + \frac{54}{19}$ hours.

Q2333. Time $= \frac{24}{17-4} + \frac{24}{17+4} = \frac{24}{13} + \frac{24}{21}$ hours.

Q2334. Time $= \frac{30}{18-5} + \frac{30}{18+5} = \frac{30}{13} + \frac{30}{23}$ hours.

Q2335. Time $= \frac{36}{19-6} + \frac{36}{19+6} = \frac{36}{13} + \frac{36}{25}$ hours.

Q2336. Time $= \frac{42}{20-2} + \frac{42}{20+2} = \frac{42}{18} + \frac{42}{22}$ hours.

Q2337. Time $= \frac{48}{21-3} + \frac{48}{21+3} = \frac{48}{18} + \frac{48}{24}$ hours.

Q2338. Time $= \frac{54}{22-4} + \frac{54}{22+4} = \frac{54}{18} + \frac{54}{26}$ hours.

Q2339. Time $= \frac{24}{23-5} + \frac{24}{23+5} = \frac{24}{18} + \frac{24}{28}$ hours.

Q2340. Time $= \frac{30}{24-6} + \frac{30}{24+6} = \frac{30}{18} + \frac{30}{30}$ hours.

Q2341. Time $= \frac{36}{15-2} + \frac{36}{15+2} = \frac{36}{13} + \frac{36}{17}$ hours.

Q2342. Time $= \frac{42}{16-3} + \frac{42}{16+3} = \frac{42}{13} + \frac{42}{19}$ hours.

Q2343. Time $= \frac{48}{17-4} + \frac{48}{17+4} = \frac{48}{13} + \frac{48}{21}$ hours.

Q2344. Time $= \frac{54}{18-5} + \frac{54}{18+5} = \frac{54}{13} + \frac{54}{23}$ hours.

Q2345. Time $= \frac{24}{19-6} + \frac{24}{19+6} = \frac{24}{13} + \frac{24}{25}$ hours.

Q2346. Time $= \frac{30}{20-2} + \frac{30}{20+2} = \frac{30}{18} + \frac{30}{22}$ hours.

Q2347. Time $= \frac{36}{21-3} + \frac{36}{21+3} = \frac{36}{18} + \frac{36}{24}$ hours.

Q2348. Time $= \frac{42}{22-4} + \frac{42}{22+4} = \frac{42}{18} + \frac{42}{26}$ hours.

Q2349. Time $= \frac{48}{23-5} + \frac{48}{23+5} = \frac{48}{18} + \frac{48}{28}$ hours.

Q2350. Time $= \frac{54}{24-6} + \frac{54}{24+6} = \frac{54}{18} + \frac{54}{30}$ hours.

Q2351. $y = \frac{D-U}{2}$.

Q2352. $y = \frac{D-U}{2}$.

Q2353. $y = \frac{D-U}{2}$.

Q2354. $y = \frac{D-U}{2}$.

Q2355. $y = \frac{D-U}{2}$.

Q2356. $y = \frac{D-U}{2}$.

Q2357. $y = \frac{D-U}{2}$.

Q2358. $y = \frac{D-U}{2}$.

Q2359. $y = \frac{D-U}{2}$.

Q2360. $y = \frac{D-U}{2}$.

Q2361. $y = \frac{D-U}{2}$.

Q2362. $y = \frac{D-U}{2}$.

Q2363. $y = \frac{D-U}{2}$.

Q2364. $y = \frac{D-U}{2}$.

Q2365. $y = \frac{D-U}{2}$.

Q2366. $y = \frac{D-U}{2}$.

Q2367. $y = \frac{D-U}{2}$.

Q2368. $y = \frac{D-U}{2}$.

Q2369. $y = \frac{D-U}{2}$.

Q2370. $y = \frac{D-U}{2}$.

Q2371. $y = \frac{D-U}{2}$.

Q2372. $y = \frac{D-U}{2}$.

Q2373. $y = \frac{D-U}{2}$.

Q2374. $y = \frac{D-U}{2}$.

Q2375. $y = \frac{D-U}{2}$.

Q2376. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2377. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2378. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2379. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2380. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2381. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2382. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2383. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2384. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2385. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2386. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2387. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2388. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2389. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2390. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2391. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2392. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2393. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2394. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2395. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2396. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2397. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2398. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2399. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Q2400. Downstream $= 30/2 = 15 \implies x + y = 15$. Upstream $= 18/3 = 6 \implies x - y = 6$. Boat $= 10.5$ km/h, Stream $= 4.5$ km/h.

Pattern 13: Equal Roots & Discriminant Proof Identities — Answer Key

Q2401. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2402. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2403. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2404. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2405. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2406. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2407. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2408. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2409. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2410. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2411. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2412. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2413. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2414. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2415. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2416. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2417. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2418. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2419. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2420. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2421. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2422. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2423. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2424. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2425. Equal roots $\implies D = 0 \implies 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0 \implies 4a(a^3 + b^3 + c^3 - 3abc) = 0$.

Q2426. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2427. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2428. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2429. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2430. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2431. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2432. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2433. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2434. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2435. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2436. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2437. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2438. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2439. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2440. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2441. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2442. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2443. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2444. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2445. $D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2) = -4(ad - bc)^2 < 0$ when $ad \neq bc$.

Q2446. $D = 4(ac+bd)^2 - 4(a^2+b^2)(c^2+d^2) = -4(ad-bc)^2 < 0$ when $ad \neq bc$.
Q2447. $D = 4(ac+bd)^2 - 4(a^2+b^2)(c^2+d^2) = -4(ad-bc)^2 < 0$ when $ad \neq bc$.
Q2448. $D = 4(ac+bd)^2 - 4(a^2+b^2)(c^2+d^2) = -4(ad-bc)^2 < 0$ when $ad \neq bc$.
Q2449. $D = 4(ac+bd)^2 - 4(a^2+b^2)(c^2+d^2) = -4(ad-bc)^2 < 0$ when $ad \neq bc$.
Q2450. $D = 4(ac+bd)^2 - 4(a^2+b^2)(c^2+d^2) = -4(ad-bc)^2 < 0$ when $ad \neq bc$.
Q2451. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2452. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2453. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2454. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2455. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2456. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2457. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2458. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2459. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2460. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2461. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2462. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2463. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2464. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2465. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2466. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2467. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2468. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2469. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2470. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2471. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2472. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2473. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2474. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2475. $D = (k+1)^2 - 4(k+4)(1) = k^2 - 2k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } -3$.
Q2476. Correct Option: (A) $D = 0$.
Q2477. Correct Option: (A) $D = 0$.
Q2478. Correct Option: (A) $D = 0$.
Q2479. Correct Option: (A) $D = 0$.
Q2480. Correct Option: (A) $D = 0$.
Q2481. Correct Option: (A) $D = 0$.
Q2482. Correct Option: (A) $D = 0$.
Q2483. Correct Option: (A) $D = 0$.
Q2484. Correct Option: (A) $D = 0$.
Q2485. Correct Option: (A) $D = 0$.
Q2486. Correct Option: (A) $D = 0$.
Q2487. Correct Option: (A) $D = 0$.
Q2488. Correct Option: (A) $D = 0$.
Q2489. Correct Option: (A) $D = 0$.
Q2490. Correct Option: (A) $D = 0$.
Q2491. Correct Option: (A) $D = 0$.
Q2492. Correct Option: (A) $D = 0$.
Q2493. Correct Option: (A) $D = 0$.
Q2494. Correct Option: (A) $D = 0$.
Q2495. Correct Option: (A) $D = 0$.
Q2496. Correct Option: (A) $D = 0$.
Q2497. Correct Option: (A) $D = 0$.
Q2498. Correct Option: (A) $D = 0$.
Q2499. Correct Option: (A) $D = 0$.
Q2500. Correct Option: (A) $D = 0$.
Q2501. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6) = 0 \implies 2p - 6 = 0 \implies p = 3$.
Q2502. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6) = 0 \implies 2p - 6 = 0 \implies p = 3$.
Q2503. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6) = 0 \implies 2p - 6 = 0 \implies p = 3$.
Q2504. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6) = 0 \implies 2p - 6 = 0 \implies p = 3$.
Q2505. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6) = 0 \implies 2p - 6 = 0 \implies p = 3$.
Q2506. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6) = 0 \implies 2p - 6 = 0 \implies p = 3$.
Q2507. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6) = 0 \implies 2p - 6 = 0 \implies p = 3$.
Q2508. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6) = 0 \implies 2p - 6 = 0 \implies p = 3$.
Q2509. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6) = 0 \implies 2p - 6 = 0 \implies p = 3$.
Q2510. $D = 36(p+1)^2 - 12(p+1)(p+9) =$

$12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2511. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2512. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2513. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2514. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2515. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2516. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2517. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2518. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2519. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2520. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2521. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2522. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2523. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2524. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2525. $D = 36(p+1)^2 - 12(p+1)(p+9) = 12(p+1)(2p-6)=0 \implies 2p-6=0 \implies p=3.$

Q2526. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2527. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2528. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2529. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2530. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2531. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2532. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2533. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2534. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2535. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2536. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2537. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2538. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2539. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2540. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2541. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2542. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2543. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2544. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2545. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2546. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2547. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2548. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2549. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2550. $D=(2k)^2-4(1)(k^2)=4k^2-4k^2=0.$ Always equal roots $x=-k$.

Q2551. $D=4p^2-4(p^2-q^2)=4q^2=(2q)^2$, a perfect square $\implies$ rational roots.

Q2552. $D=4p^2-4(p^2-q^2)=4q^2=(2q)^2$, a perfect square $\implies$ rational roots.

Q2553. $D=4p^2-4(p^2-q^2)=4q^2=(2q)^2$, a perfect square $\implies$ rational roots.

Q2554. $D=4p^2-4(p^2-q^2)=4q^2=(2q)^2$, a perfect square $\implies$ rational roots.

Q2555. $D=4p^2-4(p^2-q^2)=4q^2=(2q)^2$, a perfect square $\implies$ rational roots.

Q2556. $D=4p^2-4(p^2-q^2)=4q^2=(2q)^2$, a perfect square $\implies$ rational roots.

Q2557. $D=4p^2-4(p^2-q^2)=4q^2=(2q)^2$, a perfect square $\implies$ rational roots.

Q2558. $D=4p^2-4(p^2-q^2)=4q^2=(2q)^2$, a perfect square $\implies$ rational roots.

Q2559. $D=4p^2-4(p^2-q^2)=4q^2=(2q)^2$, a perfect square $\implies$ rational roots.

Q2560. $D=4p^2-4(p^2-q^2)=4q^2=(2q)^2$, a perfect square $\implies$ rational roots.

Q2561. $D=4p^2-4(p^2-q^2)=4q^2=(2q)^2$, a perfect square $\implies$ rational roots.

Q2562. $D=4p^2-4(p^2-q^2)=4q^2=(2q)^2$, a perfect square $\implies$ rational roots.

Q2563. $D=4p^2-4(p^2-q^2)=4q^2=(2q)^2$, a perfect

square $\implies$ rational roots.

Q2564. $D = 4p^2 - 4(p^2 - q^2) = 4q^2 = (2q)^2$, a perfect square $\implies$ rational roots.

Q2565. $D = 4p^2 - 4(p^2 - q^2) = 4q^2 = (2q)^2$, a perfect square $\implies$ rational roots.

Q2566. $D = 4p^2 - 4(p^2 - q^2) = 4q^2 = (2q)^2$, a perfect square $\implies$ rational roots.

Q2567. $D = 4p^2 - 4(p^2 - q^2) = 4q^2 = (2q)^2$, a perfect square $\implies$ rational roots.

Q2568. $D = 4p^2 - 4(p^2 - q^2) = 4q^2 = (2q)^2$, a perfect square $\implies$ rational roots.

Q2569. $D = 4p^2 - 4(p^2 - q^2) = 4q^2 = (2q)^2$, a perfect square $\implies$ rational roots.

Q2570. $D = 4p^2 - 4(p^2 - q^2) = 4q^2 = (2q)^2$, a perfect square $\implies$ rational roots.

Q2571. $D = 4p^2 - 4(p^2 - q^2) = 4q^2 = (2q)^2$, a perfect square $\implies$ rational roots.

Q2572. $D = 4p^2 - 4(p^2 - q^2) = 4q^2 = (2q)^2$, a perfect square $\implies$ rational roots.

Q2573. $D = 4p^2 - 4(p^2 - q^2) = 4q^2 = (2q)^2$, a perfect square $\implies$ rational roots.

Q2574. $D = 4p^2 - 4(p^2 - q^2) = 4q^2 = (2q)^2$, a perfect square $\implies$ rational roots.

Q2575. $D = 4p^2 - 4(p^2 - q^2) = 4q^2 = (2q)^2$, a perfect square $\implies$ rational roots.

Q2576. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2577. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2578. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2579. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2580. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2581. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2582. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2583. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2584. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2585. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2586. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2587. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2588. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2589. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2590. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2591. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2592. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2593. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2594. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2595. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2596. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2597. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2598. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2599. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Q2600. $D = k^2 - 24 = 0 \implies k = \pm 2\sqrt{6}$.

Pattern 14: Age & Multiplier Ratio Word Problems — Answer Key

Q2601. Let Jacob be x, son be y. Equations: $x + 5 = 3(y + 5) \implies x - 3y = 10$. $x - 5 = 7(y - 5) \implies x - 7y = -30$. Subtracting gives $4y = 40 \implies y = 10, x = 40$.

Q2602. Let Jacob be x, son be y. Equations: $x + 5 = 3(y + 5) \implies x - 3y = 10$. $x - 5 = 7(y - 5) \implies x - 7y = -30$. Subtracting gives $4y = 40 \implies y = 10, x = 40$.

Q2603. Let Jacob be x, son be y. Equations: $x + 5 = 3(y + 5) \implies x - 3y = 10$. $x - 5 = 7(y - 5) \implies x - 7y = -30$. Subtracting gives $4y = 40 \implies y = 10, x = 40$.

Q2604. Let Jacob be x, son be y. Equations: $x + 5 = 3(y + 5) \implies x - 3y = 10$. $x - 5 = 7(y - 5) \implies x - 7y = -30$. Subtracting gives $4y = 40 \implies y = 10, x = 40$.

Q2605. Let Jacob be x, son be y. Equations: $x + 5 = 3(y + 5) \implies x - 3y = 10$. $x - 5 = 7(y - 5) \implies x - 7y = -30$. Subtracting gives $4y = 40 \implies y = 10, x = 40$.

Q2606. Let Jacob be x, son be y. Equations: $x + 5 = 3(y + 5) \implies x - 3y = 10$. $x - 5 = 7(y - 5) \implies x - 7y = -30$. Subtracting gives $4y = 40 \implies y = 10, x = 40$.

Q2607. Let Jacob be x, son be y. Equations: $x + 5 = 3(y + 5) \implies x - 3y = 10$. $x - 5 = 7(y - 5) \implies x - 7y = -30$. Subtracting gives $4y = 40 \implies y = 10, x = 40$.

Q2608. Let Jacob be x, son be y. Equations: $x + 5 = 3(y + 5) \implies x - 3y = 10$. $x - 5 = 7(y - 5) \implies x - 7y = -30$. Subtracting gives $4y = 40 \implies y = 10, x = 40$.

Q2609. Let Jacob be x, son be y. Equations: $x + 5 = 3(y + 5) \implies x - 3y = 10$. $x - 5 = 7(y - 5) \implies x - 7y = -30$. Subtracting gives $4y = 40 \implies y = 10, x = 40$.

Q2610. Let Jacob be x, son be y. Equations: $x + 5 = 3(y + 5) \implies x - 3y = 10$. $x - 5 = 7(y - 5) \implies x - 7y = -30$. Subtracting gives $4y = 40 \implies y = 10, x = 40$.

Q2611. Let Jacob be x, son be y. Equations: $x + 5 = 3(y + 5) \implies x - 3y = 10$. $x - 5 = 7(y - 5) \implies x - 7y = -30$. Subtracting gives $4y = 40 \implies y = 10, x = 40$.

Q2612. Let Jacob be x, son be y. Equations: $x + 5 = 3(y + 5) \implies x - 3y = 10$. $x - 5 = 7(y - 5) \implies x - 7y = -30$. Subtracting gives $4y = 40 \implies y = 10, x = 40$.

Q2613. Let Jacob be x, son be y. Equations: $x + 5 = 3(y + 5) \implies x - 3y = 10$. $x - 5 = 7(y - 5) \implies x - 7y = -30$. Subtracting gives $4y = 40 \implies y = 10, x = 40$.

Q2614. Let Jacob be x, son be y. Equa-

tions: $x+5=3(y+5) \implies x-3y=10$. $x-5=7(y-5) \implies x-7y=-30$. Subtracting gives $4y=40 \implies y=10, x=40$.

Q2615. Let Jacob be x, son be y. Equations: $x+5=3(y+5) \implies x-3y=10$. $x-5=7(y-5) \implies x-7y=-30$. Subtracting gives $4y=40 \implies y=10, x=40$.

Q2616. Let Jacob be x, son be y. Equations: $x+5=3(y+5) \implies x-3y=10$. $x-5=7(y-5) \implies x-7y=-30$. Subtracting gives $4y=40 \implies y=10, x=40$.

Q2617. Let Jacob be x, son be y. Equations: $x+5=3(y+5) \implies x-3y=10$. $x-5=7(y-5) \implies x-7y=-30$. Subtracting gives $4y=40 \implies y=10, x=40$.

Q2618. Let Jacob be x, son be y. Equations: $x+5=3(y+5) \implies x-3y=10$. $x-5=7(y-5) \implies x-7y=-30$. Subtracting gives $4y=40 \implies y=10, x=40$.

Q2619. Let Jacob be x, son be y. Equations: $x+5=3(y+5) \implies x-3y=10$. $x-5=7(y-5) \implies x-7y=-30$. Subtracting gives $4y=40 \implies y=10, x=40$.

Q2620. Let Jacob be x, son be y. Equations: $x+5=3(y+5) \implies x-3y=10$. $x-5=7(y-5) \implies x-7y=-30$. Subtracting gives $4y=40 \implies y=10, x=40$.

Q2621. Let Jacob be x, son be y. Equations: $x+5=3(y+5) \implies x-3y=10$. $x-5=7(y-5) \implies x-7y=-30$. Subtracting gives $4y=40 \implies y=10, x=40$.

Q2622. Let Jacob be x, son be y. Equations: $x+5=3(y+5) \implies x-3y=10$. $x-5=7(y-5) \implies x-7y=-30$. Subtracting gives $4y=40 \implies y=10, x=40$.

Q2623. Let Jacob be x, son be y. Equations: $x+5=3(y+5) \implies x-3y=10$. $x-5=7(y-5) \implies x-7y=-30$. Subtracting gives $4y=40 \implies y=10, x=40$.

Q2624. Let Jacob be x, son be y. Equations: $x+5=3(y+5) \implies x-3y=10$. $x-5=7(y-5) \implies x-7y=-30$. Subtracting gives $4y=40 \implies y=10, x=40$.

Q2625. Let Jacob be x, son be y. Equations: $x+5=3(y+5) \implies x-3y=10$. $x-5=7(y-5) \implies x-7y=-30$. Subtracting gives $4y=40 \implies y=10, x=40$.

Q2626. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2627. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2628. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2629. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2630. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2631. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2632. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2633. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2634. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2635. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2636. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2637. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2638. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2639. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2640. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2641. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2642. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2643. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2644. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2645. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34 years, Son = 12 years.

Q2646. $x-10=12(y-10) \implies x-12y=-110$. $x+10=2(y+10) \implies x-2y=10$. Solution: Father = 34

years, Son = 12 years.

Q2647. $x - 10 = 12(y - 10) \implies x - 12y = -110$. $x + 10 = 2(y + 10) \implies x - 2y = 10$. Solution: Father = 34 years, Son = 12 years.

Q2648. $x - 10 = 12(y - 10) \implies x - 12y = -110$. $x + 10 = 2(y + 10) \implies x - 2y = 10$. Solution: Father = 34 years, Son = 12 years.

Q2649. $x - 10 = 12(y - 10) \implies x - 12y = -110$. $x + 10 = 2(y + 10) \implies x - 2y = 10$. Solution: Father = 34 years, Son = 12 years.

Q2650. $x - 10 = 12(y - 10) \implies x - 12y = -110$. $x + 10 = 2(y + 10) \implies x - 2y = 10$. Solution: Father = 34 years, Son = 12 years.

Q2651. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2652. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2653. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2654. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2655. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2656. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2657. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2658. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2659. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2660. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2661. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2662. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2663. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2664. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2665. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2666. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2667. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2668. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2669. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2670. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2671. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2672. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2673. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2674. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2675. $x = 3y + 3$ and $x + 3 = 2(y + 3) + 10 \implies y = 10, x = 33$. Father = 33 years, Son = 10 years.

Q2676. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2677. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2678. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2679. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2680. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2681. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2682. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2683. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2684. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2685. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2686. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2687. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2688. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2689. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2690. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2691. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2692. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2693. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2694. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2695. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2696. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2697. Correct Option: (A) 7 years ago ($42 - 7 = 35 =$

$7 \times (12 - 7)$).

Q2698. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2699. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2700. Correct Option: (A) 7 years ago ($42 - 7 = 35 = 7 \times (12 - 7)$).

Q2701. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2702. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2703. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2704. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2705. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2706. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2707. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2708. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2709. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2710. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2711. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2712. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2713. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2714. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2715. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2716. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2717. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2718. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2719. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2720. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2721. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2722. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2723. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2724. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2725. $x = y^2$. $x + 8 = 3(y + 8) + 4 \implies y^2 - 3y - 20 = 0 \implies y = 4, x = 16$ or 32 depending on parameters.

Q2726. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2727. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2728. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2729. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2730. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2731. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2732. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2733. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2734. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2735. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2736. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2737. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2738. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2739. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2740. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2741. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2742. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2743. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2744. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2745. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2746. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2747. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2748. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2749. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 = 2y + 32 \implies 2y = 16 \implies y = 8, x = 32$.

Q2750. $x = 4y$. $x + 16 = 2(y + 16) \implies 4y + 16 =$

$2y + 32 \implies 2y = 16 \implies y = 8, x = 32.$

Q2751. $x - y = 2, F = 2x.$

Q2752. $x - y = 2, F = 2x.$

Q2753. $x - y = 2, F = 2x.$

Q2754. $x - y = 2, F = 2x.$

Q2755. $x - y = 2, F = 2x.$

Q2756. $x - y = 2, F = 2x.$

Q2757. $x - y = 2, F = 2x.$

Q2758. $x - y = 2, F = 2x.$

Q2759. $x - y = 2, F = 2x.$

Q2760. $x - y = 2, F = 2x.$

Q2761. $x - y = 2, F = 2x.$

Q2762. $x - y = 2, F = 2x.$

Q2763. $x - y = 2, F = 2x.$

Q2764. $x - y = 2, F = 2x.$

Q2765. $x - y = 2, F = 2x.$

Q2766. $x - y = 2, F = 2x.$

Q2767. $x - y = 2, F = 2x.$

Q2768. $x - y = 2, F = 2x.$

Q2769. $x - y = 2, F = 2x.$

Q2770. $x - y = 2, F = 2x.$

Q2771. $x - y = 2, F = 2x.$

Q2772. $x - y = 2, F = 2x.$

Q2773. $x - y = 2, F = 2x.$

Q2774. $x - y = 2, F = 2x.$

Q2775. $x - y = 2, F = 2x.$

Q2776. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2777. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2778. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2779. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2780. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2781. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2782. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2783. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2784. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2785. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2786. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2787. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2788. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2789. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2790. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2791. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2792. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2793. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2794. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2795. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2796. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2797. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2798. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2799. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$

Q2800. $x + y = 45 \implies y = 45 - x.\ (x - 5)(40 - x) = 124 \implies x^2 - 45x + 324 = 0 \implies x = 36, y = 9.$